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Unit-1
Partial differentiation and its applications

- Theory
- Solved examples
- Practice assignment

'B.S. Grewal' for higher engineering mathematics is also given at the end.

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“Our greatest weakness lies in giving up. The most certain way to succeed is always to try just one more time.”

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Unit 1 Partial derivatives

If $z = f(x,y)$ be a function of 2 variables x and y then,

$\frac{\partial z}{\partial x} \rightarrow$ partial derivative of z w.r.to x (treating y as constant)

$\frac{\partial z}{\partial y} \rightarrow$ partial derivative of z w.r.to y (treating x as constant)

$\frac{\partial^2 z}{\partial x^2} \rightarrow$ sec. order partial derivative w.r.to x.

$\frac{\partial^2 z}{\partial y^2} \rightarrow$ sec. order partial derivative w.r.to y.

$\frac{\partial^2 z}{\partial x \cdot \partial y} \rightarrow$ sec. order partial derivative of z first with respect to y then w.r.to x.

Note $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \cdot \partial x}$ (i.e., $z_{xy} = z_{yx}$)

Notations

$\frac{\partial z}{\partial x} \rightarrow z_x \rightarrow p$

$\frac{\partial z}{\partial y} \rightarrow z_y \rightarrow q$

$\frac{\partial^2 z}{\partial x^2} \rightarrow z_{xx} \rightarrow r$

$\frac{\partial^2 z}{\partial y^2} \rightarrow z_{yy} \rightarrow t$

$\frac{\partial^2 z}{\partial x \cdot \partial y} \rightarrow z_{xy} \rightarrow s$.

($\hookrightarrow z_{yx}$.)

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Eg If $u = e^{xyz}$ find $\frac{\partial^3 u}{\partial x \partial y \partial z}$.

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = \frac{\partial^2}{\partial x \partial y}\left(\frac{\partial u}{\partial z}\right)$$

$$= \frac{\partial^2}{\partial x . \partial y}\left(xy . e^{xyz}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial (xy\, e^{xyz})}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}\left[x e^{xyz} + x^2 y z\, e^{xyz}\right]$$

$$= e^{xyz} + xyz\, e^{xyz} + 2xyz\, e^{xyz} + x^2 y^2 z^2 e^{xyz}$$

$$= e^{xyz}\left(1 + 3xyz + x^2 y^2 z^2\right)$$

Eg. If $z(x+y) = x^2 + y^2$, show that

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$$

Sol^n we have $z(x+y) = x^2 + y^2$ — (1)

differentiating (1) partially w.r.to x gives

$$\frac{\partial z}{\partial x}(x+y) + z = 2x \quad \text{— (2)}$$

Now w.r.to y, $\frac{\partial z}{\partial y}(x+y) + z = 2y$ — (3)

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Subtracting (1) & (3),

$$(x+y)\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right) = 2(x-y)$$

$$\Rightarrow \left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right) = \frac{2(x-y)}{(x+y)}$$

Now, Adding (2) & (3),

$$\left(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\right)(x+y) + 2z = 2(x+y)$$

$$\Rightarrow \left(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\right) = 2 - \frac{2z}{(x+y)}$$

R.H.S. of (1) $= 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right) = 4\left[1-\left(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\right)\right]$

$$= 4\left[1-2+\frac{2z}{(x+y)}\right]$$

$$= 4\left[-1+\frac{2z}{(x+y)}\right] = 4\left[-1+\frac{2(x^2+y^2)}{(x+y)^2}\right]$$

$$= 4\left[\frac{x^2+y^2-2xy}{(x+y)^2}\right] = 4\left[\frac{x-y}{x+y}\right]^2$$

$= \left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2 =$ L.H.S. Hence proved.

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Eg. If $u = \log(x^3+y^3+z^3-3xyz)$ prove that

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}.$$

Soln

$$\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)u = \left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$$

$$= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{x^3+y^3+z^3-3xyz}.$$

$$= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-z}$$

$$= \frac{3}{(x+y+z)}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)u$$

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$$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$$

$$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{3}{x+y+z}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{3}{x+y+z}\right)+\frac{\partial}{\partial y}\left(\frac{3}{x+y+z}\right)+\frac{\partial}{\partial z}\left(\frac{3}{x+y+z}\right)$$

$$= \frac{-3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2}$$

$$= \frac{-9}{(x+y+z)^2}$$

Eg. If $V = \log(\tan x + \tan y + \tan z)$

find the value of $\sin 2x \frac{\partial V}{\partial x} + \sin 2y \frac{\partial V}{\partial y} + \sin 2z \frac{\partial V}{\partial z}$

Sol^n

$$\frac{\partial V}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y + \tan z}$$

$$\frac{\partial V}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z}$$

$$\frac{\partial V}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$$

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Euler's theor. (without proof)

If u. is a homogeneous function in x and y of deg. n then

$$x.\frac{\partial u}{\partial x}+y.\frac{\partial u}{\partial y}=nu.$$

Eg. If $u=\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that, $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\tan u$.

Sol^n Given, $u=\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$

$\Rightarrow \sin u=\frac{x^2+y^2}{x+y}$ = homog. f^n of degree 1.

$\Rightarrow n=1$.

$\therefore$ Apply Euler's theor. on $\sin u$, we get

$$x\frac{\partial(\sin u)}{\partial x}+y.\frac{\partial}{\partial y}(\sin u)=1.\sin u.$$

$$\Rightarrow x.\cos u.\frac{\partial u}{\partial x}+y\cos u.\frac{\partial u}{\partial y}=\sin u.$$

$$\Rightarrow x.\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{\sin u}{\cos u}=\tan u.$$ hence proved

Eg If $u=\log\left(\frac{x^2+y^2}{x+y}\right)$ prove that $x.\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=1$

Sol^n Given. $u=\log\left(\frac{x^2+y^2}{x+y}\right)$

$$\Rightarrow e^u=\frac{x^2+y^2}{x+y}=\frac{x(1+y^2/x^2)}{(1+y/x)}$$ is homog. f^n of deg. 1.

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$\Rightarrow n = 1$

Applying Euler's theor. on e^u, we get,

$$x \frac{\partial}{\partial x}(e^u) + y \cdot \frac{\partial}{\partial y}(e^u) = 1 \cdot e^u.$$

$$\Rightarrow x \cdot e^u \cdot \frac{\partial u}{\partial x} + y \cdot e^u \cdot \frac{\partial u}{\partial y} = e^u$$

$$\Rightarrow x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 1.$$

Deductions

If $f(u)$ is a homogeneous in x and y of deg. n, ~~where~~ then

(i) $$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)}.$$

(ii) $$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

where $g(u) = n \dfrac{f(u)}{f'(u)}$.

Eg. If $u = \log\left(\frac{x^2+y^2}{x+y}\right)$, find the value of $x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy}$

Soln By deduction (ii),

$$x^2 \cdot u_{xx} + 2xy \cdot u_{xy} + y^2 \cdot u_{yy} = g(u)[g'(u) - 1]$$

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& $g(u) = \dfrac{n.f(u)}{f'(u)}$

Since $e^u = \dfrac{x^2+y^2}{x+y}$ is homogeneous of deg 1.

$\therefore$ let $f(u) = e^u$. then $g(u) = 1 \cdot e^u/e^u = 1$

Hence, $x^2 \cdot u_{xx} + 2xy\, u_{xy} + y^2 \cdot u_{yy}$

$$= 1 \cdot [0-1] = -1$$

Eg. If $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$, find $x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy}$

Sol^n we have, $f(u) = \tan u = \dfrac{x^3+y^3}{x-y}$ → homog. of deg. 2.

$\therefore$ By deduction (ii),

$$g(u) = \frac{n f(u)}{f'(u)} = \frac{2 \cdot \tan u}{\sec^2 u} = \sin 2u.$$

& $x^2 \cdot u_{xx} + 2xy\, u_{xy} + y^2 \cdot u_{yy} = g(u)[g'(u)-1]$

$$= \sin 2u [2 \cos 2u - 1]$$

$$= 2 \sin 2u \cos 2u - \sin 2u$$

$$= \sin 4u - \sin 2u.$$

Total derivative

let $z = f(x,y)$ be a function of 2 indep. variables x, & y. then we define,

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

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$\left(\text{OR } \frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}\right)$ as the total derivative of x.

(ordinary derivative in terms of partial derivatives)

Deductions

If $f(x,y)=0$ be a func^n of x & y then,

(1) $\frac{dy}{dx} = -\frac{f_x}{f_y}$ (2) $\frac{d^2y}{dx^2} = -\frac{(q^2r - 2pqs + p^2t)}{q^3}$.

Eg If $z = \sqrt{x^2+y^2}$ & $x^3+y^3+3axy = 5a^3$

find $\frac{dz}{dx}$ at $x=y=a$.

Sol^n we know,

$$dz = \frac{\partial z}{\partial x}\cdot dx + \frac{\partial z}{\partial y}\cdot dy$$

$$\Rightarrow \frac{dz}{dx} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dx} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dx}$$

$$= \left(\frac{x}{\sqrt{x^2+y^2}}\right)\cdot 1 + \left(\frac{y}{\sqrt{x^2+y^2}}\right)\cdot\frac{dy}{dx}$$

$$= \left(\frac{1}{\sqrt{x^2+y^2}}\right)\left(x + y\cdot\frac{dy}{dx}\right)$$

Given, $f = x^3+y^3+3axy-5a^3 = 0$,

$$\Rightarrow \frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{(3x^2+3ay)}{3y^2+3ax} = -\frac{(x^2+ay)}{y^2+ax} \quad (1)$$

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$$\therefore \left(\frac{dz}{dx}\right)_{x=y=a} = \left[\frac{1}{\sqrt{x^2+y^2}}\left(x+y\cdot\frac{dy}{dx}\right)\right]_{x=y=a}$$

$$= \left[\frac{1}{\sqrt{x^2+y^2}}\left(x+y\left(\frac{-x^2-ay}{y^2+ax}\right)\right)\right]_{x=y=a} \quad \text{(Using ①)}$$

$$= \frac{1}{a\sqrt{2}}\left[a-a\right] = 0$$

Change of variables

Let $u = f(x,y)$ where $x = f_1(t_1,t_2)$, $y = f_2(t_1,t_2)$ — ①

then $\frac{\partial u}{\partial t_1} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial t_1} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial t_1}$ — ②

$\frac{\partial u}{\partial t_2} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial t_2} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial t_2}$ — ③

{ As x & y are functions of t_1 & t_2 & u is a f^n of x & y $\therefore$ u becomes f^n of t_1 & t_2.

Eg ① If $u = f(x-y, y-z, z-x)$

show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Sol^n let $X = x-y$

$Y = y-z$

$Z = \cancel{z} = x.$ [sic: $z-x$]

then $u = f(X,Y,Z)$, where X, Y, Z are as above.

Since u is of the form ①, Applying ② & ③

we get, $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial X}\cdot\frac{\partial X}{\partial x} + \frac{\partial u}{\partial Y}\cdot\frac{\partial Y}{\partial x} + \frac{\partial u}{\partial Z}\cdot\frac{\partial Z}{\partial x}$

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$$= \frac{\partial u}{\partial X}\cdot 1 + \frac{\partial u}{\partial Y}\cdot 0 + \frac{\partial u}{\partial Z}\cdot(-1) = \frac{\partial u}{\partial X} - \frac{\partial u}{\partial Z}$$

Similarly,

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial X}\cdot\frac{\partial X}{\partial y} + \frac{\partial u}{\partial Y}\cdot\frac{\partial Y}{\partial y} + \frac{\partial u}{\partial Z}\cdot\frac{\partial Z}{\partial y} = \frac{\partial u}{\partial Y} - \frac{\partial u}{\partial X}.$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial X}\cdot\frac{\partial X}{\partial z} + \frac{\partial u}{\partial Y}\cdot\frac{\partial Y}{\partial z} + \frac{\partial u}{\partial Z}\cdot\frac{\partial Z}{\partial z} = \frac{\partial u}{\partial Z} - \frac{\partial u}{\partial Y}$$

Adding, we get, $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Eg If $z = f(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$

then $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$ — (1)

Sol^n:

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r} = \cos\theta\cdot\frac{\partial z}{\partial x} + \sin\theta\frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial \theta} = -r\sin\theta\cdot\frac{\partial z}{\partial x} + r\cos\theta\cdot\frac{\partial z}{\partial y}$$

R.H.S. of (1) $= \cos^2\theta\left(\frac{\partial z}{\partial x}\right)^2 + \sin^2\theta\left(\frac{\partial z}{\partial y}\right)^2$

$+ 2\cos\theta\cdot\sin\theta\,\frac{\partial z}{\partial x}\cdot\frac{\partial z}{\partial y} + \frac{1}{r^2}\left[r^2\sin^2\theta\left(\frac{\partial z}{\partial x}\right)^2\right.$

$\left. + r^2\cos^2\theta\left(\frac{\partial z}{\partial y}\right)^2 - 2r^2\cos\theta\sin\theta\,\frac{\partial z}{\partial x}\cdot\frac{\partial z}{\partial y}\right]$

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$$= \cos^2\theta \left(\frac{\partial z}{\partial x}\right)^2 + \sin^2\theta \left(\frac{\partial z}{\partial y}\right)^2 + \sin^2\theta \left(\frac{\partial z}{\partial x}\right)^2 \quad (7)$$

$$+ \cos^2\theta \left(\frac{\partial z}{\partial y}\right)^2$$

$$= \left[\cos^2\theta\left(\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right)\right]$$

$$+ \sin^2\theta\left[\left(\frac{\partial z}{\partial y}\right)^2 + \left(\frac{\partial z}{\partial x}\right)^2\right]$$

$$= (\cos^2\theta + \sin^2\theta)\left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right]$$

$$= \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \text{L.H.S of } (1).$$

Taylor's expansion of function of 2 Variables.

I form

$$f(x+h, y+k) = f(x,y) + (h f_x + k f_y) + \frac{1}{2!}(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \cdots$$

II. form (about the pt. (a,b))

$$f(x,y) = f(a,b) + \left[(x-a)(f_x)_{(a,b)} + (y-b)(f_y)_{(a,b)}\right]$$

$$+ \frac{1}{2!}\left[(x-a)^2 (f_{xx})_{(a,b)} + 2(x-a)(y-b)(f_{xy})_{(a,b)} + (y-b)^2 (f_{yy})_{(a,b)}\right] + \cdots$$

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Eg. Expand e^{xy} about $(1,1)$ upto sec deg. term.

let $f(x,y) = e^{xy}$

& $a=1, b=1$

then Using II form,

$$f(x,y) = f(1,1) + [(x-1)(f_x)_{(1,1)} + (y-1)(f_y)_{(1,1)}]$$
$$+ \frac{1}{2!}[(x-1)^2 (f_{xx})_{(1,1)} + 2(x-1)(y-1)(f_{xy})_{(1,1)}$$
$$+ (y-1)^2 (f_{yy})_{(1,1)}] + \cdots$$

$$= e + [(x-1)(ye^{xy})_{(1,1)} + (y-1)(xe^{xy})_{(1,1)}]$$
$$+ \frac{1}{2}[(x-1)^2 (y^2 e^{xy})_{(1,1)} + (y-1)^2 (x^2 e^{xy})_{(1,1)}$$
$$+ 2(x-1)(y-1)(xye^{xy})_{(1,1)}]$$
$$+ \cdots$$

$$= e(x+y-1) + \frac{e}{2}[(x-1)^2 + (y-1)^2 + 2(x-1)(y-1)]$$
$$+ \cdots$$

$$= e(x+y-1) + \frac{e}{2}[(x+y-2)^2]$$ (upto 2nd deg. term)

Eg. Expand $x^2y + 3y - 2$ in powers of $(x-1)$ & $(y+2)$ by using Taylor's series.

let $f(x,y) = x^2y + 3y - 2$

let $a = 1, \quad b = -2$ then $f(1,-2) = -10$

$f_x = 2xy, \quad f_y = x^2 + 3, \quad f_{yy} = 0, \quad f_{xx} = 2y$

$f_{xy} = 2x, \quad f_{xxx} = 0, \quad f_{yyy} = 0$

$f_{xxy} = 2, \quad f_{xyy} = 0.$

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Using,

$$f(x,y) = f(a,b) + \left[(x-a)(f_x)_{(a,b)} + (y-b)(f_y)_{(a,b)}\right] + \frac{1}{2!}\left[(x-a)^2 (f_{xx})_{(a,b)} + 2(x-a)(y-b)(f_{xy})_{(a,b)} + (y-b)^2 (f_{yy})_{(a,b)}\right]$$

$$+ \frac{1}{3!}\left[(x-a)^3 (f_{xxx})_{(a,b)} + 3(x-a)^2(y-b)(f_{xxy})_{(a,b)} + 3(x-a)(y-b)^2 (f_{xyy})_{(a,b)} + (y-b)^3 (f_{yyy})_{(a,b)}\right] + \cdots$$

$$= -10 + \left[(x-1)(-4) + (y+2)(4)\right] + \frac{1}{2!}\left[(x-1)^2(-4) + 2(x-1)(y+2)(2) + (y+2)^2 \cdot 0\right]$$

$$+ \frac{1}{3!}\left[(x-1)^3 \cdot 0 + 3(x-1)^2(y+2)(2) + 3(x-1)(y+2)^2 \cdot 0 + (y+2)^3 \cdot 0\right] + \cdots$$

$$= -10 + \left[-4x + 4 + 4y + 8\right] + \frac{1}{2}\left[-4(x-1)^2 + 4(x-1)(y+2)\right] + \frac{1}{6}\left[6(x-1)^2(y+2)\right] + \cdots$$

$$= -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2) + 0 \cdots 0$$

(Since other higher order derivatives are 0)

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Errors and Approximations

Let $u = f(x,y)$

δx = small change in x

δy = small change in y

δu = resulting change in u due to change in x & y.

&

$$\delta u = \frac{\partial u}{\partial x}\cdot \delta x + \frac{\partial u}{\partial y}\cdot \delta y \quad \text{approx.}$$

Eg Compute the approximate value of $(1.04)^{3.01}$.

Sol^n: Let $f(x,y) = x^y$

Let $x = 1$, $\delta x = 0.04$ then $x + \delta x = 1.04$

$y = 3$, $\delta y = 0.01$ then $y + \delta y = 3.01$.

then, $\delta u = \frac{\partial u}{\partial x}\delta x + \frac{\partial u}{\partial y}\cdot \delta y$

$= yx^{y-1}\cdot \delta x + x^y \log x \cdot \delta y$

$= 3(0.04) + 0 \quad = 0.12.$

Eg If $f(x,y,z) = x^2 y^3 \cdot z^{1/10}$, using partial differentiation, find the app. value of f when $x = 1.99$, $y = 3.01$, $z = 0.98$

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Sol^n: Let $x = 2$, $\delta x = -0.01$
$y = 3$, $\delta y = .01$
$z = 1$, $\delta z = -0.02$

then $\delta u = u(x+\delta x, y+\delta y, z+\delta z) - u(x,y,z)$

$$= u(1.99, 3.01, 0.98) - u(2,3,1)$$

$$= \frac{\partial u}{\partial x}\cdot\delta x + \frac{\partial u}{\partial y}\cdot\delta y + \frac{\partial u}{\partial z}\cdot\delta z$$

$$\Rightarrow u(1.99, 3.01, 0.98) = u(2,3,1) + \frac{\partial u}{\partial x}\cdot\delta x + \frac{\partial u}{\partial y}\cdot\delta y + \frac{\partial u}{\partial z}\cdot\delta z.$$

$$= u(2,3,1) + (2xy^3z^{1/10})(-0.01) + (3x^2y^2z^{1/10})(0.01) + \left(\frac{x^2y^3}{10}\cdot z^{-9/10}\right)(-0.02)$$

$$= 108 + (2(2)(3)^3)(-0.01) + 3(2^2\cdot3^2)(0.01) - \frac{(0.02)}{10}(2^2\cdot3^3)$$

$$= 107.784 \text{ app.}$$

Jacobians

Let $u(x,y)$, $v(x,y)$ be functions of x & y then then determinant $\begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix}$ is known as the Jacobian of u & v w.r.to x & y

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and is denoted by $J\left(\frac{u, v}{x, y}\right)$ OR $\frac{\partial(u,v)}{\partial(x,y)}$

Similarly for $u(x,y,z)$, $v(x,y,z)$, $w(x,y,z)$

$$J\frac{(u,v,w)}{(x,y,z)} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y & \partial u/\partial z \\ \partial v/\partial x & \partial v/\partial y & \partial v/\partial z \\ \partial w/\partial x & \partial w/\partial y & \partial w/\partial z \end{vmatrix}$$

Results ① $\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$

② $\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(w,z)} = \frac{\partial(u,v)}{\partial(w,z)}$

Eg q) If $u = x^2 - y^2$, $v = 2xy$

then $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix}$

$$= \begin{vmatrix} 2x & (-2y) \\ 2y & 2x \end{vmatrix} = 4x^2 + 4y^2$$

Also if $x = r\cos\theta$, $y = r\sin\theta$ then determine $\frac{\partial(u,v)}{\partial(r,\theta)}$.

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)} = 4(x^2+y^2) \begin{vmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{vmatrix}$$

$$= 4r^2(r\cos^2\theta + r\sin^2\theta)$$

$$= 4r^3$$

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Eg. If $u = x+y+z$, $y+z = uv$, $z = uvw$

Find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

Now, $x = u-y-z = u - uv + z - z$
$= u(1-v)$

$y = uv - z = uv(1-w)$

$z = uvw.$

$$\therefore \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v & \partial x/\partial w \\ \partial y/\partial u & \partial y/\partial v & \partial y/\partial w \\ \partial z/\partial u & \partial z/\partial v & \partial z/\partial w \end{vmatrix}$$

$$= \begin{vmatrix} 1-v & -u & 0 \\ v(1-w) & u(1-w) & -uv \\ v.w & uw & uv \end{vmatrix}$$

$$= u^2v \begin{vmatrix} 1-v & -1 & 0 \\ v & 1 & 0 \\ vw & w & 1 \end{vmatrix} = u^2v.$$

→ If $u(x,y)$ & $v(x,y)$ are connected by the relations $f_1(u,v,x,y) = 0$ & $f_2(u,v,x,y) = 0$ then

$$\frac{\partial(u,v)}{\partial(x,y)} = (-1)^2 \frac{\frac{\partial(f_1,f_2)}{\partial(x,y)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}}$$

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Similarly if we have, $u(x,y,z)$, $v(x,y,z)$, $w(x,y,z)$, connected by the relations $f_1(u,v,w,x,y,z) = 0$, $f_2(u,v,w,x,y,z) = 0$ & $f_3(u,v,w,x,y,z) = 0$ then

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \frac{(-1)^3 \dfrac{\partial(f_1,f_2,f_3)}{\partial(x,y,z)}}{\dfrac{\partial(f_1,f_2,f_3)}{\partial(u,v,w)}}$$

Eg. If $u(x,y)$, $v(x,y)$ are such that

$$u^3+v = x+y^2$$

$$u+v^3 = x^2+y$$

let $f_1(u,v,x,y) = u^3+v-x-y^2 = 0$

$f_2(u,v,x,y) = u+v^3-x^2-y = 0$

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{(-1)^2 \dfrac{\partial(f_1,f_2)}{\partial(x,y)}}{\dfrac{\partial(f_1,f_2)}{\partial(u,v)}}$$

$$= \frac{(-1)^2 \begin{vmatrix} -1 & -2y \\ -2x & -1 \end{vmatrix}}{\begin{vmatrix} 3u^2 & 1 \\ 1 & 3v^2 \end{vmatrix}} = \frac{1-4xy}{9u^2v^2-1}$$

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Eg. If $u = xyz$, $v = x^2+y^2+z^2$, $w = x+y+z$

find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.

As,

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \frac{1}{\frac{\partial(u,v,w)}{\partial(x,y,z)}}$$

& $$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} yz & xz & xy \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

$$= yz(y-z)\cdot 2 - xz(x-z)\cdot 2 + xy(x-y)\cdot 2$$

$$= 2z[y(y-z) - x(x-z)] + 2xy(x-y)$$

$$= 2[z(y^2-x^2) + x(z^2-y^2) + y(x^2-z^2)]$$

$$\therefore \frac{\partial(x,y,z)}{\partial(u,v,w)} = \frac{1}{2[z(y^2-x^2)+x(z^2-y^2)+y(x^2-z^2)]}$$

Partial derivatives using Jacobians.

If $u(x,y)$, $v(x,y)$ are connected by the relations $f_1(u,v,x,y) = 0$, $f_2(u,v,x,y) = 0$

then $$\frac{\partial u}{\partial x} = \frac{-\frac{\partial(f_1,f_2)}{\partial(x,v)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}}, \quad \frac{\partial u}{\partial y} = \frac{-\frac{\partial(f_1,f_2)}{\partial(y,v)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}}$$

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$$\frac{\partial v}{\partial x} = \frac{-\frac{\partial(f_1,f_2)}{\partial(u,x)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}} \quad , \quad \frac{\partial v}{\partial y} = \frac{-\frac{\partial(f_1,f_2)}{\partial(u,y)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}}$$

Eg If $u^2+xv^2 = x+y$

$v^2+yu^2 = x-y$

find u_x & u_y using Jacobian.

let $f_1 = u^2+xv^2-x-y$

$f_2 = v^2+yu^2-x+y$

$$\frac{\partial u}{\partial x} = \frac{-\frac{\partial(f_1,f_2)}{\partial(x,v)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}} = \frac{-\begin{vmatrix} v^2-1 & 2xv \\ -1 & 2v \end{vmatrix}}{\begin{vmatrix} 2u & 2xv \\ 2uy & 2v \end{vmatrix}}$$

$$= \frac{-(v^2+x-1)}{2u(1-xy)}$$

Similarly

$$\frac{\partial v}{\partial y} = \frac{-\frac{\partial(f_1,f_2)}{\partial(u,y)}}{\frac{\partial(f_1,f_2)}{\partial(u,v)}} = \frac{-\begin{vmatrix} 2u & -1 \\ 2uy & 1 \end{vmatrix}}{\begin{vmatrix} 2u & 2xv \\ 2yu & 2v \end{vmatrix}}$$

$$= \frac{-2(u+yu)}{4(1-xy).uv}$$

$$= \frac{-(1+y)}{2v(1-xy)}$$

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Jacobians to determine functional dependence

Two functions $f(x,y)$ & $\phi(x,y)$ are s.t.b. functionally dependent if their Jacobian vanishes identically, i.e., $\frac{\partial(f,\phi)}{\partial(x,y)} = 0$

Eg Examine the functional dependence of $u = \frac{x-y}{1+xy}$ & $v = \tan^{-1}x - \tan^{-1}y$. If dependent then find the relation.

$$\frac{\partial u}{\partial x} = \frac{(1+xy) - (x-y)\cdot y}{(1+xy)^2} = \frac{(1+y^2)}{(1+xy)^2}$$

$$\frac{\partial u}{\partial y} = \frac{(1+xy)(-1) - (x-y)\cdot x}{(1+xy)^2} = -\frac{(1+x^2)}{(1+xy)^2}$$

$$\frac{\partial v}{\partial x} = \frac{1}{(1+x^2)}, \quad \frac{\partial v}{\partial y} = \frac{-1}{1+y^2}$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{1}{(1+xy)^2}\begin{vmatrix} 1+y^2 & -(1+x^2) \\ 1/1+x^2 & \frac{-1}{(1+y^2)} \end{vmatrix} = 0$$

$\Rightarrow$ u & v are functionally dependent.

$$\tan v = \frac{\tan(\tan^{-1}x) - \tan(\tan^{-1}y)}{1+\tan(\tan^{-1}x)\cdot\tan(\tan^{-1}y)} = \frac{x-y}{1+xy} = u.$$

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Ex → Show that $u = x+y+z$, $v = x^3+y^3+z^3-3xyz$, $w = x^2+y^2+z^2-xy-yz-zx$ are functionally dependent & hence find the relation

[Ans. : $v = uw$]

Maxima & Minima for a funcⁿ of 2 or more variables. :—

Working rule.

① For stationary (critical) points, put $f_x = 0$, $f_y = 0$.

② Compute $f_{xx} \cdot f_{yy} - (f_{xy})^2$.

③ Case (i) If $f_{xx} \cdot f_{yy} - (f_{xy})^2 < 0$ then we get a saddle pt. (i.e., pt. of inflexion)

Case (ii) If $f_{xx} \cdot f_{yy} - (f_{xy})^2 > 0$.

maxima if f_{xx} (or f_{yy}) < 0

minima if f_{xx} (or f_{yy}) > 0

Case (iii) If $f_{xx} \cdot f_{yy} - (f_{xy})^2 = 0$ then further investigation is needed.

Eg Find the max. & min. values of the function,

$$f(x,y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$$

$f_x = 4x^3 - 4x + 4y$ — ①

$f_y = 4y^3 + 4x - 4y$.

$f_x = 0$ & $f_y = 0$ $\Rightarrow$ $4x^3 - 4x + 4y = 0$
$4y^3 + 4x - 4y = 0$

$\Rightarrow x^3 + y^3 = 0$

$\Rightarrow y^3 = -x^3$

$\Rightarrow y = -x$ — ②

$\therefore$ Using ① & ②,
we get $x^3 - x - x = 0$
$\Rightarrow x = 0, \pm\sqrt{2}$.

$x = 0 \Rightarrow y = 0$
$x = \sqrt{2} \Rightarrow y = -\sqrt{2}$
$x = -\sqrt{2} \Rightarrow y = \sqrt{2}$

$\therefore$ stationary points are $(0,0)$, $(\sqrt{2}, -\sqrt{2})$ & $(-\sqrt{2}, \sqrt{2})$

Now, $\underbrace{f_{xx} \cdot f_{yy} - (f_{xy})^2}_{}$ $= 16(3x^2-1)(3y^2-1) - 16$

At $(0,0)$ $\to 0$

At $(\sqrt{2}, -\sqrt{2})$ $\to$ $f_{xx} \cdot f_{yy} - (f_{xy})^2 = 384 > 0$
& $(f_{xx})_{(\sqrt{2},-\sqrt{2})} = 20 > 0$

$\therefore$ $(\sqrt{2}, -\sqrt{2})$ is pt. of minima

At $(-\sqrt{2}, \sqrt{2})$ $\to$ $f_{xx} \cdot f_{yy} - (f_{xy})^2 > 0$
& $(f_{xx})_{(-\sqrt{2},\sqrt{2})} = 20 > 0$

$\therefore$ $(-\sqrt{2}, \sqrt{2})$ is also a pt. of minima.

$\therefore$ min. value of f at $(\sqrt{2}, -\sqrt{2})$ or $(-\sqrt{2}, \sqrt{2})$ $= -8$.

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Since $f_{xx} f_{yy} - (f_{xy})^2 = 0$ at $(0,0)$

$f(0,0) = 0.$

$\exists$ a point $(1,1)$ such that $f(1,1) = 2 > 0 = f(0,0)$

and a pt. $(1,0)$ such that $f(1,0) = -1 < 0 = f(0,0)$

i.e., for $(1,1)$ f attains minimum value at $(0,0)$

& for $(1,0)$ f attains maximum value at $(0,0)$

$\therefore$ $(0,0)$ becomes saddle point

(neither maxima nor minima at $(0,0)$)

Eg. Discuss the maxima or minima of $f(x,y) = x^3 y^2 (1-x-y)$

$f_x = 3x^2y^2 - 4x^3y^2 - 3x^2y^3$

$f_y = 2x^3y - 2x^4y - 3x^3y^2$

$f_x = 0 \Rightarrow x^2y^2(3-4x-3y) = 0$

$f_y = 0 \Rightarrow x^3y\,(2-2x-3y) = 0$

$\Rightarrow x = 0,\ y = 0$

$x = 1/2,\ y = 1/3$

$\therefore$ stationary pts. are $(0,0)$, $(1/2, 1/3)$.

At $(1/2, 1/3) \rightarrow f_{xx} f_{yy} - (f_{xy})^2 = \frac{1}{144} > 0$

& $(f_{xx})_{(1/2,1/3)} = -1/9 < 0$

$\therefore$ maxima at $(1/2, 1/3)$

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At $(0,0)$ $f_{xx}f_{yy} - (f_{xy})^2 = 0$

But $f(1/3, 1/3) = \left(\frac{1}{3}\right)^6 > 0 = f(0,0)$

$f(-1,-1) = -3 < 0 = f(0,0)$

$\Rightarrow$ $(0,0)$ is saddle pt. (By the reason given in previous example)

Ans. $\boxed{\text{Max. } f_{(1/2, 1/3)} = \frac{1}{432}}$

Lagrange's method of undetermined multipliers

Let $u = f(x,y,z)$ be a function, which is to be maximized or minimized & connected by the relation $\phi(x,y,z) = 0$.

Working rule to determine stationary points

1. write $F = f + \lambda\phi$
2. obtain the equations $F_x = 0$, $F_y = 0$, $F_z = 0$
3. Solve equations in (2) along with $\phi(x,y,z) = 0$ to determine st. points of $f(x,y,z)$. ($\lambda \to$ Lagrange's multiplier)

Eg. Find the smallest and largest value of $x+2y$ on the circle $x^2+y^2 = 1$.

Soln Let $f(x,y,z) = x+2y$

$\phi(x,y,z) = x^2+y^2-1 = 0.$

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Let $F = f + \lambda\phi = (x+2y) + \lambda(x^2+y^2-1)$

$F_x = 1 + 2x\lambda \Rightarrow x = -1/2\lambda$

$F_y = 2 + 2\lambda y \Rightarrow y = -1/\lambda$

putting in $\phi(x,y,z) = 0$, we get

$$\frac{1}{4\lambda^2} + \frac{1}{\lambda^2} = 1 \Rightarrow \lambda = \pm\frac{\sqrt{5}}{2}$$

$\lambda = \sqrt{5}/2 \Rightarrow x = -1/\sqrt{5}$ & $y = -2/\sqrt{5}$

$\lambda = -\sqrt{5}/2 \Rightarrow x = 1/\sqrt{5}$ & $y = 2/\sqrt{5}$

So stationary pts. are $(-1/\sqrt{5}, -2/\sqrt{5})$,
$(1/\sqrt{5}, 2/\sqrt{5})$.

~~At~~ $f(-1/\sqrt{5}, -2/\sqrt{5}) = -\sqrt{5}$

$f(1/\sqrt{5}, 2/\sqrt{5}) = \sqrt{5}$

$\therefore$ max. $f = \sqrt{5}$ at $(1/\sqrt{5}, 2/\sqrt{5})$
min. $f = -\sqrt{5}$ at $(-1/\sqrt{5}, -2/\sqrt{5})$

Eg 7. Find the volume of the greatest rectangular box that can be inscribed inside the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Sol^n: Let the edges of the box by x, y, z. then

Volume, $V = xyz$.

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Let $f = xyz$

$$\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$$

$$F = f + \lambda\phi = xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$$

$$F_x = yz + \frac{2\lambda x}{a^2} = 0 \Rightarrow \frac{a^2 yz}{x} = -2\lambda$$

$$F_y = xz + \frac{2y\lambda}{b^2} = 0 \Rightarrow \frac{xzb^2}{y} = -2\lambda$$

$$F_z = xy + \frac{2\lambda z}{c^2} = 0 \Rightarrow \frac{xyc^2}{z} = -2\lambda$$

i.e., $$\frac{a^2yz}{x} = \frac{xzb^2}{y} = \frac{xyc^2}{z}$$

$$\left.\begin{aligned} &\frac{xzb^2}{y} = \frac{xyc^2}{z} \Rightarrow \frac{y^2}{b^2} = \frac{z^2}{c^2} \\ &\frac{a^2yz}{x} = \frac{xyc^2}{z} \Rightarrow \frac{z^2}{c^2} = \frac{x^2}{a^2} \end{aligned}\right] \Rightarrow \frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} \quad \text{— (1)}$$

Using $\phi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, $\frac{x^2}{a^2} + \frac{x^2}{a^2} + \frac{x^2}{a^2} = 1$

$$\Rightarrow x = \frac{a}{\sqrt{3}} \quad \text{— (2)}$$

$$\& \therefore y = \frac{b}{\sqrt{3}}, \; z = \frac{c}{\sqrt{3}}$$

(By (1) & (2))

$$\therefore V = xyz = \frac{abc}{3\sqrt{3}}$$ Ans.

Formation of partial differential equations

① By eliminating arbitrary constant.
② By eliminating arbitrary function.

Eg Derive a partial diff eqn by eliminating arbitrary constant.

(a) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ — ①

Soln we have $z = \frac{1}{2}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$

$p = z_x = \frac{1}{2}\left(\frac{2x}{a^2}\right) = \frac{x}{a^2} \Rightarrow a^2 = \frac{x}{p}$

$q = z_y = \frac{1}{2}\left[\frac{2y}{b^2}\right] = \frac{y}{b^2} \Rightarrow b^2 = \frac{y}{q}$.

∴ putting in ①, $2z = \frac{x^2}{x/p} + \frac{y^2}{y/q}$

$\Rightarrow \boxed{2z = xp + yq}$ → required partial diff. eqn.

(b) $(x-a)^2 + (y-b)^2 + z^2 = 1$ — ②

differentiating w.r.to x, $2(x-a) + 2z.p = 0$
differentiating w.r.to y, $2(y-b) + 2z.q = 0$

$\Rightarrow (x-a) = -zp$
$(y-b) = -zq$

putting in ②,
$(-zp)^2 + (-zq)^2 + z^2 = 1$
$\Rightarrow \boxed{z^2(p^2+q^2+1) = 1}$

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Eg (a) $z = f(x+ay) + g(x-ay)$

$p = z_x = f'(x+ay) + g'(x-ay)$

$q = z_y = f'(x+ay) \cdot a - a g'(x-ay)$

$z_{xx} = f''(x+ay) + g''(x-ay)$

$z_{yy} = a^2 f''(x+ay) + a^2 g''(x-ay)$

$= a^2 \left(f''(x+ay) + g''(x-ay) \right)$

$= a^2 z_{xx}$.

$\therefore \boxed{z_{yy} = a^2 z_{xx}}$ Ans.

(b) $z = (x+y) \cdot \phi(x^2-y^2)$

$p = z_x = \phi(x^2-y^2) + (x+y)\phi'(x^2-y^2) \cdot (2x)$ — (1)

$q = z_y = \phi(x^2-y^2) + (x+y)\phi'(x^2-y^2)(-2y)$ (2)

(1) $\Rightarrow p - \phi(x^2-y^2) = (x+y)\phi'(x^2-y^2)(2x)$

$\Rightarrow p - \dfrac{z}{x+y} = (x+y)\phi'(x^2-y^2)(2x)$ — (3)

(2) $\Rightarrow q - \phi(x^2-y^2) = (x+y)\phi'(x^2-y^2)(-2y)$

$\Rightarrow q - \dfrac{z}{x+y} = (x+y)\phi'(x^2-y^2)(-2y)$ — (4)

Dividing (3) & (4), $\dfrac{p - z/(x+y)}{q - z/(x+y)} = \dfrac{-x}{y}$

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$\Rightarrow py(x+y) - zy = -xq(x+y) + xz$

$\Rightarrow py(x+y) = -xq(x+y) + z(x+y)$

$\Rightarrow \boxed{py + xq = z}$

Lagrange's method for solving linear partial diff. eqn.

A linear partial diff. eqn. is of the form $P(x,y,z)\cdot p + Q(x,y,z)\cdot q = R(x,y,z)$. —(1)

Working rule to find the soln of (1)

(i) write $\dfrac{dx}{P} = \dfrac{dy}{Q} = \dfrac{dz}{R}$

(ii) Solve eqn in (i), to get $u = a$ & $v = b$ as solutions

(iii) write the complete soln as $\phi(u,v) = 0$ OR $u = f(v)$.

Eg. obtain the solution of following linear partial diff. eqns.

(a) $y^2z\cdot p - x^2z\,q = x^2y$.

clearly It is of the form (1) with,
$P = y^2z,\quad Q = -x^2z,\quad R = x^2y$

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$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{y^2 z} = \frac{dy}{-x^2 z} = \frac{dz}{x^2 y}$$

Considering $\frac{dx}{y^2 \cancel{z}} = \frac{dy}{-x^2 \cancel{z}} \Rightarrow \frac{dx}{y^2} = \frac{dy}{-x^2}$

$\Rightarrow -x^2 dx = y^2 dy$

$\Rightarrow \underbrace{x^3 + y^3}_{u} = a$ (Integrating)

Next, $\frac{dy}{-x^2 z} = \frac{dz}{x^2 y}$

$\Rightarrow \frac{dy}{-z} = \frac{dz}{y}$

$\Rightarrow \underbrace{y^2 + z^2}_{v} = b$

$\therefore$ Complete solution is $\phi(u, v) = 0$

i.e., $\phi(x^3 + y^3, y^2 + z^2) = 0$.

(b) $(x^2 - y^2 - z^2)p + 2xy \cdot q = 2xz$

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} \quad -(1)$$

$\frac{dy}{2xy} = \frac{dz}{2xz} \Rightarrow \frac{dy}{y} = \frac{dz}{z}$

$\Rightarrow \log y - \log z = \log a$

$\Rightarrow \underbrace{y/z}_{u} = a$

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~~for~~ by multiplying & dividing each ratio in with x, y, z respectively in num. & denom. of (1), we get another fraction as,

$$\frac{x\,dx + y\,dy + z\,dz}{x(x^2-y^2-z^2) + y(2xy) + z(2xz)}$$

$$= \frac{x\,dx + y\,dy + z\,dz}{x(x^2+y^2+z^2)}$$ (Numerator is total derivative of ~~$x\,dx+y\,dy+z\,dz$~~ $x^2+y^2+z^2$)

$$= \frac{d(x^2+y^2+z^2)}{2x(x^2+y^2+z^2)}$$

$$\therefore \quad \frac{dz}{2xz} = \frac{d(x^2+y^2+z^2)}{2x(x^2+y^2+z^2)}$$

$$\Rightarrow \quad \frac{dz}{z} = \frac{d(x^2+y^2+z^2)}{(x^2+y^2+z^2)} \Rightarrow \log z = \log(x^2+y^2+z^2) + \log b$$

$$\Rightarrow \quad \frac{z}{x^2+y^2+z^2} = b$$

$\therefore$ Complete solⁿ is $\phi\left(y/z, \dfrac{z}{x^2+y^2+z^2}\right) = 0$

(c) $x^2(y-z).\,p + y^2(z-x)\,q = z^2(x-y)$

$$\Rightarrow \quad \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

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$$= \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{xy - zx + yz - yx + zx - yz}$$

$$= \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{0}$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

$$\Rightarrow \log x + \log y + \log z = \log a$$

$$\Rightarrow xyz = a$$

Next $\dfrac{\frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2}}{0}$ & similarly,

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = b$$

Complete solⁿ is $\phi\left(xyz, \frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) = 0.$

Non-linear partial diff. eqn. of order 1

The partial diff. eqn in which p & q occur other than 1st degree or in product is known as Non-linear partial diff. eqn.

Eg a) $p^2 + q^2 = 1$

b) $pq + p + q = 0.$

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Charpit's method (for solving non linear partial diff. eqn.)

Let $f(x, y, z, p, q) = 0$ be a non linear partial diff. eqn.

Write the charpit's subsidiary equations as,

$$\frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z}$$

and use $dz = p\,dx + q\,dy$.

Eg Solve the following non-linear partial diff. eqns.

(a) $(p^2+q^2) \cdot y = qz$

Let $f(x, y, z, p, q) = (p^2+q^2)y - qz = 0$

charpit's equations are,

$$\frac{dx}{-2py} = \frac{dy}{-2qy+z} = \frac{dz}{-p(2py) - q(2qy) + zq} = \frac{dp}{-pq} = \frac{dq}{p^2+q^2-q^2}$$

$$\Rightarrow \frac{dx}{-2py} = \frac{dy}{z-2qy} = \frac{dz}{-qz} = \frac{dp}{-pq} = \frac{dq}{p^2}$$

Using last two fractions, we get

$$p^2+q^2 = c^2 \quad -(1)$$

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Given $(p^2+q^2)y = qz$

$\Rightarrow \quad c^2 y = qz \quad$ (by ①)

$\Rightarrow \quad q = \dfrac{c^2 y}{z}$

$p^2 = c^2 - q^2 = c^2 - \dfrac{c^4 y^2}{z^2} = \dfrac{c^2(z^2 - y^2c^2)}{z^2}$

$\Rightarrow p = \dfrac{c}{z}\sqrt{z^2-y^2c^2}$

Using $dz = p\,dx + q\,dy$, we get

$dz = \dfrac{c}{z}\sqrt{z^2-y^2c^2}\,dx + \dfrac{c^2y}{z}\,dy$

$\Rightarrow z.dz = \left(c\sqrt{z^2-y^2c^2}\right)dx + (c^2y)\,dy$

$\Rightarrow z\,dz - c^2y\,dy = \left(c\sqrt{z^2-y^2c^2}\right)dx$

$\Rightarrow \dfrac{d(z^2-y^2c^2)}{2\sqrt{z^2-y^2c^2}} = c\,dx \quad \left\{ \because z\,dz - c^2y\,dy = d\dfrac{(z^2-y^2c^2)}{2} \right.$ Using total derivati

$\Rightarrow \boxed{\sqrt{z^2-y^2c^2} = cx + b}$

(b) $2xz - px^2 - 2qxy + pq = 0 \quad$ —①

charpit's equations are,

$\dfrac{dx}{x^2-q} = \dfrac{dy}{2xy-p} = \dfrac{dz}{-p(-x^2+q)-q(-2xy+p)} = \dfrac{dp}{2z-2qy}$

$= \dfrac{dq}{\cancel{-2qx}+\cancel{2qx}}$

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$$\Rightarrow \frac{dx}{x^2-q} = \frac{dy}{2xy-p} = \frac{dz}{px^2-2pq+2qxy} = \frac{dp}{2z-2qy} = \frac{dq}{0}$$

$\Rightarrow dq = 0$

$\Rightarrow q = a$

put $q = a$ in (1) to get $p = \frac{2axy-2xz}{a-x^2}$

$$\Rightarrow p = \frac{2x(z-ay)}{x^2-a}$$

Using $dz = pdx + qdy$

$$= \frac{2x(z-ay)}{x^2-a}dx + a\,dy$$

$$\Rightarrow \frac{dz-a\,dy}{(z-ay)} = \frac{2x\,dx}{x^2-a}$$

$$\Rightarrow \frac{d(z-ay)}{(z-ay)} = \frac{2x\,dx}{x^2-a}, \quad \text{(On Integrating)}$$

$$\Rightarrow \log(z-ay) = \log(x^2-a) + \log b$$

$$\Rightarrow \boxed{\frac{z-ay}{x^2-a} = b}$$

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Some standard forms of non linear partial diff. eqn's

① $f(p,q)=0$ (i.e, if diff. eqn is indep. of x, y, z).

It's soln is given by $z = ax+by+c$ where $f(a,b)=0$.

Eg. $p-q=1$.

$f(p,q) = p-q-1=0$.

$f(a,b) = a-b-1=0$

$\Rightarrow b = a-1$.

$\therefore$ soln is $z = ax+by+c$

$$\Rightarrow \boxed{z = ax+(a-1)\cdot y + c.}$$

② $f(z,p,q)=0$ (i.e., eqn's not containing x and y)

(i) Assume $u = x+ay$

& put $p = \frac{dz}{du}$, $q = a\frac{dz}{du}$ in the given eqn.

(ii) Solve resulting ordinary diff. eqn in z & u.

(iii) Replace u by $x+ay$.

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Eg $p(1+q) = qz$ — ①

It is of the form $f(z,p,q)=0$

let $u = x+ay$, $p = \frac{dz}{du}$, $q = \frac{a\,dz}{du}$

then ① becomes

$$\frac{dz}{du}\left(1+\frac{a\,dz}{du}\right) = \frac{a\,dz}{du}\cdot z$$

$$\Rightarrow \quad \frac{a\,dz}{du} = az-1$$

$$\Rightarrow \quad \frac{a.dz}{az-1} = du$$

On integrating, $\log(az-1) = u+b$

$= x+ay+b$. Ans.

③ $f(x,p) = f(y,q)$ (i.e., terms containing x & p can be separated from the terms containing y & q.)

Eg. $p^2+q^2 = x+y$

$$\Rightarrow p^2 - x = y - q^2$$

let $p^2 - x = y - q^2 = a$

$p^2 - x = a$ and $y - q^2 = a$

$\Rightarrow p = \sqrt{a+x}$ and $q = \sqrt{y-a}$

$$dz = p\,dx + q\,dy = \left(\sqrt{a+x}\right)dx + \left(\sqrt{y-a}\right)dy$$

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On integrating,

$$z = \frac{2}{3}(x+a)^{3/2} + \frac{2}{3}(y-a)^{3/2} + b$$ Ans.

4) If $z = px + qy + f(p,q)$ (where $f(p,q)$ is any fn of p & q.

↳ Clairaut's eqn.

Its solution is given by

$z = ax + by + f(a,b)$ (i.e., replace p by a & q by b)

where a & b are arbitrary constants

Eg. a) $z = px + qy + \sqrt{1+p^2+q^2}$

Soln is $z = ax + by + \sqrt{1+a^2+b^2}$.

b) $z = px + qy + 2\sqrt{pq}$

Soln is $z = ax + by + 2\sqrt{ab}$.

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PUSH YOURSELF
BECAUSE, NO ONE
ELSE IS GOING
TO DO IT FOR YOU.

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Mansi sharma
BVCOE.

ASSIGNMENT 1 (Unit-1)
{*From previous year papers*}

All students are required to complete this assignment before 25^{th} feb.

Partial differentiation

Q.1 (i) If $\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1$, show that

$$u_x^2 + u_y^2 + u_z^2 = 2(xu_x + yu_y + zu_z).$$

(ii) Show that at any point on the surface $x^x y^y z^z = c$ where $x = y = z$, we have $z_{xy} = \frac{-1}{x\log(ex)}$.

Q.2 If $u = \log(x^3 + y^3 - x^2y - y^2x)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 u = -\frac{4}{(x+y)^2}$.

Q.3 If $u = \log(x^3+y^3+z^3-3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$.

Q.4 If $w = r^m : r = x^2+y^2+z^2$, then show $\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = m(m+1)r^{m-2}$.

Q.5 If $u = f(r)$ where $r^2 = x^2 + y^2 + z^2$, show that,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r}f'(r).$$

Q.6 If $u = f(r)$ and $x = r\cos\theta, y = r\sin\theta$, show that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r).$$

Q.7 If $e^{-z/(x^2-y^2)} = x - y$ then determine the value of $yz_x + xz_y$.

Q.8 If $u = lx + my, v = mx - ly$, show that,

$$\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial x}{\partial u}\right) = \frac{l^2}{l^2+m^2} \quad and \quad \left(\frac{\partial y}{\partial v}\right)\left(\frac{\partial v}{\partial y}\right) = \frac{l^2+m^2}{l^2}.$$

Change of independent variables

Q.9 If $u = u(x-y, y-z, z-x)$ then show that $u_x + u_y + u_z = 0$.

Q.10 If $v = v(2x-3y, 3y-4z, 4z-2x)$ then show that $6v_x + 4v_y + 3v_z = 0$.

Q.11 Change the equation in cartesian coordinates $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ in to polar coordinates r and θ [Hint.Let $x = r\cos\theta, y = r\sin\theta$ and apply change of variable to find u_{xx}, u_{yy}].

Q.12 Transform the equation $\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$ into $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$ by changing the variables u and v to x and y by means of the relations $x = u\cos\theta - v\sin\theta, y = u\sin\theta + v\cos\theta$.

Q.13 If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then find $x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + z^2\frac{\partial^2 u}{\partial z^2}$.

Euler's theorem

Q.14 Verify Euler's theorem for the function $x^{1/3}y^{-1/4}\tan^{-1}\frac{y}{x}$.

Q.15 If $u = \log\frac{x^2+y^2}{x+y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1$ and hence evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

Q.16 If $u = \sin^{-1}\frac{x^3+y^3}{\sqrt{x}+\sqrt{y}}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{5}{2}\tan u$.

Q.17 Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = -2\cot u$, where $u = \cos^{-1}\left(\frac{x^3+y^3+z^3}{ax+by+cz}\right)$ and hence evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.[Use deductions]

Q.18 If $u = \tan^{-1}\frac{x^3+y^3}{x-y}$, show that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 2\cos 3u \sin u$.

Q.19 If $z = xf(y/x) + g(y/x)$, show that $x^2z_{xx} + 2xyz_{xy} + y^2z_{yy} = 0$.

Q.20 If z is a homogeneous function of degree n then show that

$$x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} = n(n-1)z.$$

[Hint. differentiate Euler's equation first w.r.t. x then w.r.t. y.]

Q.21 If $z = x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$, then prove that $z_{xy} = z_{yx}$ and evaluate $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2}$.

Errors and approximations

Q.22 Use Taylor's theorem to expand $f(x, y) = e^x \log(1 + y)$ in powers of x and y.

Q.23 Obtain Taylor's expansion of $\frac{1}{1+x-y}$ in powers of $(x - 1)$ and $(y - 1)$.

Q.24 Compute $\tan^{-1}\left(\frac{0.9}{1.1}\right)$ by Taylor's expansion.

Q.25 Compute the value of $[(3.8)^2 + 2(2.1)^3]^{1/5}$ using the theory of approximation. $[\delta u = 0.01]$

Q.26 If $f(x, y, z) = x^2y^3z^{1/10}$, using partial differentiation, find the app. value of f at $(1.99, 3.01, 0.98)$. [107.784]

Q.27 In a plane triangle ABC find the maximum value of $\cos A \cos B \cos C$. [Hint. Use $A + B + C = \pi$ and then put $f_A = 0, f_B = 0$ to find angles A and B, ans.$= 1/8$]

Maxima and minima

Q.28 Evaluate the extreme values of $x^4 + y^4 + 4xy - 2y^2$.

Q.29 Discuss the maxima and minima for the following functions,

(i) $2(x - y)^2 - x^4 - y^4$.

(ii) $x^3y^2(1 - x - y)$.

Q.30 Find the dimensions of the rectangular box with maximum volume inscribed by a solid ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.[Hint Use Lagrange's method of undetermined multipliers with $V = f(x, y, z) = xyz$ and $\phi(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$]

Q.31 Find the dimensions of rectangular box(without top) with a given volume so that the material used is maximum.

Q.32 Find the minimum distance between origin and a point $p(x, y, z)$, which lie on the plane $ax + by + cz + d = 0$

Q.33 Find the minimum distance between origin and a point $p(x, y, z)$, which lie on the line of intersection of two planes $x+y+z = 1$ and $2x+y+3z = 6$.

Q.34 The temperature T at any point (x, y, z) of space is given by $T = 400xyz^2$. Find the highest temperature at the surface of the sphere $x^2 + y^2 + z^2 = 1$.

Q.35 Find the extreme values of $\sqrt{x^2 + y^2}$ when $13x^2 - 10xy + 13y^2 = 72$.

Q.36 Find the minimum value of $x^2+y^2+z^2$ subject to the condition $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 0$.

Q.37 If $u = ax^2+by^2+cz^2$ where $x^2+y^2+z^2 = 1$ and $lx+my+nz = 0$ prove that the stationary values of u satisfy the equation $\frac{l^2}{a-u} + \frac{m^2}{b-u} + \frac{n^2}{c-u}$. [Hint use $F = f + \lambda\phi_1 + \mu\phi_2$]

Q.38 Find the extreme point of $x^3+y^3+z^3$ subject to the conditions $x+y+z = 1$ and $x^2 + y^2 = a^2$.

Jacobians

Q.39 If $J = \frac{\partial(u,v)}{\partial(x,y)}$ and $J^* = \frac{\partial(x,y)}{\partial(u,v)}$, show that $JJ^* = 1$.

Q.40 If $u = x + y^2, v = y + z^2, w = z + x^2$ then using Jacobians, show that $\frac{\partial x}{\partial u} = -\frac{1}{1+8xyz}$.

Q.41 If $u = xyz, v = x^2 + y^2 + z^2, w = x + y + z$, then find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.

Q.42 If $u^3+v+w = x+y^2+z^2, u+v^3+w = x^2+y+z^2, u+v+w^3 = x^2+y^2+z$ find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

Q.43 Calculate $\frac{\partial(u,v)}{\partial(r,\theta)}$, if $u = 2axy, v = a(x^2-y^2)$ where $x = r\cos\theta, y = r\sin\theta$.

Q.44 Show that $u = x^2+y^2+2xy+2x+2y, v = e^{x+y}$ are dependent and find the relation between them.[Ans.$u = (\log v)^2 + 2\log v$]

Q.45 Investigate the functional dependence of $u = \frac{x-y}{1+xy}$ and $v = \tan^{-1} x - \tan^{-1} y$, if dependent, find the relation.

Partial differential equations

Q.46 Form the partial differential equation by eliminating arbitrary function from $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$.

Q.47 Find the differential equation of all spheres whose centres lie on the z-axis.

Q.48 Form a partial differential equation from the following,

(i) $x^2 + y^2 = (z - c)^2 \tan^2 \alpha$.

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

(iii) $z = f(y/x) + \phi(xy)$.

(iv) $lx + my + nz = \phi(x^2 + y^2 + z^2)$.

(v) $z = (x + y)\phi(x^2 - y^2)$.

Q.49 Obtain the general solution of the following **linear** partial diff. equations,
(i) $y^2zp - x^2zq = x^2y$

(ii) $x(y - z)p + y(z - x)q = z(x - y)$.

(iii) $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

(iv) $\frac{y-z}{yz}p + \frac{z-x}{zx}q = \frac{x-y}{xy}$.

(v) $y^2p - xyq = x(z - 2y)$.

(vi) $x^2p + y^2q - z^2 = 0$ [first write it as $Pp + Qq = R$]

Q.50 Solve the following partial differential equations,

(i) $z = p^2 + q^2$.

(ii) $p(1 - q^2) = q(1 - z)$.

(iii) $z^2 = pqxy$.

(iv) $p^2 + q^2 = x^2 + y^2$.

(v) $z = px + qy + c\sqrt{1 + p^2 + q^2}$.

(vi) $p^2x^2 = z(z - qy)$.

(vii) $p^2 + q^2 = z^2(x + y)$.

If a function is continuous at all points of a region, then it is said to be *continuous in that region.* A function which is not continuous at a point is said to be *discontinuous* at that point.

Obs. *Usually, the limit is the same irrespective of the path along which the point (x, y) approaches (a, b) and*

$$\underset{x\to a}{Lt}\left\{\underset{y\to b}{Lt} f(x,y)\right\}=\underset{y\to b}{Lt}\left\{\underset{x\to a}{Lt} f(x,y)\right\}$$

But it is not always so, as the following examples show :

$$\text{Lt}\left(\frac{x-y}{x+y}\right) \text{ as } (x,y)\to(0,0) \text{ along the line } y=mx$$

$$=\underset{x\to 0}{\text{Lt}}\frac{x-mx}{x+mx}=\frac{1-m}{1+m} \text{ which is different for lines with different slopes.}$$

Also $\underset{x\to 0}{\text{Lt}}\left[\underset{y\to 0}{\text{Lt}}\left(\frac{x-y}{x+y}\right)\right]=\underset{x\to 0}{\text{Lt}}\left(\frac{x}{x}\right)=1$, whereas $\underset{y\to 0}{\text{Lt}}\left[\underset{x\to 0}{\text{Lt}}\left(\frac{x-y}{x+y}\right)\right]=\underset{y\to 0}{\text{Lt}}\left(\frac{-y}{y}\right)=1.$

$\therefore$ As (x, y) is made to approach $(0, 0)$ along different paths, $f(x, y)$ approaches different limits. Hence the two repeated limits are not equal and $f(x, y)$ is discontinuous at the origin.

Also the function is not defined at $(0, 0)$ since $f(x, y) = 0/0$ for $x = 0, y = 0$.

(4) *As in the case of functions of one variable, the following results hold :*

I. *If* $\underset{\substack{x\to a\\ y\to b}}{Lt} f(x,y)=l$ and $\underset{\substack{x\to a\\ y\to b}}{Lt} g(x,y)=m,$

then (*i*) *If* $\underset{\substack{x\to a\\ y\to b}}{Lt}[f(x,y)\pm g(x,y)]=l\pm m$ (*ii*) $\underset{\substack{x\to a\\ y\to b}}{Lt}[f(x,y)\,.\,g(x,y)]=l\,.\,m$

(*iii*) $\underset{\substack{x\to a\\ y\to b}}{Lt}[f(x,y)/g(x,y)]=l/m$ $(m\neq 0)$

II. *If* $f(x, y), g(x, y)$ *are continuous at* (a, b) *then so also are the functions*

$f(x, y) \pm g(x, y), f(x, y)\,.\,g(x, y)$ and $f(x, y)/g(x, y)$

provided $g(x, y) \neq 0$ in the last case.

PROBLEMS 5.1

Evaluate the following limits :

1. $\underset{\substack{x\to 1\\ y\to 2}}{\text{Lt}}\frac{2x^2y}{x^2+y^2+1}$ 2. $\underset{\substack{x\to 0\\ y\to 0}}{\text{Lt}}\frac{xy}{x^2+y^2}$ 3. $\underset{\substack{x\to \infty\\ y\to 2}}{\text{Lt}}\frac{xy+1}{x^2+2y^2}$ 4. $\underset{\substack{x\to 1\\ y\to 1}}{\text{Lt}}\frac{x(y-1)}{y(x-1)}$

5. If $f(x,y)=\frac{x-y}{2x+y}$, show that $\underset{x\to 0}{\text{Lt}}\left[\underset{y\to 0}{\text{Lt}} f(x,y)\right]\neq\underset{y\to 0}{\text{Lt}}\left[\underset{x\to 0}{\text{Lt}} f(x,y)\right]$

   Also show that the function is discountinuous at the origin.

6. Show that the function $f(x, y) = x^2 + 2y$, $(x, y) \neq (1, 2)$

   $3(x, y) = (1, 2)$ $= 0$

   is discontinuous at $(1, 2)$.

7. Investigate the continuity of the function

   $f(x, y) = xy/(x^2 + y^2)$, $(x, y) \neq (0, 0)$

   $= 0$ $(x, y) = (0, 0)$

   at the origin.

Note. In whatever follows, all the functions considered are continuous and their partial derivatives (as defined below) exist.

5.2 PARTIAL DERIVATIVES

Let $z = f(x, y)$ be a function of two variables x and y.

If we keep y as constant and vary x alone, then z is a function of x only. The derivative of z with respect to x, treating y as constant, is called the *partial derivative of z with respect to x* and is denoted by one of the symbols

$$\frac{\partial z}{\partial x}, \frac{\partial f}{\partial x}, f_x(x, y), D_x f. \qquad \text{Thus } \frac{\partial z}{\partial x} = \underset{\delta x \to 0}{\text{Lt}} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}$$

Similarly, the derivative of z with respect to y, keeping x as constant, is called the *partial derivative of z with respect to y* and is denoted by one of the symbols.

$$\frac{\partial z}{\partial y}, \frac{\partial f}{\partial y}, f_y(x, y), D_y f. \qquad \text{Thus } \frac{\partial z}{\partial y} = \underset{\partial y \to 0}{\text{Lt}} \frac{f(x, y + \delta y) - f(x, y)}{\delta y}$$

Similarly, if z is a function of three or more variables $x_1, x_2, x_3, \ldots$ *the partial derivative of z with respect to x_1, is obtained by differentiating z with respect to x_1, keeping all other variables constant* and is written as $\partial z/\partial x_1$.

In general f_x and f_y are also functions of x and y and so these can be differentiated further partially with respect to x and y.

Thus $$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} \text{ or } \frac{\partial^2 z}{\partial x^2} \text{ or } f_{xx}, \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y} \text{ or } \frac{\partial^2 f}{\partial x \partial y} \text{ or } f_{yx}{}^*$$

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x} \text{ or } \frac{\partial^2 f}{\partial y \partial x} \text{ or } f_{xy}, \text{ and } \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} \text{ or } \frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy}.$$

It can easily be verified that, in all ordinary cases,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}.$$

Sometimes we use the following notation

$$\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial^2 z}{\partial x^2} = r, \frac{\partial^2 z}{\partial x \partial y} = s, \frac{\partial^2 z}{\partial y^2} = t.$$

Example 5.1. *Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$.*

Solution. We have $z = x^3 + y^3 - 3axy$.

$$\therefore \quad \frac{\partial z}{\partial x} = 3x^2 + 0 - 3ay(1) = 3x^2 - 3ay, \text{ and } \frac{\partial z}{\partial y} = 0 + 3y^2 - 3ax(1) = 3y^2 - 3ax$$

Also $$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}(3x^2 - 3ay) = 6x, \quad \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}(3x^2 - 3ay) = -3a$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}(3y^2 - 3ax) = 6y, \text{ and } \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}(3y^2 - 3ax) = -3a.$$

We observe that $\dfrac{\partial^2 z}{\partial y \partial x} = \dfrac{\partial^2 z}{\partial x \partial y}$.

Example 5.2. *If $u = x^2 \tan^{-1} \dfrac{y}{x} - y^2 \tan^{-1} \dfrac{x}{y}$,*

show that $$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}.$$ (Mumbai, 2008 S)

and $$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$ (Madras, 2000)

Solution. We have $$\frac{\partial u}{\partial y} = x^2 \cdot \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x} - \left\{2y \cdot \tan^{-1} \frac{x}{y} + y^2 \cdot \frac{1}{1 + (x/y)^2} \cdot \left(-\frac{x}{y^2}\right)\right\}$$

$$= \frac{x^3}{x^2 + y^2} - 2y \tan^{-1} \frac{x}{y} + \frac{xy^2}{x^2 + y^2} = x - 2y \tan^{-1} \frac{x}{y}.$$

*It is important to note that in the subscript notation the subscripts are written in the same order in which we differentiate whereas in the '∂' notation the order is opposite.

$\therefore$ $$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left\{x - 2y \tan^{-1}\frac{x}{y}\right\} = 1 - 2y \cdot \frac{1}{1+(x/y)^2} \cdot \frac{1}{y} = 1 - \frac{2y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2}.$$

Similarly, $$\frac{\partial u}{\partial x} = 2x \tan^{-1} y/x - y$$

and $$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left\{2x \tan^{-1}\frac{y}{x} - y\right\} = \frac{x^2-y^2}{x^2+y^2}.$$ Hence the result.

Example 5.3. *If $z = f(x + ct) + \phi(x - ct)$, prove that*

$$\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$$ *(J.N.T.U., 2006 ; V.T.U., 2003 S)*

Solution. We have $\frac{\partial z}{\partial x} = f'(x + ct) \cdot \frac{\partial}{\partial x}(x + ct) + \phi'(x - ct)\frac{\partial}{\partial x}(x - ct) = f'(x + ct) + \phi'(x - ct)$

and $$\frac{\partial^2 z}{\partial x^2} = f''(x + ct) + \phi''(x - ct) \quad ...(i)$$

Again $$\frac{\partial z}{\partial t} = f'(x + ct)\frac{\partial}{\partial t}(x + ct) + \phi'(x - ct)\frac{\partial}{\partial t}(x - ct) = cf'(x - ct) - c\phi'(x - ct)$$

and $$\frac{\partial^2 z}{\partial t^2} = c^2 f''(x + ct) + c^2 \phi''(x - ct) = c^2 [f''(x + ct) + \phi''(x - ct)] \quad ...(ii)$$

From (*i*) and (*ii*), it follows that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$.

Obs. This is an important partial differential equation, known as *wave equation* (§ 18.4).

Example 5.4. *If $\theta = t^n e^{-r^2/4t}$, what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$?*

(Nagpur, 2009 ; Kurukshetra, 2006 ; U.P.T.U., 2006)

Solution. We have $\frac{\partial \theta}{\partial r} = t^n \cdot e^{-r^2/4t} \cdot \left(\frac{-2r}{4t}\right) = -\frac{r}{2} t^{n-1} e^{-r^2/4t}$

$\therefore$ $$r^2 \frac{\partial \theta}{\partial r} = -\frac{r^3}{2} t^{n-1} \cdot e^{-r^2/4t}$$

and $$\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = -\frac{3r^2}{2} t^{n-1} e^{-r^2/4t} - \frac{r^3}{2} t^{n-1} \cdot e^{-r^2/4t}\left(-\frac{2r}{4t}\right)$$

$\therefore$ $$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \left(-\frac{3}{2} t^{n-1} + \frac{r^2}{4} t^{n-2}\right) e^{-r^2/4t}$$

Also $$\frac{\partial \theta}{\partial t} = nt^{n-1} \cdot e^{-r^2/4t} + t^n \cdot e^{-r^2/4t} \cdot \frac{r^2}{4t^2} = \left(nt^{n-1} + \frac{1}{4} r^2 t^{n-2}\right) e^{-r^2/4t}$$

Since $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$,

$\therefore$ $$\left(-\frac{3}{2} t^{n-1} + \frac{1}{4} r^2 t^{n-2}\right) e^{-r^2/4t} = \left(nt^{n-1} + \frac{1}{4} r^2 t^{n-2}\right) e^{-r^2/4t} \quad \text{or} \quad \left(n + \frac{3}{2}\right) t^{n-1} e^{-r^2/4t} = 0.$$

Hence $n = -3/2$.

Example 5.5. *If $v = (x^2 + y^2 + z^2)^{-1/2}$, prove that*

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0.$$ *(Laplace equation)** *(V.T.U., 2006 ; Osmania, 2003 S)*

*See footnote p. 18.

Solution. We have $\frac{\partial v}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2} \cdot 2x = -x(x^2+y^2+z^2)^{-3/2}$

and $$\frac{\partial^2 v}{\partial x^2} = -1[1 \cdot (x^2+y^2+z^2)^{-3/2} + x(-3/2)(x^2+y^2+z^2)^{-5/2} \cdot 2x]$$
$$= -(x^2+y^2+z^2)^{-5/2}[x^2+y^2+z^2-3x^2] = (x^2+y^2+z^2)^{-5/2}(2x^2-y^2-z^2)$$

Similarly, $$\frac{\partial^2 v}{\partial y^2} = (x^2+y^2+z^2)^{-5/2}(-x^2+2y^2-z^2) \text{ and } \frac{\partial^2 v}{\partial z^2} = (x^2+y^2+z^2)^{-5/2}(-x^2-y^2+2z^2)$$

Hence $$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = (x^2+y^2+z^2)^{-5/2} \cdot (0) = 0.$$

Obs. *A function v satisfying the Laplace equation is said to be a* **harmonic function.**

Example 5.6. *If $u = \log(x^3+y^3+z^3-3xyz)$, show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.*

(P.T.U., 2010 ; Anna, 2009 ; Bhopal, 2008 ; U.P.T.U., 2006)

Solution. We have $\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$, $\frac{\partial u}{\partial y} = \frac{3y^2-3zx}{x^3+y^3+z^3-3xyz}$, $\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$

$$\therefore \quad \frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z} = \frac{3(x^2+y^2+z^2-xy-yz-zx)}{x^3+y^3+z^3-3xyz}$$
$$= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)} = \frac{3}{x+y+z} \qquad \textit{(V.T.U., 2009)}$$

Now $$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$$
$$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right) = \left(\frac{3}{x+y+z}\right)$$
$$= -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} = -\frac{9}{(x+y+z)^2}.$$

Example 5.7. *If $\frac{x^2}{a^2+u}+\frac{y^2}{b^2+u}+\frac{z^2}{c^2+u} = 1$, prove that*

$$\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2 = 2\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}\right) \qquad \textit{(U.P.T.U., 2003)}$$

Solution. We have $x^2(a^2+u)^{-1}+y^2(b^2+u)^{-1}+z^2(c^2+u)^{-1} = 1$ ...(*i*)

Differentiating (*i*) partially w.r.t. x, we get

$$2x(a^2+u)^{-1} - x^2(a^2+u)^{-2}\frac{\partial u}{\partial x} - y^2(b^2+u)^{-2}\frac{\partial u}{\partial y} - z^2(c^2+u)^{-2}\frac{\partial u}{\partial z} = 0$$

or $$\frac{2x}{a^2+u} = \left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{y^2}{(c^2+u)^2}\right\}\frac{\partial u}{\partial x}$$

or $$\frac{\partial u}{\partial x} = \frac{2x}{(a^2+u)v} \text{ where } v = \frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{y^2}{(c^2+u)^2}$$

Similarly differentiating (*i*) partially w.r.t. y, we get

$$\frac{2y}{b^2+u} = \left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right\}\frac{\partial u}{\partial y} \text{ or } \frac{\partial u}{\partial y} = \frac{2y}{(b^2+u)v}$$

Similarly, differentiating (i) partially w.r.t. z, we get

$$\frac{2z}{(b^2+u)} = \left\{\frac{x^2}{(a^2+u)^2} + \frac{y^2}{(b^2+u)^2} + \frac{z^2}{(c^2+u)^2}\right\}\frac{\partial u}{\partial z} \text{ or } \frac{\partial u}{\partial z} = \frac{2z}{(c^2+u)v}$$

$$\therefore \quad \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = \frac{4}{v^2}\left\{\frac{x^2}{(a^2+u)^2} + \frac{y^2}{(b^2+u)^2} + \frac{y^2}{(c^2+u)^2}\right\} = \frac{4}{v} \qquad ...(ii)$$

Also
$$2\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}\right) = 2\left\{\frac{2x^2}{(a^2+u)v} + \frac{2y^2}{(b^2+u)v} + \frac{2z^2}{(c^2+u)v}\right\}$$

$$= \frac{4}{v}\left\{\frac{x^2}{(a^2+u)} + \frac{y^2}{(b^2+u)} + \frac{z^2}{(c^2+u)}\right\} = \frac{4}{v} \qquad \text{[By } (i)\text{] } ...(iii)$$

Hence the equality of (ii) and (iii) proves the result.

Example 5.8. *If* $u = x^y$, *show that* $\dfrac{\partial^3 u}{\partial x^2 \partial y} = \dfrac{\partial^3 u}{\partial x \partial y \partial x}$. *(Anna, 2009)*

Solution. We have $\dfrac{\partial u}{\partial y} = x^y \log_e x$ and $\dfrac{\partial^2 u}{\partial x \partial y} = yx^{y-1} \,.\, \log x + x^y \,.\, \dfrac{1}{x} = x^{y-1}(y \log x + 1)$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2 \partial y} = \frac{\partial}{\partial x}[x^{y-1}(y \log x + 1)] \qquad ...(i)$$

Again
$$\frac{\partial u}{\partial x} = yx^{y-1} \text{ and } \frac{\partial^2 u}{\partial y \partial x} = 1 \,.\, x^{y-1} + y\left(\frac{1}{x}x^y \log x\right) = x^{y-1}(1 + y \log x)$$

$$\therefore \quad \frac{\partial^3 u}{\partial x \partial y \partial x} = \frac{\partial}{\partial x}[x^{y-1}(y \log x + 1)] \qquad ...(ii)$$

From (i) and (ii) follows the required result.

PROBLEMS 5.2

1. Evaluate $\partial z/\partial x$ and $\partial z/\partial y$, if
   (i) $z = x^2 y - x \sin xy$; (ii) $z = \log(x^2 + y^2)$;
   (iii) $z = \tan^{-1}\{(x^2+y^2)/(x+y)\}$; (iv) $x + y + z = \log z$.
2. If $z(x+y) = x^2 + y^2$, show that $\left(\dfrac{\partial z}{\partial x} - \dfrac{\partial z}{\partial y}\right)^2 = 4\left(1 - \dfrac{\partial z}{\partial x} - \dfrac{\partial z}{\partial y}\right)$. *(V.T.U., 2003)*
3. If $z = e^{ax+by} f(ax - by)$, prove that $b\dfrac{\partial z}{\partial x} + a\dfrac{\partial z}{\partial y} = 2abz$. *(V.T.U., 2010)*
4. Given $u = e^{r\cos\theta}\cos(r\sin\theta)$, $v = e^{r\cos\theta}\sin(r\sin\theta)$; prove that $\dfrac{\partial u}{\partial r} = \dfrac{1}{r}\dfrac{\partial v}{\partial \theta}$ and $\dfrac{\partial v}{\partial r} = -\dfrac{1}{r}\dfrac{\partial u}{\partial \theta}$.
5. If $z = \tan(y + ax) - (y - ax)^{3/2}$, show that $\dfrac{\partial^2 z}{\partial x^2} = a^2\dfrac{\partial^2 z}{\partial y^2}$. *(Mumbai, 2009)*
6. Verify that $f_{xy} = f_{yx}$, when f is equal to (i) $\sin^{-1}(y/x)$; (ii) $\log x \tan^{-1}(x^2 + y^2)$.
7. If $f(x, y) = (1 - 2xy + y^2)^{-1/2}$, show that $\dfrac{\partial}{\partial x}\left[(1-x^2)\dfrac{\partial f}{\partial x}\right] + \dfrac{\partial}{\partial y}\left[y^2\dfrac{\partial f}{\partial y}\right] = 0$. *(Rohtak, 2006 S)*
8. Prove that $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ if (i) $u = \tan^{-1}\left[\dfrac{2xy}{x^2 - y^2}\right]$. (ii) $u = \log(x^2 + y^2) + \tan^{-1}(y/x)$. *(Anna, 2009)*
9. If $v = \dfrac{1}{\sqrt{t}}e^{-x^2/4a^2 t}$, prove that $\dfrac{\partial v}{\partial t} = a^2\dfrac{\partial^2 v}{\partial x^2}$.

10. The equation $\frac{\partial u}{\partial t} = \mu \frac{\partial^2 u}{\partial x^2}$ refers to the conduction of heat along a bar without radiation, show that if $u = Ae^{-gx} \sin (nt - gx)$, where A, g, n are positive constants then $g = \sqrt{(n/2\mu)}$.
11. Find the value of n so that the equation $V = r^n (3 \cos^2 \theta - 1)$ satisfies the relation
$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta}\left(\sin \theta \frac{\partial V}{\partial \theta}\right) = 0.$$
12. If $z = \log (e^x + e^y)$, show that $rt - s^2 = 0$ where $r = \partial^2 z/\partial x^2$, $s = \partial^2 z/\partial x \partial y$, $t = \partial^2 z/\partial y^2$.
13. If $u = \frac{y}{z} + \frac{z}{x}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.
14. Let $r^2 = x^2 + y^2 + z^2$ and $V = r^m$, prove that $V_{xx} + V_{yy} + V_{zz} = m(m+1)r^{m-2}$. (*Raipur, 2005*)
15. If $v = \log (x^2 + y^2 + z^2)$, prove that $(x^2 + y^2 + z^2)\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) = 2$.
16. If $v = x^y . y^x$, prove that $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = v(x + y + \log v)$. (*Anna, 2005*)
17. If $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log ex)^{-1}$. (*Bhopal, 2008*)
18. If $u = e^{xyz}$, find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$. (*Rajasthan, 2005 ; Osmania, 2003 S*)

5.3 WHICH VARIABLE IS TO BE TREATED AS CONSTANT

(1) Consider the equation $x = r \cos \theta$, $y = r \sin \theta$ ...(1)

To find $\partial r/\partial x$, we need a relation between r and x. Such a relation will contain one more variable θ or y, for we can eliminate only one variable out of four from the relations (1). Thus the two possible relations are

$$r = x \sec \theta \quad ...(2) \qquad \text{and} \qquad r^2 = x^2 + y^2 \quad ...(3)$$

Now we can find $\partial r/\partial x$ either from (2) by treating θ as constant or from (3) by regarding y as constant. And there is no reason to suppose that the two values of $\partial r/\partial x$ so found, are equal. *To avoid confusion as to which variable is regarded constant, we introduce the following :*

Notation : $(\partial r/\partial x)_\theta$ *means the partial derivative of r with respect to x keeping* θ *constant in* a relation expressing r as a function of x and θ.

Thus from (2), $(\partial r/\partial x)_\theta = \sec \theta$.

When no indication is given regarding the variable to be kept constant, then according to convention $(\partial/\partial x)$ always means $(\partial/\partial x)_y$ and $\partial/\partial y$ means $(\partial/\partial y)_x$. Similarly, $\partial/\partial r$ means $(\partial/\partial r)_\theta$ and $\partial/\partial\theta$ means $(\partial/\partial\theta)_r$.

(2) In thermodynamics, we come across ten variables such as p (pressure), v (volume), T (temperature), W (work), ϕ (entropy) etc. Any one of these can be expressed as a function of other two variables *e.g.*, $T = f(p, v)$, $T = g(p, \phi)$

As we shall see, these respectively give rise to the following results :

$$dT = \frac{\partial T}{\partial p} dp + \frac{\partial T}{\partial v} dv \qquad ...(i)$$

$$dT = \frac{\partial T}{\partial p} dp + \frac{\partial T}{\partial \phi} d\phi \qquad ...(ii)$$

Now, $\partial T/\partial p$ appearing in (*i*), has been obtained from T as function of p and v, treating v as constant, we write it as $(\partial T/\partial p)_v$.

Similarly, $\partial T/\partial p$ occurring in (*ii*), is written as $(\partial T/\partial p)_\phi$.

Example 5.9. *If $u = f(r)$ and $x = r \cos \theta$, $y = r \sin \theta$, prove that*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r).$$ (*S.V.T.U., 2008 ; Rajasthan, 2006 ; U.P.T.U., 2005*)

Solution. We have $\frac{\partial u}{\partial x} = f'(r) \cdot \frac{\partial r}{\partial x}$ and $\frac{\partial^2 u}{\partial x^2} = f''(r) \cdot \left(\frac{\partial r}{\partial x}\right)^2 + f'(r) \cdot \frac{\partial^2 r}{\partial x^2}$

Similarly, $$\frac{\partial^2 u}{\partial y^2} = f''(r) \cdot \left(\frac{\partial r}{\partial y}\right)^2 + f'(r) \cdot \frac{\partial^2 r}{\partial y^2}$$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) \cdot \left[\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2\right] + f'(r)\left[\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2}\right]$$

Now to find $\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}$ etc., we write $r = (x^2 + y^2)^{1/2}$

$$\therefore \quad \frac{\partial r}{\partial x} = \frac{1}{2}(x^2 + y^2)^{-1/2} \cdot 2x = \frac{x}{r} \quad \text{and} \quad \frac{\partial^2 r}{\partial x^2} = \frac{r \cdot 1 - x \cdot \partial r/\partial x}{r^2} = \frac{r - x^2/r}{r^2} = \frac{y^2}{r^3}.$$

Similarly, $$\frac{\partial r}{\partial y} = \frac{y}{x} \text{ and } \frac{\partial^2 r}{\partial y^2} = \frac{x^2}{r^3}.$$

Substituting the values of $\partial r/\partial x$ etc. in (*i*), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r)\left[\frac{x^2}{r^2} + \frac{y^2}{r^2}\right] + f'(r)\left[\frac{y^2}{r^3} + \frac{x^2}{r^3}\right] = f''(r) + \frac{1}{r}f'(r).$$

Example 5.10. *If $x = e^{r\cos\theta}\cos(r\sin\theta)$ and $y = e^{r\cos\theta}\sin(r\sin\theta)$, prove that* $\frac{\partial x}{\partial \theta} = -r\frac{\partial y}{\partial r}, \frac{\partial y}{\partial \theta} = r\frac{\partial x}{\partial r}$.

Hence show that $\frac{\partial^2 x}{\partial \theta^2} + r\frac{\partial x}{\partial r} + r^2\frac{\partial^2 x}{\partial r^2} = 0.$

Solution. We have $x = e^{r\cos\theta}\cos(r\sin\theta)$

$$\therefore \quad \frac{\partial x}{\partial \theta} = e^{r\cos\theta}(-r\sin\theta) \cdot \cos(r\sin\theta) + e^{r\cos\theta}[-\sin(r\sin\theta)] \cdot r\cos\theta$$
$$= -re^{r\cos\theta}[\sin\theta\cos(r\sin\theta) + \cos\theta\sin(r\sin\theta)]$$
$$= -re^{r\cos\theta}\sin(\theta + r\sin\theta) \quad \text{...(i)}$$

and $$\frac{\partial x}{\partial r} = e^{r\cos\theta} \cdot \cos\theta \cdot \cos(r\sin\theta) - e^{r\cos\theta}\sin\theta\,(r\sin\theta)\sin\theta$$
$$= e^{r\cos\theta}\cos(\theta + r\sin\theta) \quad \text{...(ii)}$$

Similarly, $y = e^{r\cos\theta}\sin(r\sin\theta)$ gives

$$\frac{\partial y}{\partial \theta} = re^{r\cos\theta}\cos(\theta + r\sin\theta) \quad \text{...(iii)}$$

and $$\frac{\partial y}{\partial r} = e^{r\cos\theta}\sin(\theta + r\sin\theta) \quad \text{...(iv)}$$

From (*i*) and (*iv*), $\frac{\partial x}{\partial \theta} = -r\frac{\partial y}{\partial r}$ ...(*v*)

From (*ii*) and (*iii*), $\frac{\partial y}{\partial \theta} = r\frac{\partial x}{\partial r}$ ...(*vi*)

From (*v*), $$\frac{\partial^2 x}{\partial \theta^2} = -r\frac{\partial^2 y}{\partial \theta\partial r} = -r\frac{\partial^2 y}{\partial r\partial \theta}$$

From (*vi*), $\frac{\partial x}{\partial r} = \frac{1}{r}\frac{\partial y}{\partial \theta}$ which gives $\frac{\partial^2 x}{\partial r^2} = -\frac{1}{r^2}\frac{\partial y}{\partial \theta} + \frac{1}{r}\frac{\partial^2 y}{\partial r\partial \theta}$

$$\therefore \quad \frac{\partial^2 x}{\partial \theta^2} + r\frac{\partial x}{\partial r} + r^2\frac{\partial^2 x}{\partial r^2} = -r\frac{\partial^2 y}{\partial r\partial \theta} + \frac{\partial y}{\partial \theta} - \frac{\partial y}{\partial \theta} + r\frac{\partial^2 y}{\partial r\partial \theta} = 0.$$

PROBLEMS 5.3

1. If $x = r\cos\theta, y = r\sin\theta$, show that (i) $\dfrac{\partial r}{\partial x} = \dfrac{\partial x}{\partial r}$ (ii) $\dfrac{1}{r}\dfrac{\partial x}{\partial\theta} = r\dfrac{\partial\theta}{\partial x}$, (iii) $\left(\dfrac{\partial r}{\partial x}\right)^2 + \left(\dfrac{\partial r}{\partial y}\right)^2 = 1.$ *(Burdwan, 2003)*

2. If $x^2 = au + bv, y^2 = au - bv$, prove that $\left(\dfrac{\partial u}{\partial x}\right)_y \cdot \left(\dfrac{\partial x}{\partial u}\right)_v = \dfrac{1}{2} = \left(\dfrac{\partial v}{\partial y}\right)_x \cdot \left(\dfrac{\partial y}{\partial v}\right)_u.$

3. If $u = lx + my, v = mx - ly$, show that $\left(\dfrac{\partial u}{\partial x}\right)_y \cdot \left(\dfrac{\partial x}{\partial u}\right)_v = \dfrac{l^2}{l^2+m^2}, \left(\dfrac{\partial y}{\partial v}\right)_x \left(\dfrac{\partial v}{\partial y}\right)_u = \dfrac{l^2+m^2}{l^2}.$

4. If $x = r\cos\theta, y = r\sin\theta$, prove that

(i) $\dfrac{\partial^2 r}{\partial x^2} + \dfrac{\partial^2 r}{\partial y^2} = \dfrac{1}{r}\left[\left(\dfrac{\partial r}{\partial x}\right)^2 + \left(\dfrac{\partial r}{\partial y}\right)^2\right]$ (ii) $\dfrac{\partial^2\theta}{\partial x^2} + \dfrac{\partial^2\theta}{\partial y^2} = 0 \quad (x \neq 0, y \neq 0).$

5. If $z = x\log(x + r) - r$ where $r^2 = x^2 + y^2$, prove that $\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial y^2} = \dfrac{1}{x+y}, \dfrac{\partial^3 z}{\partial x^3} = -\dfrac{x}{r^3}.$ *(Mumbai, 2008)*

6. If $u = f(r)$ where $r = \sqrt{(x^2+y^2+z^2)}$, show that $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = f''(r) + \dfrac{2}{r}f'(r).$

5.4 (1) HOMOGENEOUS FUNCTIONS

An expression of the form $a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \ldots + a_ny^n$ in which every term is of the nth degree, is called a homogeneous function of degree n. This can be rewritten as

$$x^n[a_0 + a_1(y/x) + a_2(y/x)^2 + \ldots + a_n(y/x)^n].$$

Thus *any function $f(x, y)$ which can be expressed in the form $x^n\phi(y/x)$, is called a* **homogeneous function** of degree n in x and y.

For instance, $x^3\cos(y/x)$ is a homogeneous function of degree 3, in x and y.

In general, a function $f(x, y, z, t, \ldots)$ is said to be a homogeneous function of degree n in $x, y, z, t, \ldots$, if it can be expressed in the form $x^n\phi(y/x, z/x, t/x, \ldots)$.

(2) Euler's theorem on homogeneous functions*. *If u be a homogeneous function of degree n in x and y, then*

$$\mathbf{x}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \mathbf{y}\frac{\partial \mathbf{u}}{\partial \mathbf{y}} = \mathbf{nu}.$$

Since u is a homogeneous function of degree n in x and y, therefore,

$$u = x^n f(y/x)$$

$$\therefore \quad \frac{\partial u}{\partial x} = nx^{n-1}f\left(\frac{y}{x}\right) + x^n f'\left(\frac{y}{x}\right)\cdot y\left(-\frac{1}{x^2}\right) = nx^{n-1}f\left(\frac{y}{x}\right) - yx^{n-2}f'\left(\frac{y}{x}\right)$$

and $$\frac{\partial u}{\partial y} = x^n f'\left(\frac{y}{x}\right)\cdot\frac{1}{x} = x^{n-1}f'\left(\frac{y}{x}\right). \text{ Hence } x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nx^n f\left(\frac{y}{x}\right) = nu.$$

In general, if u be a homogeneous function of degree n in $x, y, z, t, \ldots$, then,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + t\frac{\partial u}{\partial t} \ldots = nu.$$

Example 5.11. *Show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2u\log u$ where $\log u = (x^3 + y^3)/(3x + 4y)$.*

Solution. Since $$z = \log u = \frac{x^3+y^3}{3x+4y} = x^2\cdot\frac{1+(y/x)^3}{3+4(y/x)},$$

* After an enormously creative Swiss mathematician *Leonhard Euler* (1707–1783). He studied under John *Bernoulli* and became a professor of mathematics in St. Petersburg, Russia. Even after becoming totally blind in 1771, he contributed to almost all branches of mathematics.

$\therefore$ z is a homogeneous function of degree 2 in x and y.

By Euler's theorem, we get

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z \quad ...(i)$$

But $$\frac{\partial z}{\partial x} = \frac{1}{u}\frac{\partial u}{\partial x} \text{ and } \frac{\partial z}{\partial y} = \frac{1}{u}\frac{\partial u}{\partial y}$$

Hence (i) becomes

$$x \cdot \frac{1}{u}\frac{\partial u}{\partial x} + y \cdot \frac{1}{u}\frac{\partial u}{\partial y} = 2 \log u \quad \text{or} \quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u \log u.$$

Example 5.12. *If* $u = \sin^{-1} \dfrac{x + 2y + 3z}{x^8 + y^8 + z^8}$, *find the value of* $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z}$. *(U.P.T.U., 2004)*

Solution. Here u is not a homogeneous function. We therefore, write

$$\omega = \sin u = \frac{x + 2y + 3z}{x^8 + y^8 + z^8} = x^{-7} \cdot \frac{1 + 2(y/x) + 3(z/x)}{1 + (y/x)^8 + (z/x)^8}$$

Thus ω is a homogeneous function of degree -7 in x, y, z. Hence by Euler's theorem

$$x\frac{\partial \omega}{\partial x} + y\frac{\partial \omega}{\partial y} + z\frac{\partial \omega}{\partial z} = (-7)\,\omega \quad ...(i)$$

But $$\frac{\partial \omega}{\partial x} = \cos u\,\frac{\partial u}{\partial x}, \frac{\partial \omega}{\partial y} = \cos u\,\frac{\partial u}{\partial y}, \frac{\partial \omega}{\partial z} = \cos u\,\frac{\partial u}{\partial z}$$

$\therefore$ (i) becomes $\quad x \cos u\dfrac{\partial u}{\partial x} + y \cos u\dfrac{\partial u}{\partial y} + z \cos u\dfrac{\partial u}{\partial z} = -7 \sin u \quad$ or $\quad x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = -7 \tan u.$

Example 5.13. *If* $u = \dfrac{x^3y^3z^3}{x^3 + y^3 + z^3} + \log\left(\dfrac{xy + yz + zx}{x^2 + y^2 + z^2}\right)$, *find the value of* $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z}$.

(Mumbai, 2009)

Solution. Let $\quad v = \dfrac{x^3y^3z^3}{x^3 + y^3 + z^3}$ and $w = \log\left(\dfrac{xy + yz + zx}{x^2 + y^2 + z^2}\right) \quad ...(i)$

so that $\quad u = v + w$

Since $v = x^6 \dfrac{(y/x)^3(z/x)^3}{1 + (y/x)^3 + (z/x)^3}$, therefore v is a homogeneous function of degree 6 in x, y, z.

Hence by Euler's theorem $\quad x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} + z\dfrac{\partial v}{\partial z} = 6v \quad ...(ii)$

Since $w = \log\left\{\dfrac{\frac{y}{x} + \frac{y}{x}\cdot\frac{z}{x} + \frac{z}{x}}{1 + \left(\frac{y}{x}\right)^2 + \left(\frac{z}{x}\right)^2}\right\}$ therefore w is a homogeneous function of degree zero in x, y, z.

Hence by Euler's theorem $\quad x\dfrac{\partial w}{\partial x} + y\dfrac{\partial w}{\partial y} + z\dfrac{\partial w}{\partial z} = 0 \quad ...(iii)$

Addint (ii) and (iii), we obtain

$$x\left(\frac{\partial v}{\partial x} + \frac{\partial w}{\partial x}\right) + y\left(\frac{\partial v}{\partial y} + \frac{\partial w}{\partial y}\right) + z\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial z}\right) = 6v$$

or $$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 6\,\frac{x^3y^3z^3}{x^3 + y^3 + z^3} \quad \text{[By } (i)\text{]}$$

Example 5.14. *If z is a homogeneous function of degree n in x and y, show that*

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial z^2} = n(n-1)\, z.$$ *(Anna, 2009 ; V.T.U., 2007 ; U.P.T.U., 2006)*

Solution. By Euler's theorem, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$ ...(*i*)

Differentiating (*i*) partially w.r.t. x, we get $x \frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial x} + y \frac{\partial^2 z}{\partial x \partial y} = n \frac{\partial z}{\partial x}$

i.e., $$x \frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 z}{\partial x \partial y} = (n-1) \frac{\partial z}{\partial x}$$ ...(*ii*)

Again differentiating (*i*) partially w.r.t. y, we get $x \frac{\partial^2 z}{\partial y \partial x} + \frac{\partial z}{\partial y} + y \frac{\partial^2 z}{\partial y^2} = n \frac{\partial z}{\partial y}$

i.e., $$x \frac{\partial^2 z}{\partial x \partial y} + y \frac{\partial^2 z}{\partial y^2} = (n-1) \frac{\partial z}{\partial y}$$ ...(*iii*)

Multiplying (*ii*) by x and (*iii*) by y and adding, we get

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = (n-1)\left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right) = n(n-1)\, z.$$ [By (*i*)]

Example 5.15. *If $u = \sin^{-1} \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$.*

(Rajasthan, 2006 ; Calicut, 2005)

and $$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4 \cos^3 u}.$$ *(P.T.U., 2006)*

Solution. Here u is not a homogeneous function but $z = \sin u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$ is a homogeneous function of degree 1/2 in x and y.

$\therefore$ By Euler's theorem, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{2} z$

or $$x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} = \frac{1}{2} \sin u$$

Thus $$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$$ ...(*i*)

Differentiating (*i*) w.r.t. x partially, we get

$$x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y \frac{\partial^2 u}{\partial x \partial y} = \frac{1}{2} \sec^2 u \frac{\partial u}{\partial x} \quad \text{or} \quad x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = \left(\frac{1}{2} \sec^2 u - 1 \right) \frac{\partial u}{\partial x}$$ ...(*ii*)

Again differentiating (*i*) w.r.t. y partially, we get

$$x \frac{\partial^2 u}{\partial y \partial x} + y \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = \frac{1}{2} \sec^2 u \frac{\partial u}{\partial y} \quad \text{or} \quad x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2} \sec^2 u - 1 \right) \frac{\partial u}{\partial x}$$ ...(*iii*)

Multiplying (*ii*) by x and (*iii*) by y and adding, we obtain

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2} \sec^2 u - 1 \right)\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right)$$

or $$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2} \sec^2 u - 1 \right)\left(\frac{1}{2} \tan u \right)$$ [By (*i*)]

$$= \frac{1}{4} \frac{\sin u}{\cos^3 u} - \frac{1}{2} \frac{\sin u}{\cos u} = -\frac{\sin u\,(2\cos^2 u - 1)}{4 \cos^3 u}$$

Hence $$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4 \cos^3 u}.$$

PROBLEMS 5.4

1. Verify Euler's theorem, when (*i*) $f(x, y) = ax^2 + 2hxy + by^2$
(*ii*) $f(x, y) = x^2 (x^2 - y^2)^3/(x^2 + y^2)^3$.
(*iii*) $f(x, y) = 3x^2 yz + 5xy^2z + 4z^4$ (*J.N.T.U., 1999*)

2. If $u = \sin^{-1} \dfrac{x}{y} + \tan^{-1} \dfrac{y}{x}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 0$. (*Hazaribagh, 2009 ; Osmania, 2003 S*)

3. If $u = \sin^{-1} \dfrac{x^2 + y^2}{x + y}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \tan u$ (*Bhopal, 2009 ; V.T.U., 2003*)

4. If $\sin u = \dfrac{x^2y^2}{x + y}$, show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 3 \tan u$. (*Kottayam, 2005 ; V.T.U., 2003 S*)

5. If $u = \cos^{-1} \dfrac{x + y}{\sqrt{x} + \sqrt{y}}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = -\dfrac{1}{2} \cot u$. (*V.T.U., 2004*)

6. Show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2u \log u$, where $u = e^{x^2 + y^2}$ (*P.T.U., 2010*)

7. If $z = f(y/x) + \sqrt{(x^2 + y^2)}$, show that $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = \sqrt{x^2 + y^2}$. (*Mumbai, 2008*)

8. If $u = \dfrac{x}{y + z} + \dfrac{y}{z + x} + \dfrac{z}{x + y}$, show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 0$. (*V.T.U., 2000 S*)

9. If $\sin u = \dfrac{x + 2y + 3z}{\sqrt{(x^8 + y^8 + z^8)}}$, show that $xu_x + yu_y + zu_z + 3 \tan u = 0$. (*S.V.T.U., 2009 ; U.T.U., 2009*)

10. If $z = x\phi\left(\dfrac{y}{x}\right) + \psi\left(\dfrac{y}{x}\right)$, prove that $x^2\dfrac{\partial^2 z}{\partial x^2} + 2xy\dfrac{\partial^2 z}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = 0$. (*S.V.T.U., 2009 ; U.P.T.U., 2006*)

11. If $u = \tan^{-1} \dfrac{x^3 + y^3}{x + y}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \sin 2u$. (*P.T.U., 2009 S*)

and $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = 2 \cos 3u \sin u$. (*Mumbai, 2009 ; Bhopal, 2008 ; S.V.T.U., 2007*)

12. Given $z = x^n f_1(y/x) + y^{-n} f_2(x/y)$, prove that $x^2\dfrac{\partial^2 z}{\partial x^2} + 2xy\dfrac{\partial^2 z}{\partial x \partial y} + y^2\dfrac{\partial^2 z}{\partial y^2} + x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = n^2 z$.
(*Kurukshetra, 2009 S ; Rohtak, 2003*)

13. If $u = x^2 \tan^{-1}(y/x) - y^2 \tan^{-1}(x/y)$, evaluate $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2}$. (*U.T.U., 2009 ; Hissar, 2005 S*)

14. If $u = \tan^{-1}(y^2/x)$, prove that $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = -\sin^2 u \cdot \sin 2u$. (*Bhillai, 2005 ; P.T.U., 2005*)

15. If $u = \operatorname{cosec}^{-1} \left(\dfrac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}\right)^{1/2}$, prove that $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = \dfrac{\tan u}{12}\left(\dfrac{13}{12} + \dfrac{\tan^2 u}{12}\right)$.
(*Mumbai, 2008 ; Rohtak, 2006 S*)

5.5 (1) TOTAL DERIVATIVE

If $u = f(x, y)$, where $x = \phi(t)$ and $y = \psi(t)$, then we can express u as a function of t alone by substituting the values of x and y in $f(x, y)$. Thus we can find the ordinary derivative du/dt which is called the *total derivative* of u to distinguish it from the partial derivatives $\partial u/\partial x$ and $\partial u/\partial y$.

Now to find du/dt without actually substituting the values of x and y in $f(x, y)$, we establish the following **Chain rule :**

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \qquad ...(i)$$

Proof. We have $u = f(x, y)$

Giving increment δt to t, let the corresponding increments of x, y and u be δx, δy and δu respectively.

Then $\quad u + \delta u = f(x + \delta x, y + \delta y)$

Subtracting, $\quad \delta u = f(x + \delta x, y + \delta y) - f(x, y)$

$$= \{f(x + \delta x, y + \delta y) - f(x, y + \delta y)\} + \{f(x, y + \delta y) - f(x, y)\}$$

$$\therefore \quad \frac{\delta u}{\delta t} = \frac{f(x + \delta x, y + \delta y) - f(x, y + \delta y)}{\delta x} \cdot \frac{\delta x}{\delta t} + \frac{f(x, y + \delta y) - f(x, y)}{\delta y} \cdot \frac{\delta y}{\delta t}$$

Taking limits as $\delta t \to 0$, δx and δy also $\to 0$, we have

$$\frac{du}{dt} = \underset{\delta y \to 0}{\text{Lt}} \left[\underset{\delta y \to 0}{\text{Lt}} \left\{ \frac{f(x + \delta x, y + \delta y) - f(x, y + \delta y)}{\delta x} \right\} \right] \frac{dx}{dt} + \underset{\delta y \to 0}{\text{Lt}} \left\{ \frac{f(x, y + \delta y) - f(x, y)}{\delta y} \right\} \frac{dy}{dt}$$

$$= \underset{\delta y \to 0}{\text{Lt}} \left\{ \frac{\partial f(x, y + \delta y)}{\delta y} \right\} \cdot \frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y} \cdot \frac{dy}{dt}$$

[Supposing $\partial f(x, y)/\partial x$ to be a continuous function of y]

$$= \frac{\partial f(x, y)}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y} \cdot \frac{dy}{dt} \text{ which is the desired formula.}$$

Cor. Taking $t = x$, (*i*) becomes, $\dfrac{du}{dx} = \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} \cdot \dfrac{dy}{dx}$ ...(*ii*)

Obs. If $u = f(x, y, z)$, where x, y, z are all functions of a variable t, then **Chain rule** is

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \qquad ...(iii)$$

(2) Differentiation of implicit functions. If $f(x, y) = c$ be an implicit relation between x and y which defines as a differentiable function of x, then (*ii*) becomes

$$0 = \frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx}$$

This gives the *important formula* $\dfrac{\mathbf{dy}}{\mathbf{dx}} = -\dfrac{\partial \mathbf{f}}{\partial \mathbf{x}} \Big/ \dfrac{\partial \mathbf{f}}{\partial \mathbf{y}}$ $\qquad \left[\dfrac{\partial f}{\partial y} \neq 0\right]$

for the first differential coefficient of an implicit function.

Example 5.16. *Given $u = \sin(x/y)$, $x = e^t$ and $y = t^2$, find du/dt as a function of t. Verify your result by direct substitution.*

Solution. We have $\dfrac{du}{dt} = \dfrac{\partial u}{\partial x} \cdot \dfrac{dx}{dt} + \dfrac{\partial u}{\partial y} \cdot \dfrac{dy}{dt} = \left(\cos \dfrac{x}{y}\right) \dfrac{1}{y} \cdot e^t + \left(\cos \dfrac{x}{y}\right)\left(-\dfrac{x}{y^2}\right) 2t$

$$= \cos(e^t/t^2) \cdot e^t/t^2 - 2\cos(e^t/t^2) \cdot e^t/t^3 = \{(t-2)/t^3\} e^t \cos(e^t/t^2)$$

Also $u = \sin(x/y) = \sin(e^t/t^2)$

$$\therefore \quad \frac{du}{dt} = \cos\left(\frac{e^t}{t^2}\right) \cdot \frac{t^2 e^t - e^t \cdot 2t}{t^4} = \frac{t-2}{t^3} e^t \cos\left(\frac{e^t}{t^2}\right) \text{ as before.}$$

Example 5.17. *If x increases at the rate of 2 cm/sec at the instant when $x = 3$ cm. and $y = 1$ cm., at what rate must y be changing in order that the function $2xy - 3x^2 y$ shall be neither increasing nor decreasing?*

Solution. Let $\quad u = 2xy - 3x^2 y$, so that

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} = (2y - 6xy) \frac{dx}{dt} + (2x - 3x^2) \frac{dy}{dt} \qquad ...(i)$$

when $x = 3$ and $y = 1$, $dx/dt = 2$, and u is neither increasing nor decreasing, *i.e.*, $du/dt = 0$.

$\therefore$ (*i*) becomes $\qquad 0 = (2 - 6 \times 3)\, 2 + (2 \times 3 - 3 \times 9) \dfrac{dy}{dt}$

or $\qquad \dfrac{dy}{dt} = -\dfrac{32}{21}$ cm/sec. Thus y is decreasing at the rate of 32/21 cm/sec.

Example 5.18. *If $u = x \log xy$ where $x^3 + y^3 + 3xy = 1$, find du/dx.* (V.T.U., 2009)

Solution. From $f(x, y) = x^3 + y^3 + 3xy - 1$, we have

$$\frac{dy}{dx} = -\frac{\partial f}{\partial x}\Big/\frac{\partial f}{\partial y} = -\frac{3x^2 + 3y}{3y^2 + 3x} = -\frac{x^2 + y}{y^2 + x} \quad ...(i)$$

Also $$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dx} = (1 \cdot \log xy + x \cdot 1/x) + (x/y)\cdot dy/dx.$$

Hence $du/dx = 1 + \log xy - x(x^2 + y)/y(y^2 + x)$ [By (*i*)]

Example 5.19. *If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$.* (U.P.T.U., 2005)

Solution. Let $$v = \frac{y-x}{xy} = \frac{1}{x} - \frac{1}{y} \quad \text{and} \quad w = \frac{z-x}{xz} = \frac{1}{x} - \frac{1}{z} \quad ...(i)$$

so that $u = u(v, w)$

$$\therefore \quad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial v}\cdot\frac{\partial v}{\partial x} + \frac{\partial u}{\partial w}\cdot\frac{\partial w}{\partial x} = \frac{\partial u}{\partial v}\left(-\frac{1}{x^2}\right) + \frac{\partial u}{\partial w}\left(-\frac{1}{x^2}\right) \quad \text{[Using (i)]}$$

or $$x^2\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial v} - \frac{\partial u}{\partial w} \quad ...(ii)$$

Also $$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial v}\cdot\frac{\partial u}{\partial y} + \frac{\partial u}{\partial w}\cdot\frac{\partial w}{\partial y} = \frac{\partial u}{\partial v}\left(\frac{1}{y^2}\right) + \frac{\partial u}{\partial w}(0) \quad \text{[Using (i)]}$$

or $$y^2\frac{\partial u}{\partial y} = \frac{\partial u}{\partial v} \quad ...(iii)$$

Similarly $$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial v}\ \frac{\partial v}{\partial z} + \frac{\partial u}{\partial w}\ \frac{\partial w}{\partial z} = \frac{\partial u}{\partial v}(0) + \frac{\partial u}{\partial w}\left(\frac{1}{z^2}\right) \quad \text{[Using (i)]}$$

or $$z^2\frac{\partial u}{\partial z} = \frac{\partial u}{\partial w} \quad ...(iv)$$

Adding (*ii*), (*iii*) and (*iv*), we have

$$x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0.$$

Example 5.20. *Formula for the second differential coefficient of an implicit function.*
If $f(x, y) = 0$, show that

$$\frac{d^2y}{dx^2} = -\frac{q^2r - 2pqs + p^2t}{q^3}$$ (Kurukshetra, 2006)

Solution. We have $$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} = -\frac{p}{q} \quad ...(i)$$

$$\therefore \quad \frac{d^2y}{dx^2} = -\frac{d}{dx}\left(\frac{dy}{dx}\right) = -\frac{d}{dx}\left(\frac{p}{q}\right) = -\frac{q(dp/dx) - p(dq/dx)}{q^2} \quad ...(ii)$$

Using the notations : $r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial p}{\partial x}$, $s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial q}{\partial x}$, $t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial q}{\partial y}$,

we have $$\frac{dp}{dx} = \frac{\partial p}{\partial x} + \frac{\partial p}{\partial y}\cdot\frac{dy}{dx} = r + s(-p/q) = -\frac{qr - ps}{q}$$

and $$\frac{dq}{dx} = \frac{\partial q}{\partial x} + \frac{\partial q}{\partial y}\cdot\frac{dy}{dx} = s + t(-p/q) = \frac{qs - pt}{q}$$

Substituting the values of dp/dx and dq/dx in (*ii*), we get

$$\frac{d^2y}{dx^2} = -\frac{1}{q^2}\left[q\left(\frac{qr-ps}{q}\right) - p\left(\frac{qs-pt}{q}\right)\right] = -\frac{q^2r - 2pqs + p^2t}{q^3}.$$

PROBLEMS 5.5

1. If $z = u^2 + v^2$ and $u = at^2$, $v = 2at$, find dz/dt. (*P.T.U., 2005*)
2. If $u = \tan^{-1}(y/x)$ where $x = e^t - e^{-t}$, and $y = e^t + e^{-t}$, find du/dt. (*V.T.U., 2003*)
3. Find the value of $\frac{du}{dt}$ given $u = y^2 - 4ax$, $x = at^2$, $y = 2at$. (*Anna, 2009*)
4. At a given instant the sides of a rectangle are 4 ft. and 3 ft. respectively and they are increasing at the rate of 1.5 ft./sec. and 0.5 ft./sec. respectively, find the rate at which the area is increasing at that instant.
5. If $z = 2xy^2 - 3x^2y$ and if x increases at the rate of 2 cm. per second and it passes through the value $x = 3$ cm., show that if y is passing through the value $y = 1$ cm., y must be decreasing at the rate of $2\frac{2}{15}$ cm. per second, in order that z shall remain constant.
6. If $u = x^2 + y^2 + z^2$ and $x = e^{2t}$, $y = e^{2t}\cos 3t$, $z = e^{2t}\sin 3t$. Find $\frac{du}{dt}$ as a total derivative and verify the result by direct substitution.
7. If $\phi(cx - az, cy - bz) = 0$, show that $\frac{a\partial z}{\partial x} = \frac{b\partial z}{\partial y} = c$.
8. If $f(x, y) = 0$, $\phi(y, z) = 0$, show that $\frac{\partial f}{\partial y}\cdot\frac{\partial \phi}{\partial z}\cdot\frac{dz}{dx} = \frac{\partial f}{\partial x}\cdot\frac{\partial \phi}{\partial y}$.
9. If the curves $f(x, y) = 0$ and $\phi(y, z) = 0$ touch, show that at the point of contact, $\frac{\partial f}{\partial x}\cdot\frac{\partial \phi}{\partial y} = \frac{\partial f}{\partial y}\cdot\frac{\partial \varphi}{\partial \gamma}$.
10. If $f(x, y) = 0$, show that $\left(\frac{\partial f}{\partial y}\right)^3\frac{d^2y}{dx^2} = 2\left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right)\left(\frac{\partial^2 f}{\partial x\partial y}\right) - \left(\frac{\partial f}{dy}\right)^2\frac{\partial^2 f}{\partial x^2} - \left(\frac{\partial f}{\partial x}\right)^2\left(\frac{\partial^2 f}{\partial y^2}\right)$.

5.6 CHANGE OF VARIABLES

If $\quad u = f(x, y)$ ...(1)

where $\quad x = \phi(s, t)$ and $y = \Psi(s, t)$ ...(2)

it is often necessary to change expressions involving $u, x, y, \partial u/\partial x, \partial u/\partial y$ etc. to expressions involving $u, s, t, \partial u/\partial s, \partial u/\partial t$ etc.

The necessary formulae for the change of variables are easily obtained. If t is regarded as a constant, then x, y, u will be functions of s alone. Therefore, by (*i*) of page 208, we have

$$\frac{\partial \mathbf{u}}{\partial \mathbf{s}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\cdot\frac{\partial \mathbf{x}}{\partial \mathbf{s}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\cdot\frac{\partial \mathbf{y}}{\partial \mathbf{s}} \quad ...(3)$$

where the ordinary derivatives have been replaced by the partial derivatives because x, y are functions of two variables s and t.

$\because$ Similarly, regarding s as constant, we obtain $\frac{\partial \mathbf{u}}{\partial \mathbf{t}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\cdot\frac{\partial \mathbf{x}}{\partial \mathbf{t}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\cdot\frac{\partial \mathbf{y}}{\partial \mathbf{t}}$ ...(4)

On solving (3) and (4) as simultaneous equations in $\partial u/\partial x$ and $\partial u/\partial y$, we get their values in terms of $\partial u/\partial s$, $\partial u/\partial t, u, s, t$.

If instead of the equations (2), s and t are given in terms of x and y, say: $s = \xi(x, y)$ and $t = \eta(x, y)$, ...(5)

then it is easier to use the formulae $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial x}\cdot\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial x}$ ...(6)

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} \qquad ...(7)$$

The higher derivatives of u can be found by repeated application of formulae (3) and (4) or of (6) and (7).

Example 5.21. *If $u = F(x - y, y - z, z - x)$, prove that*

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

(V.T.U., 2010 ; U.T.U. 2009 ; U.P.T.U., 2003)

Solution. Put $x - y = r,\ y - z = s$ and $z - x = t$, so that $u = f(r, s, t)$.

$$\therefore \quad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$$

$$= \frac{\partial u}{\partial r} . (1) + \frac{\partial x}{\partial s} . (0) + \frac{\partial u}{\partial t} . (-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \qquad ...(i)$$

Similarly,
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \qquad ...(ii)$$

and
$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial z} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \qquad ...(iii)$$

Adding (*i*), (*ii*) and (*iii*), we get the required result.

Example 5.22. *If $z = f(x, y)$ and $x = e^u \cos v,\ y = e^u \sin v$, prove that $x \dfrac{\partial z}{\partial v} + y \dfrac{\partial z}{\partial u} = e^{2u} \dfrac{\partial z}{\partial y}$*

and
$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial u}{\partial v}\right)^2\right]$$

(Mumbai, 2009)

Solution. We have $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x} \cdot \dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y} \cdot \dfrac{\partial y}{\partial u}$

$$= \frac{\partial z}{\partial x}(e^u \cos v) + \frac{\partial z}{\partial y}(e^u \sin v) \qquad ...(i)$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$= \frac{\partial z}{\partial x}(-e^u \sin v) + \frac{\partial x}{\partial y}(e^u \cos v) \qquad ...(ii)$$

$$\therefore \quad x \frac{\partial x}{\partial v} + y \frac{\partial z}{\partial u} = (e^u \cos v)\left[-e^u \sin v \frac{\partial z}{\partial x} + e^u \cos v \frac{\partial z}{\partial y}\right] + (e^u \sin v)\left[e^u \cos v \frac{\partial z}{\partial x} + e^u \sin v \frac{\partial z}{\partial y}\right]$$

$$= (e^{2u} \cos^2 v + e^{2u} \sin^2 v) \frac{\partial z}{\partial y} = e^{2u} \frac{\partial z}{\partial y}$$

Now squaring (*i*) and (*ii*) and adding, we get

$$\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = e^{2u}\left(\cos v \frac{\partial z}{\partial x} + \sin v \frac{\partial z}{\partial y}\right)^2 + e^{2u}\left(-\sin v \frac{\partial z}{\partial x} + \cos v \frac{\partial z}{\partial y}\right)^2$$

or
$$e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right] = \cos^2 v \left(\frac{\partial z}{\partial x}\right)^2 + \sin^2 v \left(\frac{\partial z}{\partial y}\right)^2 + 2 \sin v \cos v \frac{\delta z}{\delta z} \frac{\delta z}{\delta y}$$

$$+ \sin^2 v \left(\frac{\partial z}{\partial x}\right)^2 + \cos^2 v \left(\frac{\partial z}{\partial y}\right)^2 - 2 \sin v \cos v \frac{\partial z}{\partial x} \frac{\partial z}{\partial y}$$

$$= (\cos^2 v + \sin^2 v)\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2$$

Hence $$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right].$$

Example 5.23. *If $x + y = 2e^{\theta} \cos \phi$ and $x - y = 2ie^{\theta} \sin \phi$, show that*

$$\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \phi^2} = 4xy \frac{\partial^2 u}{\partial x \partial y}$$ *(Nagpur, 2009 ; U.P.T.U., 2002)*

Solution. We have $x = e^{\theta}(\cos \phi + i \sin \phi) = e^{\theta} . e^{i\phi}$ [See p. 205]

and $y = e^{\theta}(\cos \phi - i \sin \phi) = e^{\theta} . e^{-i\phi}$

Here u is a composite function of θ and ϕ.

$$\therefore \quad \frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$= \frac{\partial u}{\partial x} . (e^{\theta} . e^{i\phi}) + \frac{\partial u}{\partial y}(e^{\theta} . e^{-i\phi}) = x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$$

or $$\frac{\partial}{\partial \theta} = x\frac{\partial}{\partial u} + y\frac{\partial}{\partial y} \qquad ...(i)$$

Also $$\frac{\partial u}{\partial \phi} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \phi} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \phi} = \frac{\partial u}{\partial x} . (e^{\theta} . ie^{i\phi}) + \frac{\partial u}{\partial y}(e^{\theta} . - ie^{-i\phi}) = ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}$$

$$\frac{\partial}{\partial \phi} = ix\frac{\partial}{\partial x} - iy\frac{\partial}{\partial y} \qquad ...(ii)$$

Using the operator (i), we have

$$\frac{\partial^2 u}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial \theta}\right) = \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

$$= x\frac{\partial}{\partial x}\left(x\frac{\partial u}{\partial x}\right) + x\frac{\partial}{\partial x}\left(y\frac{\partial u}{\partial y}\right) + y\frac{\partial}{\partial y}\left(x\frac{\partial u}{\partial x}\right) + y\frac{\partial}{\partial y}\left(y\frac{\partial u}{\partial y}\right)$$

$$= x\left(x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}\right) + xy\frac{\partial^2 u}{\partial x \partial y} + yx\frac{\partial^2 u}{\partial y \partial x} + y\left(y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y}\right)$$

$$= x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} \qquad ...(iii)$$

Similarly using (ii), $$\frac{\partial^2 u}{\partial \phi^2} = \frac{\partial}{\partial \phi}\left(\frac{\partial u}{\partial \phi}\right) = \left(ix\frac{\partial}{\partial x} - iy\frac{\partial}{\partial y}\right)\left(ix\frac{\partial u}{\partial x} - iy\frac{\partial u}{\partial y}\right)$$

$$= - x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} - y^2\frac{\partial^2 u}{\partial y^2} - x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} \qquad ...(iv)$$

Adding (iii) and (iv), we get $$\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \phi^2} = 4xy\frac{\partial^2 u}{\partial x \partial y}$$

Example 5.24. *Transform the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar coordinates.* *(P.T.U., 2010)*

Solution. We have $x = r \cos \theta, y = r \sin \theta$ and $r = \sqrt{(x^2 + y^2)}, \theta = \tan^{-1}(y/x)$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{(x^2 + y^2)}} = \cos \theta \text{ and } \frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2} = -\frac{\sin \theta}{r}$$

Thus, $$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r}\frac{\partial u}{\partial \theta}$$

i.e., $$\frac{\partial}{\partial x} = \cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta} \qquad \text{Similarly, } \frac{\partial}{\partial y} = \sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}.$$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\left(\cos\theta\frac{\partial u}{\partial r} - \frac{\sin\theta}{r}\frac{\partial u}{\partial\theta}\right)$$

$$= \cos^2\theta \frac{\partial^2 u}{\partial r^2} - \frac{2\sin\theta\cos\theta}{r}\frac{\partial^2 u}{\partial r\partial\theta} + \frac{\sin^2\theta}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\sin^2\theta}{r}\frac{\partial u}{\partial r} + \frac{2\sin\theta\cos\theta}{r^2}\frac{\partial u}{\partial\theta} \qquad ...(i)$$

and $$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \left(\sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta}\right)\left(\sin\theta\frac{\partial u}{\partial r} + \frac{\cos\theta}{r}\frac{\partial u}{\partial\theta}\right)$$

$$= \sin^2\theta\frac{\partial^2 u}{\partial r^2} + \frac{2\sin\theta\cos\theta}{r}\frac{\partial^2 u}{\partial r\partial\theta} + \frac{\cos^2\theta}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\cos^2\theta}{r}\frac{\partial u}{\partial r} - \frac{2\sin\theta\cos\theta}{r^2}\frac{\partial u}{\partial r} \qquad ...(ii)$$

Adding (*i*) and (*ii*), we get $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = \dfrac{\partial^2 u}{\partial r^2} + \dfrac{1}{r^2}\dfrac{\partial^2 u}{\partial\theta^2} + \dfrac{1}{r}\dfrac{\partial u}{\partial r}$.

Hence the transformed equation is $\dfrac{\partial^2 u}{\partial r^2} + \dfrac{1}{r}\dfrac{\partial u}{\partial r} + \dfrac{1}{r^2}\dfrac{\partial^2 u}{\partial\theta^2} = 0$.

PROBLEMS 5.6

1. If $z = f(x, y)$ and $x = e^u + e^{-v}$, $y = e^{-u} - e^v$, prove that $\dfrac{\partial z}{\partial u} - \dfrac{\partial z}{\partial v} = x\dfrac{\partial z}{\partial x} - y\dfrac{\partial z}{\partial y}$. *(V.T.U., 2006)*

2. If $u = f(r, s)$, $r = x + at$, $s = y + bt$ and x, y, t are independent variables, show that $\dfrac{\partial u}{\partial t} = a\dfrac{\partial u}{\partial x} + b\dfrac{\partial u}{\partial y}$.

3. If $\phi(z/x^3, y/x) = 0$, prove that $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = 3z$. *(Mumbai, 2007)*

4. If $u = f(x, y)$ and $x = r\cos\theta$, $y = r\sin\theta$, prove that $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2 = \left(\dfrac{\partial u}{\partial r}\right)^2 + \dfrac{1}{r^2}\left(\dfrac{\partial u}{\partial\theta}\right)^2$.
(V.T.U., 2010 ; Madras 2006 ; Rohtak, 2005)

5. If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\dfrac{1}{2}\dfrac{\partial u}{\partial x} + \dfrac{1}{3}\dfrac{\partial u}{\partial y} + \dfrac{1}{4}\dfrac{\partial u}{\partial z} = 0$. *(U.P.T.U., 2006 ; Raipur, 2005)*

6. If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, prove that $\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = 0$ *(Mumbai, 2008 S)*

7. If $u = f(r, s, t)$ and $r = x/y$, $s = y/z$, $t = z/x$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 0$. *(Anna, 2009 ; Kurukshetra, 2006)*

8. If $x = u + v + w$, $y = vw + wu + uv$, $z = uvw$ and F is a function of x, y, z, show that
$$u\frac{\partial F}{\partial u} + v\frac{\partial F}{\partial v} + w\frac{\partial F}{\partial w} = x\frac{\partial F}{\partial x} + 2y\frac{\partial F}{\partial y} + 3z\frac{\partial F}{\partial z}.$$

9. Given that $u(x, y, z) = f(x^2 + y^2 + z^2)$ where $x = r\cos\theta\cos\phi$, $y = r\cos\theta\sin\phi$ and $z = r\sin\theta$, find $\dfrac{\partial u}{\partial\theta}$ and $\dfrac{\partial u}{\partial\phi}$.

10. If the three thermodynamic variables P, V, T are connected by a relation $f(P, V, T) = 0$, show that
$$\left(\frac{\partial P}{\partial T}\right)_V\left(\frac{\partial T}{\partial V}\right)_P\left(\frac{\partial V}{\partial P}\right)_T = -1.$$

11. If by the substitution $u = x^2 - y^2$, $v = 2xy$, $f(x, y) = \theta(u, v)$, show that
$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 4(x^2 + y^2)\left(\frac{\partial^2\theta}{\partial u^2} + \frac{\partial^2\theta}{\partial v^2}\right).$$ *(Anna, 2003)*

12. Transform $\dfrac{\partial^2 z}{\partial x^2} + 2xy^2\dfrac{\partial z}{\partial x} + 2(y - y^3)\dfrac{\partial z}{\partial y} + x^2y^2z = 0$ by the substitution $x = uv$, $y = 1/v$. Hence show that z is the same function of u and v as of x and y.

5.7 (1) JACOBIANS

If u and v are functions of two independent variables x and y, then the determinant

$\begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix}$ is called the *Jacobian** of u, v with respect to x, y

and is written as $\dfrac{\partial(u, v)}{\partial(x, y)}$ or $J\left(\dfrac{u, v}{x, y}\right)$.

Similarly the Jacobian of u, v, w with respect to x, y, z is

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y & \partial u/\partial z \\ \partial v/\partial x & \partial v/\partial y & \partial v/\partial z \\ \partial w/\partial x & \partial w/\partial y & \partial w/\partial z \end{vmatrix}$$

Likewise, we can define Jacobians of four or more variables. An important application of Jacobians is in connection with the change of variables in multiple integrals (§ 7.7).

(2) Properties of Jacobians. We give below two of the important properties of Jacobians. For simplicity, the properties are stated in terms of two variables only, but these are evidently true in general.

I. If $J = \partial(u, v)/\partial(x, y)$ and $J' = \partial(x, y)/\partial(u, v)$ then $JJ' = 1$.

Let $u = f(x, y)$ and $v = g(x, y)$.

Suppose, on solving for x and y, we get $x = \phi(u, v)$ and $y = \psi(u, v)$.

Then
$$\left.\begin{aligned} \frac{\partial u}{\partial u} &= 1 = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial u}, \\ \frac{\partial u}{\partial v} &= 0 = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial v}, \\ \frac{\partial v}{\partial u} &= 0 = \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial u}, \\ \frac{\partial v}{\partial v} &= 1 = \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial v}, \end{aligned}\right\} \quad ...(i)$$
and

$$\therefore \quad JJ' = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix} \times \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial y/\partial x & \partial v/\partial y \end{vmatrix} \times \begin{vmatrix} \partial x/\partial u & \partial y/\partial u \\ \partial x/\partial v & \partial y/\partial v \end{vmatrix}$$

(Interchanging rows and columns of the 2nd determinant).

$$= \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1. \qquad \text{[By virtue of } (i)\text{]}$$

II. **Chain rule for Jacobians.** *If u, v are functions of r, s and r, s are functions of x, y, then*

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)}\cdot\frac{\partial(r, s)}{\partial(x, y)}.$$

$$\frac{\partial(u, v)}{\partial(r, s)}\cdot\frac{\partial(r, s)}{\partial(x, y)} = \begin{vmatrix} \partial u/\partial r & \partial u/\partial s \\ \partial v/\partial r & \partial v/\partial s \end{vmatrix} \times \begin{vmatrix} \partial r/\partial x & \partial s/\partial x \\ \partial r/\partial y & \partial s/\partial y \end{vmatrix}$$

[Interchanging rows and columns of the 2nd det.]

$$= \begin{vmatrix} \dfrac{\partial u}{\partial r}\cdot\dfrac{\partial r}{\partial x} + \dfrac{\partial u}{\partial s}\cdot\dfrac{\partial s}{\partial x} & \dfrac{\partial u}{\partial r}\cdot\dfrac{\partial r}{\partial y} + \dfrac{\partial u}{\partial s}\cdot\dfrac{\partial s}{\partial y} \\ \dfrac{\partial v}{\partial r}\cdot\dfrac{\partial r}{\partial x} + \dfrac{\partial v}{\partial s}\cdot\dfrac{\partial s}{\partial x} & \dfrac{\partial v}{\partial r}\cdot\dfrac{\partial r}{\partial y} + \dfrac{\partial v}{\partial s}\cdot\dfrac{\partial s}{\partial y} \end{vmatrix} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix} = \frac{\partial(u, v)}{\partial(x, y)}.$$

* Called after the German mathematician *Carl Gustav Jacob Jacobi* (1804–1851), who made significant contributions to mechanics, partial differential equations, astronomy, elliptic functions and the calculus of variations.

Example 5.25. *(i) In polar coordinates, $x = r\cos\theta$, $y = r\sin\theta$, show that*

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r.$$ *(U.P.T.U., 2006 ; V.T.U., 2004 : Andhra, 2000)*

(ii) In cylindrical coordinates (Fig. 8.28), $x = \rho\cos\phi$, $y = \rho\sin\phi$, $z = z$, show that

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \rho.$$

(iii) In spherical polar coordidnates (Fig. 8.29), $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2\sin\theta.$$ *(Anna, 2009 ; Hazaribagh, 2009 ; Rohtak, 2003)*

Solution. (*i*) We have $\quad \frac{\partial x}{\partial r} = \cos\theta,\ \frac{\partial x}{\partial \theta} = -r\sin\theta$ and $\frac{\partial y}{\partial r} = \sin\theta,\ \frac{\partial y}{\partial \theta} = -r\cos\theta$

$$\therefore \quad \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

(*ii*) We have $\quad \frac{\partial x}{\partial \rho} = \cos\phi,\ \frac{\partial x}{\partial \phi} = -\rho\sin\phi,\ \frac{\partial x}{\partial z} = 0,$

$$\frac{\partial y}{\partial \rho} = \sin\phi,\ \frac{\partial y}{\partial \phi} = \rho\cos\phi,\ \frac{\partial y}{\partial z} = 0 \text{ and } \frac{\partial z}{\partial \rho} = 0,\ \frac{\partial z}{\partial \phi} = 0,\ \frac{\partial z}{\partial z} = 1$$

$$\therefore \quad \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \begin{vmatrix} \cos\phi & -\rho\sin\theta & 0 \\ \sin\phi & \rho\cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho.$$

(*iii*) We have $\quad \frac{\partial x}{\partial r} = \sin\theta\cos\theta,\ \frac{\partial x}{\partial \theta} = r\cos\theta\cos\phi,\ \frac{\partial x}{\partial \phi} = -r\sin\theta\sin\phi,$

$$\frac{\partial y}{\partial r} = \sin\theta\sin\phi,\ \frac{\partial y}{\partial \theta} = r\cos\theta\sin\phi,\ \frac{\partial y}{\partial \phi} = r\sin\theta\cos\phi,$$

and $$\frac{\partial z}{\partial r} = \cos\theta,\ \frac{\partial z}{\partial \theta} = -r\sin\theta,\ \frac{\partial z}{\partial \phi} = 0.$$

$$\therefore \quad \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix} = r^2\sin\theta.$$

Example 5.26. *If $y_1 = \frac{x_2x_3}{x_1}$, $y_2 = \frac{x_3x_1}{x_2}$, $y_3 = \frac{x_1x_2}{x_3}$, show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4.* *(U.P.T.U., 2006)*

Solution. We have $\frac{\partial y_1}{\partial x_1} = -\frac{x_2x_3}{x_1^2},\ \frac{\partial y_1}{\partial x_2} = \frac{x_3}{x_1},\ \frac{\partial y_1}{\partial x_3} = \frac{x_2}{x_1}$

$$\frac{\partial y_2}{\partial x_1} = \frac{x_3}{x_2},\ \frac{\partial y_2}{\partial x_2} = -\frac{x_3x_1}{x_2^2},\ \frac{\partial y_2}{\partial x_3} = \frac{x_1}{x_2}$$

and $$\frac{\partial y_3}{\partial x_1} = \frac{x_2}{x_3},\ \frac{\partial y_3}{\partial x_2} = \frac{x_1}{x_3},\ \frac{\partial y_3}{\partial x_3} = -\frac{x_1x_2}{x_3^2}$$

$$\therefore \quad \frac{\partial(y_1y_2y_3)}{\partial(x_1x_2x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix} \begin{vmatrix} \frac{-x_2x_3}{x_1^2} & \frac{x_3}{x_1} & \frac{x_2}{x_1} \\ \frac{x_3}{x_2} & \frac{-x_3x_1}{x_2^2} & \frac{x_1}{x_1} \\ \frac{x_2}{x_3} & \frac{x_1}{x_3} & \frac{-x_1x_2}{x_3^2} \end{vmatrix}$$

$$= \frac{1}{x_1^2x_2^2x_3^2}\begin{vmatrix} -x_2x_3 & x_3x_1 & x_1x_2 \\ x_2x_3 & -x_3x_1 & x_1x_2 \\ x_2x_3 & x_3x_1 & -x_1x_2 \end{vmatrix} = -\frac{x_1^2x_2^2x_3^2}{x_1^2x_2^2x_3^2}\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= -1(1-1) - 1(-1-1) + 1(1+1) = 0 + 2 + 2 = 4.$$

Example 5.27. *In $u = x + 3y^2 - z^2$, $v = 4x^2yz$, $w = 2z^2 - xy$, evaluate $\partial(u, v, w)/\partial(x, y, z)$ at $(1, -1, 0)$.* *(V.T.U., 2006)*

Solution. $$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 6y & -3z^2 \\ 8xyz & 4x^2z & 4x^2y \\ -y & -x & 4z \end{vmatrix}$$

$\therefore$ At the point $(1, -1, 0)$ $$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 1 & -6 & 0 \\ 0 & 0 & -4 \\ 1 & -1 & 0 \end{vmatrix} = 4(-1+6) = 20.$$

Example 5.28. *If $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$, find $\frac{\partial(u, v)}{\partial(r, \theta)}$.* *(V.T.U., 2009 ; Madras, 2006)*

Solution. We have $$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)}$$

Since $u = x^2 - y^2$, $u = 2xy$

$$\therefore \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2) \qquad ...(ii)$$

Since $x = r\cos\theta$, $y = r\sin\theta$,

$$\therefore \quad \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r \qquad ...(iii)$$

Hence, $$\frac{\partial(u, v)}{\partial(r, \theta)} = 4(x^2 + y^2).\, r = 4(r^2\cos^2\theta + r^2\sin^2\theta).\, r = 4r^3$$ [Using (*ii*) & (*iii*)]

(3) Jacobian of Implicit functions. *If u_1, u_2, u_3 instead of being given explicitly in terms x_1, x_2, x_3, be connected with them equations such as*

$$f_1(u_1, u_2, u_3, x_1, x_2, x_3) = 0,\ f_2(u_1, u_2, u_3, x_1, x_2, x_3) = 0,\ f_3(u_1, u_2, u_3, x_1, x_2, x_3) = 0, \text{ then}$$

$$\frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = (-1)^3\frac{\partial(f_1, f_2, f_3)}{\partial(x_1, x_2, x_3)} \div \frac{\partial(f_1, f_2, f_3)}{\partial(u_1, u_2, u_3)}$$

Obs. This result can be easily generalised. It bears analogy to the result $\frac{dy}{dx} = -\frac{\partial f}{\partial x}\Big/\frac{\partial f}{\partial y}$, where x, y are connected by the relation $f(x, y) = 0$.

Example 5.29. *If $u = x, y, z$, $v = x^2 + y^2 + z^2$, $w = x + y + z$, find $\partial(x, y, z)/\partial(u, v, w)$.* *(U.P.T.U., 2003)*

Solution. Let $\quad f_1 = u - x\,y\,z,\ f_2 = v - x^2 - y^2 - z^2,\ f_3 = w - x - y - z.$

We have $$\frac{\partial(x, y, z)}{\partial(u, v, w)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} \div \frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} \quad ...(i)$$

Now, $$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} \partial f_1/\partial x & \partial f_1/\partial y & \partial f_1/\partial z \\ \partial f_2/\partial x & \partial f_2/\partial y & \partial f_2/\partial z \\ \partial f_3/\partial x & \partial f_3/\partial y & \partial f_3/\partial z \end{vmatrix} = \begin{vmatrix} -yz & -xz & -xy \\ -2x & -2y & -2z \\ -1 & -1 & -1 \end{vmatrix}$$

$$= -2(x-y)(y-z)(z-x) \quad ...(ii)$$

and $$\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad ...(iii)$$

Substituting values from (*ii*) and (*iii*) in (*i*), we get

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = (-1) \times 1/[-2(x-y)(y-z)(z-x)] = 1/2(x-y)(y-z)(z-x).$$

(4) Functional relationship. *If u_1, u_2, u_3 be functions of x_1, x_2, x_3 then the necessary and sufficient condition for the existence of a functional relationship of the form $f(u_1, u_2, u_3) = 0$, is*

$$J\left(\frac{u_1, u_2, u_3}{x_1, x_2, x_3}\right) = 0.$$

Example 5.30. *If $u = x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}$, $v = \sin^{-1} x + \sin^{-1} y$, show that u, v are functionally related and find the relationship.* (*Kurukshetra, 2006*)

Solution. We have $$\frac{\partial u}{\partial x} = \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}}, \frac{\partial u}{\partial y} = \frac{-xy}{\sqrt{(1-y^2)}} + \sqrt{(1-x^2)}$$

and $$\frac{\partial v}{\partial x} = \frac{1}{\sqrt{(1-x^2)}}, \frac{\partial v}{\partial y} = \frac{1}{\sqrt{(1-y^2)}}$$

$$\therefore \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}}, & \sqrt{(1-x^2)} - \frac{xy}{\sqrt{(1-y^2)}} \\ \frac{1}{\sqrt{(1-x^2)}}, & \frac{1}{\sqrt{(1-y^2)}} \end{vmatrix}$$

$$= 1 - \frac{xy}{\sqrt{[(1-x^2)(1-y^2)]}} - 1 + \frac{xy}{\sqrt{[(1-x^2)(1-y^2)]}} = 0$$

Hence u and v are functionally related *i.e.*, they are not independent.

We have $$v = \sin^{-1} x + \sin^{-1} y = \sin^{-1} [x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}]$$

i.e., $$u = \sin v$$

which is the required relationship between u and v.

PROBLEMS 5.7

1. If $x = r\cos\theta, y = r\sin\theta$, evaluate $\partial(r, \theta)/\partial(x, y)$, $\partial(x, y)/\partial(r, \theta)$ and prove that $\{\partial(r, \theta)/\partial(x, y)\}.\{\partial(x, y)/\partial(r, \theta)\} = 1$. (*V.T.U., 2010*)
2. If $x = u(1-v), y = uv$, prove that $JJ' = 1$. (*V.T.U., 2000 S*)
3. If $x = a\cosh\xi\cos\eta, y = a\sinh\xi\sin\eta$, show that $\partial(x, y)/\partial(\xi, \eta) = \frac{1}{2}a^2(\cosh 2\xi - \cos 2\eta)$ (*S.V.T.U., 2007*)
4. If $x = e^u \sec v, y = e^u \tan v$, find $J = \partial(u,v)/\partial(x, y)$, $J = \partial(x, y)/\partial(u, v)$. Hence show $JJ' = 1$. (*V.T.U., 2007 S*)
5. If $u = x^2 - 2y^2, v = 2x^2 - y^2$ where $x = r\cos\theta, y = r\sin\theta$, show that $\frac{\partial(u, v)}{\partial(r, \theta)} = 6r^3 \sin 2\theta$.
6. If $u = x^2 + y^2 + z^2, v = xy + yz + zx, w = x + y + z$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$. (*U.T.U., 2009; V.T.U., 2008*)

7. If $F = xu + v - y$, $G = u^2 + vy + w$, $H = zu - v + vw$, compute $\partial(F, G, H)/\partial(u, v, w)$.
8. If $u = x + y + z$, $uv = y + z$, $uvw = z$, show that $\partial(x, y, z)/\partial(u, v, w) = u^2v$.
(*Kurukshetra, 2009; P.T.U., 2009 S; V.T.U., 2003*)
9. If $u^3 + v^3 = x + y$ and $u^2 + v^2 = x^3 + y^3$, show that $\dfrac{\partial(u,v)}{\partial(x,y)} = \dfrac{1}{2}\dfrac{y^2 - x^2}{uv(u-v)}$ (*U.P.T.U., 2006 MCA*)
10. If $u = \dfrac{x+y}{1-xy}$ and $v = \tan^{-1}x + \tan^{-1}y$, find $\dfrac{\partial(u,v)}{\partial(x,y)}$. Are u and v functionally related. If so, find this relationship.
(*Nagpur, 2008*)
11. If $u = 3x + 2y - z$, $v = x - 2y + z$ and $w = x(x + 2y - z)$, show that they are functionally related, and find the relation.
(*Nagpur, 2009*)

5.8 (1) GEOMETRICAL INTERPRETATION

If $P(x, y, z)$ be the coordinates of a point referred to axes OX, OY, OZ then the equation $z = f(x, y)$ represents a surface. (Fig. 5.1)

Let a plane $y = b$ parallel to the XZ-plane pass through P cutting the surface along the curve APB given by

$$z = f(x, b).$$

As y remains equal to b and x varies then P moves along the curve APB and $\partial z/\partial x$ is the ordinary derivative of $f(x, b)$ w.r.t. x.

Hence $\partial z/\partial x$ *at P is the tangent of the angle which the tangent at P to the section of the surface* $z = f(x, y)$ *by a plane through P parallel to the plane XOZ, makes with a line parallel to the x-axis.*

Similarly, $\partial z/\partial y$ *at P is the tangent of the angle which the tangent at P to the curve of intersection of the surface* $z = f(x, y)$ *and the plane* $x = a$, *makes with a line parallel to the y-axis.*

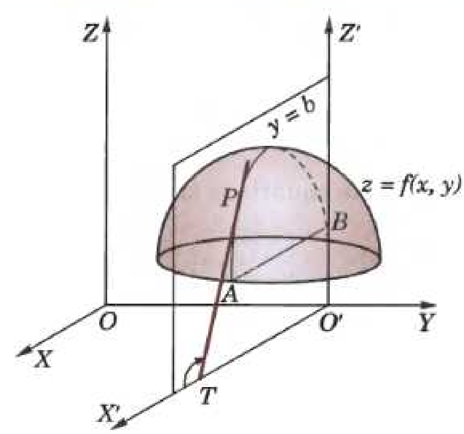

Fig. 5.1

(2) Tangent plane and Normal to a surface. Let $P(x, y, z)$ and $Q(x + \delta x, y + \delta y, z + \delta z)$ be two neighbouring points on the surface $F(x, y, z) = 0$. (Fig. 5.2) ...(*i*)

Let the arc PQ be δs and the chord PQ be δc, so that (as for plane curves)

$$\underset{Q \to P}{\text{Lt}} (\delta s/\delta c) = 1.$$

The direction cosines of PQ are $\dfrac{\delta x}{\delta c}, \dfrac{\delta y}{\delta c}, \dfrac{\delta z}{\delta c}$ *i.e.,* $\dfrac{\delta x}{\delta s}\cdot\dfrac{\delta s}{\delta c}, \dfrac{\delta y}{\delta s}\cdot\dfrac{\delta s}{\delta c}, \dfrac{\delta z}{\delta s}\cdot\dfrac{\delta s}{\delta c}$

When $\delta s \to 0$, $Q \to P$ and PQ tends to tangent line PT. Then noting that the coordinates of any point on arc PQ are functions of s only, the direction cosines of PT are

$$\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \qquad ...(ii)$$

Differentiating (*i*) with respect to s, we obtain $\dfrac{\partial F}{\partial x}\cdot\dfrac{dx}{ds} + \dfrac{\partial F}{\partial y}\cdot\dfrac{dy}{ds} + \dfrac{\partial F}{\partial z}\cdot\dfrac{dz}{ds} = 0.$

This shows that the tangent line whose direction cosines are given by (*ii*), is perpendicular to the line having direction ratios

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \qquad ...(iii)$$

Since we can take different curves joining Q to P, we get a number of tangent lines at P and the line having direction ratios (*iii*) will be perpendicular to *all* these tangent lines at P. Thus all the tangent lines at P lie in a plane through P perpendicular to line (*iii*).

Hence *the equation of the tangent plane to (i) at the point P is*

$$\mathbf{\frac{\partial F}{\partial x}(X - x) + \frac{\partial F}{\partial y}(Y - y) + \frac{\partial F}{\partial z}(Z - z) = 0}$$

where (X, Y, Z) are the current coordinates of any point on this tangent plane.

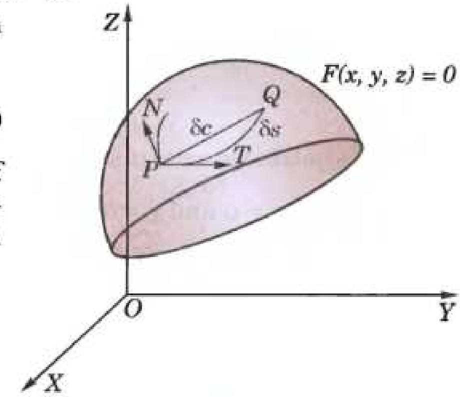

Fig. 5.2

Also *the equation of the normal to the surface at P (i.e., the line through P,* perpendicular to the tangent plane at *P*) is

$$\frac{\mathbf{X-x}}{\partial \mathbf{F}/\partial \mathbf{x}}=\frac{\mathbf{Y-y}}{\partial \mathbf{F}/\partial \mathbf{y}}=\frac{\mathbf{Z-z}}{\partial \mathbf{F}/\partial \mathbf{z}}.$$

Example 5.31. *Find the equations of the tangent plane and the normal to the surface* $z^2 = 4(1 + x^2 + y^2)$ *at* (2, 2, 6).

Solution. We have $F(x, y, z) = 4x^2 + 4y^2 - z^2 + 4$.

$\therefore$ $\partial F/\partial x = 8x$, $\partial F/\partial y = 8y$, $\partial F/\partial z = -2z$, and at the point (2, 2, 6)

$\partial F/\partial x = 16$, $\partial F/\partial y = 16$, $\partial F/\partial z = -12$

Hence the equation of the tangent plane at (2, 2, 6) is $16(X-2) + 16(Y-2) - 12(Z-6) = 0$

i.e., $4X + 4Y - 3Z + 2 = 0$ ...(*i*)

Also the equation of the normal at (2, 2, 6) [being perpendicular to (*i*)] is

$$\frac{X-2}{4}=\frac{Y-2}{4}=\frac{Z-6}{-3}.$$

PROBLEMS 5.8

Find the equations of the tangent plane and normal to each of the following surfaces at the given points :

1. $2x^2 + y^2 = 3 - 2z$ at $(2, 1, -3)$ (*Assam, 1998*)
2. $x^3 + y^3 + 3xyz = 3$ at $(1, 2, -1)$ (*Osmania, 2003 S*)
3. $xyz = a^2$ at (x_1, y_1, z_1).
4. $2xz^2 - 3xy - 4x = 7$ at $(1, -1, 2)$.
5. Show the plane $3x + 12y - 6z - 17 = 0$ touches the conicoid $3x^2 - 6y^2 + 9z^2 + 17 = 0$. Find also the point of contact.
6. Show that the plane $ax + by + cz + d = 0$ touches the surface $px^2 + qy^2 + 2z = 0$, if $\frac{a^2}{p}+\frac{b^2}{q}+2cd=0$.
7. Find the equation of the normal to the surface $x^2 + y^2 + z^2 = a^2$. (*P.T.U., 2009 S*)

5.9 TAYLOR'S THEOREM FOR FUNCTIONS OF TWO VARIABLES

Considering $f(x + h, y + k)$ as a function of a single variable x, we have by Taylor's theorem*

$$f(x+h, y+k) = f(x, y+k) + h\frac{\partial f(x, y+k)}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 f(x, y+k)}{\partial x^2} + \dots \quad \dots(i)$$

Now expanding $f(x, y + k)$ as a function of y only,

$$f(x, y+k) = f(x, y) + k\frac{\partial f(x, y)}{\partial y} + \frac{k^2}{2!}\frac{\partial^2 f(x, y)}{\partial y^2} + \dots$$

$\therefore$ (*i*) takes the form $f(x+h, y+k) = f(x, y) + k\dfrac{\partial f(x, y)}{\partial y} + \dfrac{k^2}{2!}\dfrac{\partial^2 f(x, y)}{\partial y^2} + \dots$

$$+ h\frac{\partial}{\partial x}\left\{f(x, y) + k\frac{\partial f(x, y)}{\partial y} + \frac{k^2}{2!}\frac{\partial^2 f(x, y)}{\partial y^2} + \dots\right\} + \frac{h^2}{2!}\frac{\partial^2}{\partial x^2}\left\{f(x, y) + k\frac{\partial (x, y)}{\partial y} + \dots\right\}$$

Hence, $$f(x+h, y+k) = f(x, y) + h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y} + \frac{1}{2!}\left(h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right) + \dots \quad \dots(1)$$

In symbols we write it as $f(x+h, y+k) = f(x, y) + \left(h\dfrac{\partial}{\partial x} + k\dfrac{\partial}{\partial y}\right)f + \dfrac{1}{2!}\left(h\dfrac{\partial}{\partial x} + k\dfrac{\partial}{\partial y}\right)^2 f + \dots$

Taking $x = a$ and $y = b$, (1) becomes

$$f(a+h, b+k) = f(a, b) + [hf_x(a, b) + kf_y(a, b)] + \frac{1}{2!}[h^2 f_{xx}(a, b) + 2hkf_{xy}(a, b) + k^2 f_{yy}(a, b)] + \dots$$

*See footnote on page 145.

Putting $a + h = x$ and $b + k = y$ so that $h = x - a$, $k = y - b$, we get

$$\mathbf{f(x, y) = f(a, b) + [(x - a)\, f_x (a, b) + (y - b)\, f_y (a, b)]}$$

$$\mathbf{+ \frac{1}{2!} [(x - a)^2 f_{xx} (a, b) + 2(x - a)(y - b)\, f_{xy} (a, b) + (y - b)^2 f_{yy}(a, b)] + ...} \quad ...(2)$$

This is Taylor's expansion of $f(x, y)$ in powers of $(x - a)$ and $(y - b)$. It is used to expand $f(x, y)$ in the neighbourhood of (a, b).

Cor. Putting $a = 0, b = 0$, in (2), we get

$$\mathbf{f(x, y) = f(0, 0) + [xf_x(0, 0) + yf_y (0, 0)] + \frac{1}{2!} [x^2 f_{xx} (0, 0) + 2xy\, f_{xy} (0, 0) + y^2 f_{yy} (0, 0)] + ...} \quad ...(3)$$

This is Maclaurin's expansion of $f(x, y)$.

Example 5.32. *Expand $e^x \log (1 + y)$ in powers of x and y upto terms of third degree.*

(*V.T.U., 2010 ; P.T.U., 2009 ; J.N.T.U., 2006*)

Solution. Here

$$\begin{aligned}
f(x, y) &= e^x \log (1 + y) & \therefore \quad f(0, 0) &= 0\\
f_x(x, y) &= e^x \log (1 + y) & f_x(0, 0) &= 0\\
f_y(x, y) &= e^x \frac{1}{1+y} & f_y(0, 0) &= 1\\
f_{xx}(x, y) &= e^x \log (1 + y) & f_{xx}(0, 0) &= 0\\
f_{xy}(x, y) &= e^x \frac{1}{1+y} & f_{xy}(0, 0) &= 1\\
f_{yy}(x, y) &= - e^x (1 + y)^{-2} & f_{yy}(0, 0) &= -1\\
f_{xxx}(x, y) &= e^x \log (1 + y) & f_{xxx}(0, 0) &= 0\\
f_{xxy}(x, y) &= e^x \frac{1}{1+y} & f_{xxy}(0, 0) &= 1\\
f_{xyy}(x, y) &= - e^x (1 + y)^{-2} & f_{xyy}(0, 0) &= -1\\
f_{yyy}(x, y) &= 2e^x (1 + y)^{-3} & f_{yyy}(0, 0) &= 2
\end{aligned}$$

Now Maclaurin's expansion of $f(x, y)$ gives

$$f(x, y) = f(0, 0) + xf_x(0, 0) + yf_y(0, 0) + \frac{1}{2!} \{x^2 f_{xx}(0, 0) + 2xy\, f_{xy}(0, 0) + y^2 f_{yy}(0, 0)\}$$

$$+ \frac{1}{3!} \{x^3 f_{xxx}(0, 0) + 3x^2 y f_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)\} + ...$$

$$\therefore \quad e^x \log (1 + y) = 0 + x(0) + y(1) + \frac{1}{2!} \{x^2 (0) + 2xy(1) + y^2 (-1)\}$$

$$+ \frac{1}{3!} \{x^3(0) + 3x^2y(1) + 3xy^2(-1) + y^3(2)\} + ...$$

$$= y + xy - \frac{1}{2} y^2 + \frac{1}{2} (x^2y - xy^2) + \frac{1}{3} y^3 + ...$$

Example 5.33. *Expand $x^2y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ using Taylor's theorem.*

(*P.T.U., 2010 ; V.T.U., 2008 ; U.P.T.U., 2006 ; Anna, 2005*)

Solution. Taylor's expansion of $f(x, y)$ in powers of $(x - a)$ and $(x - b)$ is given by

$$f(x, y) = f(a, b) + [(x - a) f_x (a, b) + (y - b) f_y(a, b)] + \frac{1}{2!} [(x - a)^2 f_{xx} (a, b)$$

$$+ 2 (x - a)(y - b) f_{xy}(a, b) + (y - b)^2 f_{yy}(a, b)] + \frac{1}{3!} [(x - a)^3 f_{xxx}(a, b)$$

$$+ 3 (x - a)^2 (y - b) f_{xxy}(a, b) + 3 (x - a) (y - b)^2 f_{xyy} (a, b)$$

$$+ (y - b)^3 f_{yyy} (a, b)] + ... \quad ...(i)$$

Hence $a = 1$, $b = -2$ and $f(x, y) = x^2y + 3y - 2$

$\therefore$ $f(1, -2) = -10, f_x = 2xy, f_x(1, -2) = -4; f_y = x^2 + 3, f_y(1, -2) = 4; f_{xx} = 2y,$
$f_{xx}(1, -2) = -4; f_{xy} = 2x, f_{xy}(1, -2) = 2; f_{yy} = 0, f_{yy}(1, -2) = 0; f_{xxx} = 0, f_{xxx} = (1, -2) = 0;$
$f_{xxy}(1, -2) = 2, f_{xyy}(1, -2) = 0, f_{yyy}(1, -2) = 0$

All partial derivatives of higher order vanish.

Substituting these in (*i*), we get

$$x^2y + 3y - 2 = -10 + [(x-1)(-4) + (y+2)4] + \frac{1}{2}[(x-1)^2(-4) + 2(x-1)(y+2)(2)$$
$$+ (y+2)^2(0)] + \frac{1}{6}[(x-1)^3(0) + 3(x-1)^2(y+2)(2) + 3(x-1)(y+2)^2(0) + (y+2)^3(0)]$$
$$= -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2).$$

Example 5.34. *Expand $f(x, y) = \tan^{-1}(y/x)$ in powers of $(x-1)$ and $(y-1)$ upto third-degree terms. Hence compute $f(1.1, 0.9)$ approximately.* *(V.T.U., 2010 ; J.N.T.U., 2006 ; U.P.T.U., 2006)*

Solution. Here $a = 1, b = 1$ and $f(1, 1) = \tan^{-1}(1) = \pi/4$.

$$f_x = \frac{-y}{x^2+y^2} \qquad f_x(1,1) = -\frac{1}{2}; \qquad f_y = \frac{x}{x^2+y^2}, \qquad f_y(1,1) = \frac{1}{2}$$

$$f_{xx} = \frac{2xy}{(x^2+y^2)^2}, \qquad f_{xx}(1,1) = \frac{1}{2}; \qquad f_{xy} = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad f_{xy}(1,1) = 0$$

$$f_{yy} = \frac{-2xy}{(x^2+y^2)^2}, \qquad f_{yy}(1,1) = -\frac{1}{2};$$

$$f_{xxx} = \frac{2y^3 - 6x^2y}{(x^2+y^2)^3}, \qquad f_{xxx}(1,1) = -\frac{1}{2}; \qquad f_{xxy} = \frac{2x^3 - 6xy^2}{(x^3+y^2)^3}, \qquad f_{xxy}(1,1) = -\frac{1}{2}$$

$$f_{xyy} = \frac{6x^2y - 2y^3}{(x^2+y^2)^3}, \qquad f_{xyy}(1,1) = \frac{1}{2}; \qquad f_{yyy} = \frac{6xy^2 - 2x^3}{(x^2+y^2)^3}, \qquad f_{yyy}(1,1) = \frac{1}{2}$$

Taylor's expansion of $f(x, y)$ in powers of $(x-1)$ and $(y-1)$ is given by

$$f(x,y) = f(1,1) + \frac{1}{1!}\{(x-1)f_x(1,1) + (y-1)f_y(1,1)\} + \frac{1}{2!}\{(x-1)^2 f_{xx}(1,1) + 2(x-1)(y-1)$$
$$f_{xy}(1,1) + (y-1)^2 f_{yy}(1,1) + \frac{1}{3!}\{(x-1)^3 f_{xxx}(1,1) + 3(x-1)^2(y-1)f_{xxy}(1,1)$$
$$+ 3(x-1)(y-1)^2 f_{xyy}(1,1) + (y-1)^3 f_{yyy}(1,1)\} + \ldots$$

$$\therefore \quad \tan^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{4} + \left\{(x-1)\left(-\frac{1}{2}\right) + (y-1)\frac{1}{2}\right\} + \frac{1}{2!}\left\{(x-1)^2\frac{1}{2} + 2(x-1)(y-1)(0) + (y-1)^2\left(-\frac{1}{2}\right)\right\}$$
$$+ \frac{1}{3!}\left\{(x-1)^3\left(-\frac{1}{2}\right) + 3(x-1)^2(y-1)\left(-\frac{1}{2}\right) + 3(x-1)(y-1)^2\frac{1}{2} + (y-1)^3\frac{1}{2}\right\} + \ldots$$
$$= \frac{\pi}{4} - \frac{1}{2}\{(x-1)-(y-1)\} + \frac{1}{4}\{(x-1)^2 - (y-1)^2\} - \frac{1}{12}\{(x-1)^3 + 3(x-1)^2(y-1)$$
$$- 3(x-1)(y-1)^2 - (y-1)^3\} + \ldots$$

Putting $x = 1.1$ and $y = 0.9$, we get

$$f(1.1, 0.9) = \frac{\pi}{4} - \frac{1}{2}(0.2) + \frac{1}{4}(0) - \frac{1}{12}\{(0.1)^3 - 3(0.1)^3 - 3(0.1)^3 - (-0.1)^3\}$$
$$= 0.7854 - 0.1000 + 0.0003 = 0.6857.$$

5.10 (1) ERRORS AND APPROXIMATIONS

Let $f(x, y)$ be a continuous function of x and y. If δx and δy be the increments of x and y, then the new value of $f(x, y)$ will be $f(x + \delta x, y + \delta y)$. Hence

$$\delta f = f(x + \delta x, y + \delta y) - f(x, y).$$

Expanding $f(x + \delta x, y + \delta y)$ by Taylor's theorem and supposing δx, δy to be so small that their products, squares and higher powers can be neglected, we get

$$\delta \mathbf{f} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \mathbf{f}}{\partial \mathbf{y}} \delta \mathbf{y}, \text{ approximately.}$$

Similarly if f be a function of several variables $x, y, z, t, \ldots$, then

$$\delta f = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z + \frac{\partial f}{\partial t} \delta t + \ldots \text{ approximately.}$$

These formulae are very useful in correcting the effect of small errors in measured quantities.

(2) Total Differential

If u is a function of two variables x and y, the *total differential* of u is defined as

$$\mathbf{du} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \mathbf{dx} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}} \mathbf{dy} \qquad \ldots(1)$$

The differentials dx and dy are respectively the increments δx and δy in x and y. If x and y are not independent variables but functions of another variable t even then the formula (1) holds and we write $dx = \frac{dx}{dt} dt$ and $dy = \frac{dy}{dt} dt$. Similar definition can be given for a function of three or more variables.

Example 5.35. *The diameter and altitude of a can in the shape of a right circular cylinder are measured as 4 cm and 6 cm respectively. The possible error in each measurement is 0.1 cm. Find approximately the maximum possible error in the values computed for the volume and the lateral surface.*

Solution. Let x be the diameter and y the height of the can. Then its volume $V = \frac{\pi}{4} x^2 y$

$$\therefore \qquad \delta V = \frac{\partial V}{\partial x} \delta x + \frac{\partial V}{\partial y} \delta y = \frac{\pi}{4} (2xy\delta x + x^2 \delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm.

$$\therefore \qquad \delta V = \frac{\pi}{4}(2 \times 4 \times 6 \times 0.1 + 4^2 \times 0.1) = 1.6\pi \text{ cm}^3$$

Also its lateral surface $S = \pi xy$

$$\therefore \qquad \delta S = \pi(y\, \delta x + x\, \delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm., we have $\delta S = \pi(6 \times 0.1 + 4 \times 0.1) = \pi \text{ cm}^2$.

Example 5.36. *The period of a simple pendulum is $T = 2\pi \sqrt{(l/g)}$, find the maximum error in T due to the possible error upto 1% in l and 2.5% in g.* (U.P.T.U., 2004)

Solution. We have $T = 2\pi \sqrt{(l/g)}$

or
$$\log T = \log 2\pi + \frac{1}{2} \log l - \frac{1}{2} \log g$$

$$\therefore \qquad \frac{1}{T}\, \delta \mathrm{T} = 0 + \frac{1}{2} \frac{1}{l}\, \delta l - \frac{1}{2} \frac{1}{g}\, \delta g$$

$$\frac{\delta T}{T} 100 = \frac{1}{2}\left(\frac{\delta l}{l} 100 - \frac{\delta g}{g} 100\right) = \frac{1}{2}(1 \pm 2.5) = 1.75 \text{ or } -0.75$$

Thus the maximum error in $T = 1.75\%$

Example 5.37. *A balloon is in the form of right circular cylinder of radius 1.5 m and length 4 m and is surmounted by hemispherical ends. If the radius is increased by 0.01 m and length by 0.05 m, find the percentage change in the volume of balloon.* (U.P.T.U., 2005)

Solution. Let the volume of the balloon (Fig. 5.3) be V, so that

$$V = \pi r^2 h + \frac{2}{3}\pi r^3 + \frac{2}{3}\pi r^3 = \pi r^2 h + \frac{4}{3}\pi r^3$$

$$\therefore \qquad \delta V = 2\pi r \delta r h + \pi r^2 \delta h + \frac{4}{3}\pi 3 r^2 \delta r$$

or
$$\frac{\delta V}{V} = \frac{\pi r[2h\delta r + r\delta h + 4r\delta r]}{\pi r^2 h + \frac{4}{3}\pi r^3}$$

$$= \frac{2(h+2r)\,\delta r + r\delta h}{rh + \frac{4}{3}r^2} = \frac{2(4+3)(.01) + 1.5(.05)}{1.5 \times 4 + \frac{4}{3}(1.5)^2}$$

$$= \frac{0.14 + 0.075}{6+3} = \frac{0.215}{9}$$

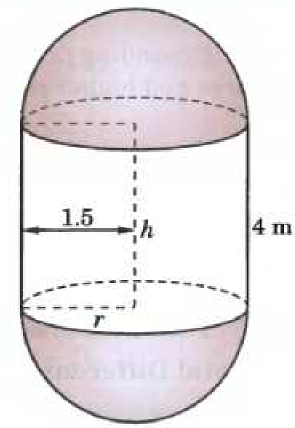

Fig. 5.3

Hence, the percentage change in $V = 100\,\dfrac{\delta V}{V} = \dfrac{21.5}{9} = 2.39\%$

Example 5.38. *In estimating the cost of a pile of bricks measured as 2 m × 15 m × 1.2 m, the tape is stretched 1% beyond the standard length. If the count is 450 bricks to 1 cu. m. and bricks cost ₹ 530 per 1000, find the approximate error in the cost.* (V.T.U., 2001)

Solution. Let x, y and z m be the length, breadth and height of the pile so that its volume $V = xyz$

or
$$\log V = \log x + \log y + \log z \;\therefore\; \frac{\delta V}{V} = \frac{\delta x}{x} + \frac{\delta y}{y} + \frac{\delta z}{z}$$

Since
$$V = 2 \times 15 \times 1.2 = 36 \text{ m}^3, \text{ and } \frac{\delta x}{x} = \frac{\delta y}{y} = \frac{\delta z}{z} = \frac{1}{100}$$

$$\therefore \qquad \delta V = 36\left(\frac{3}{100}\right) = 1.08 \text{ m}^3.$$

Number of bricks in $\delta V = 1.08 \times 450 = 486$

Thus error in the cost $= 486 \times \dfrac{530}{1000} =$ ₹ 257.58 which is a loss to the brick seller.

Example 5.39. *The height h and semi-vertical angle α of a cone are measured and from them A, the total area of the surface of the cone including the base is calculated. If h and α are in error by small quantities δh and $\delta\alpha$ respectively, find the corresponding error in the area. Show further that if $\alpha = \pi/6$, an error of + 1% in h will be approximately compensated by an error of − 0.33 degrees in α.*

Solution. If r be the base radius and l the slant height of the cone, (Fig. 5.4), then total area

$$A = \text{area of base} + \text{area of curved surface}$$
$$= \pi r^2 + \pi r l = \pi r(r + l)$$
$$= \pi h \tan\alpha\,(h\tan\alpha + h\sec\alpha)$$
$$= \pi h^2(\tan^2\alpha + \tan\alpha\sec\alpha)$$

$$\therefore \qquad \delta A = \frac{\delta A}{\delta h}\delta h + \frac{\delta A}{\delta\alpha}\delta\alpha$$
$$= 2\pi h(\tan^2\alpha + \tan\alpha\sec\alpha)\delta h$$
$$+ \pi h^2(2\tan\alpha\sec^2\alpha + \sec^3\alpha + \tan\alpha\sec\alpha\tan\alpha)\delta\alpha$$

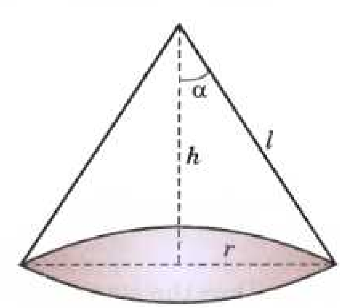

Fig. 5.4

which gives the error in the area A.

Putting $\delta h = h/100$ and $\alpha = \pi/6$, we get

$$\delta A = 2\pi h\left[\left(\frac{1}{\sqrt{3}}\right)^2 + \frac{1}{\sqrt{3}}\cdot\frac{2}{\sqrt{3}}\right]\frac{h}{100} + \pi h^2\left[2.\frac{1}{\sqrt{3}}\cdot\frac{4}{3} + \frac{8}{3\sqrt{3}} + \frac{1}{\sqrt{3}}\cdot\frac{2}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}\right]\delta\alpha$$

$$= \frac{2\pi h^2}{100} + 2\sqrt{3}\pi h^2\ \delta\alpha$$

The error in h will be compensated by the error in α, when

$$\delta A = 0 \text{ i.e., } \frac{2\pi h^2}{100} + 2\sqrt{3}\pi h^2\ \delta\alpha = 0$$

or
$$\delta\alpha = -\frac{1}{100\sqrt{3}} \text{ radians} = -\frac{.01}{1.732} \times 57.3° = -0.33°.$$

Example 5.40. *Show that the approximate change in the angle A of a triangle ABC due to small changes δa, δb, δc in the sides a, b, c respectively, is given by*

$$\delta A = \frac{a}{2\Delta}(\delta a - \delta b \cos C - \delta c \cos B)$$

where Δ is the area of the triangle. Verify that δA + δB + δC = 0.

Solution. We know that $a^2 = b^2 + c^2 - 2bc \cos A$

so that
$$2a\delta a = 2b\delta b + 2c\delta c - 2(c\delta b \cos A - b\delta c \cos A + bc \sin A\ \delta A)$$

$$\therefore \quad bc \sin A\ \delta A = a\delta a - (b - c\cos A)\ \delta b - (c - b \cos A)\ \delta c$$

or
$$2\ \Delta\delta A = a\delta a - (c\cos A + a\cos C - c\cos A)\ \delta b - (a \cos B + b\cos A - b\cos A)\ \delta c$$

[$\because$ $b = c\cos A + a\cos C$ etc. ...(*i*)]

$$= a\delta a - a\cos C\ \delta b - a\cos B\ \delta c$$

or
$$\delta A = \frac{a}{2\Delta}(\delta a - \delta b \cos C - \delta c \cos B)$$

By symmetry, we have

$$\delta B = \frac{b}{2\Delta}(\delta b - \delta c \cos A - \delta a \cos C)$$

$$\delta C = \frac{c}{2\Delta}(\delta c - \delta a \cos B - \delta b \cos A)$$

$$\therefore \quad \delta A + \delta B + \delta C = \frac{1}{2\Delta}(a - b\cos C - c\cos B)\ \delta a + (b - c\cos A - a\cos C)\ \delta b + (c - a\cos B - b\cos A)]$$

$$= \frac{1}{2\Delta}[(a - a)\ \delta a + (b - b)\ \delta b + (c - c)\ \delta c] = 0$$ [By (*i*)]

Example 5.41. *If the sides of a plane triangle ABC vary in such a way that its circumradius remains constant, prove that* $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$.

Solution. The circumradius R of ΔABC is given by

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$\therefore \quad a = 2R\sin A$$ [$\because$ R is constant

Taking differentials, $da = 2R\cos A\ dA$ or $\frac{da}{\cos A} = 2RdA$

Similarly, $\frac{db}{\cos B} = 2RdB,\ \frac{dc}{\cos C} = 2RdC$

$$\therefore \quad \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R\ (dA + dB + dC)$$

Now $A + B + C = \pi$, gives $dA + dB + dC = 0$ ...(*i*)

Thus
$$\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$$ [By (*i*)]

PROBLEMS 5.9

1. Expand the following functions as far as terms of third degree :
   (i) $\sin x \cos y$ (*V.T.U., 2009*) (ii) $e^x \sin y$ at $(-1, \pi/4)$ (*Anna, 2009*)
   (iii) $xy^2 + \cos xy$ about $(1, \pi/2)$. (*Hissar, 2005 S ; V.T.U., 2003*)
2. Expand $f(x, y) = x^y$ in powers of $(x - 1)$ and $(y - 1)$. (*U.T.U., 2009*)
3. If $f(x, y) = \tan^{-1} xy$, compute $f(0.9, -1.2)$ approximately.
4. If the kinetic energy $k = wv^2/2g$, find approximately the change in the kinetic energy as w changes from 49 to 49.5 and v changes from 1600 to 1590. (*V.T.U., 2006*)
5. Find the possible percentage error in computing the resistance r from the formula $1/r = 1/r_1 + 1/r_2$, if r_1, r_2 are both in error by 2%.
6. The voltage V across a resistor is measured with an error h, and the resistance R is measured with an error k. Show that the error in calculating the power $W(V, R) = V^2/R$ generated in the resistor, is $VR^{-2}(2Rh - Vk)$. (*V.T.U., 2009*)
7. Find the percentage error in the area of an ellipse if one per cent error is made in measuring the major and minor axes. (*V.T.U., 2011*)
8. The time of oscillation of a simple pendulum is given by the equation $T = 2\pi\sqrt{(l/g)}$. In an experiment carried out to find the value of g, errors of 1.5% and 0.5% are possible in the values of l and T respectively. Show that the error in the calculated value of g is 0.5%. (*Cochin, 2005*)
9. If $pv^2 = k$ and the relative errors in p and v are respectively 0.05 and 0.025, show that the error in k is 10%. (*Mysore, 1999*)
10. If the H.P. required to propel a steamer varies as the cube of the velocity and square of the length. Prove that a 3% increase in velocity and 4% increase in length will require an increase of about 17% in H.P.
11. The range R of a projectile which starts with a velocity v at an elevation α is given by $R = (v^2 \sin 2\alpha)/g$. Find the percentage error in R due to an error of 1% in v and an error of $\frac{1}{2}$% in α. (*Kurukshetra, 2009*)
12. In estimating the cost of a pile of bricks measured as 6 m × 50 m × 4 m, the tape is stretched 1% beyond the standard length. If the count is 12 bricks in 1 m^3 and bricks cost ₹ 100 per 1000, find the approximate error in the cost. (*U.T.U., 2010 ; U.P.T.U., 2005*)
13. The deflection at the centre of a rod of length l and diameter d supported at its ends, loaded at the centre with a weight w varies at wl^3d^{-4}. What is the increase in the deflection corresponding to p% increase in w, q% decrease in l and r% increase in d ?
14. The work that must be done to propel a ship of displacement D for a distance s in time t is proportional to $(s^2D^{2/3}/t^2)$. Find approximately the increase of work necessary when the displacement is increased by 1%, the time is diminished by 1% and the distance diminished by 2%.
15. The indicated horse power I of an engine is calculated from the formula $I = PLAN/33{,}000$, where $A = \pi d^2/4$. Assuming that error of r per cent may have been made in measuring P, L, N and d, find the greatest possible error in I.
16. The torsional rigidity of a length of wire is obtained from the formula $N = 8\pi Il/t^2r^4$. If l is decreased by 2%, r is increased by 2%, t is increased by 1.5%, show that the value of N is diminished by 13% approximately. (*V.T.U., 2003*)
17. If $x^2 + y^2 + z^2 - 2xyz = 1$, show that $\dfrac{dx}{\sqrt{(1-x^2)}} + \dfrac{dy}{\sqrt{(1-y^2)}} + \dfrac{dz}{\sqrt{(1-z^2)}} = 0$.
   [**Hint.** $2(x - yz)\,dx + 2(y - zx)\,dy + 2(z - xy)\,dz = 0$. Also $(x - yz)^2 = (1 - y^2)(1 - z^2)$, ...]

5.11 (1) MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

Def. *A function $f(x, y)$ is said to have a* **maximum** *or* **minimum** *at $x = a, y = b$, according as*

$$f(a + h, b + k) < \text{or} > f(a, b),$$

for all positive or negative small values of h and k.

In other words, if $\Delta = f(a + h, b + k) - f(a, b)$, *is of the same sign for all small values of h, k, and if this sign is negative, then $f(a, b)$ is a maximum. If this sign is positive, $f(a, b)$ is a minimum.*

Considering $z = f(x, y)$ as a surface, maximum value of z occurs at the top of an elevation (*e.g.*, a dome) from which the surface descends in every direction and a minimum value occurs at the bottom of a depression (*e.g.*, a bowl) from which the surface ascends in every direction. Sometimes the maximum or minimum value may form a *ridge* such that the surface descends or ascends in all directions except that of the ridge. Besides these, we have such a point of the surface, where the tangent plane is horizontal and the surface looks like leather seat on the horse's back [Fig. 5.5 (*c*)] which falls for displacement in certain directions and rises for displacements in other directions. Such a point is called a **saddle point.**

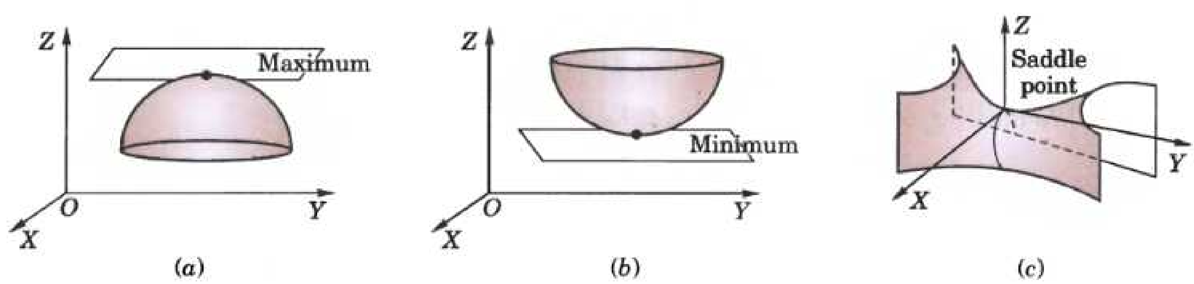

Fig. 5.5

Note. *A maximum or minimum value of a function is called its* **extreme value.**

(2) Conditions for f(x, y) to be maximum or minimum

Using Taylor's theorem page 235, we have $\Delta = f(a + h, b + k) - f(a, b)$

$$= \left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}\right)_{a,b} + \frac{1}{2!}\left(h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right) + \ldots \quad \ldots(i)$$

For small values of h amd k, the second and higher order terms are still smaller and hence may be neglected. Thus

$$\text{sign of } \Delta = \text{sign of } [hf_x(a, b) + kf_y(a, b)].$$

Taking $h = 0$ we see that the right hand side changes sign when k changes sign. Hence $f(x, y)$ cannot have a maximum or a minimum at (a, b) unless $f_y(a, b) = 0$.

Similarly taking $k = 0$, we find that $f(x, y)$ cannot have a maximum or minimum at (a, b) unless $f_x(a, b) = 0$.

Hence *the necessary conditions for $f(x, y)$ to have a maximum or minimum at (a, b) are that*

$$\mathbf{f_x(a, b) = 0, \; f_y(a, b) = 0.}$$

If these conditions are satisfied, then for small value of h and k, (i) gives

$$\text{sign of } \Delta = \text{sign of } \left[\frac{1}{2!}(h^2 r + 2hks + k^2 t)\right] \text{ where } r = f_{xx}(a, b),\ s = f_{xy}(a, b) \text{ and } t = f_{yy}(a, b).$$

Now $\quad h^2r + 2hks + k^2t = \dfrac{1}{r}\left[(h^2r^2 + 2hkrs + k^2rt)\right] = \dfrac{1}{r}\left[(hr + ks)^2 + k^2(rt - s^2)\right]$

Thus sign of Δ = sign of $\dfrac{1}{2r}\left\{(hr + ks)^2 + k^2(rt - s^2)\right\}$ $\quad \ldots(ii)$

In (ii), $(hr + ks)^2$ is always positive and $k^2(rt - s^2)$ will be positive if $rt - s^2 > 0$. In this case, Δ will have the same sign as that of r for all values of h and k.

Hence if $rt - s^2 > 0$, then *$f(x, y)$ has a maximum or a minimum at (a, b) according as $r <$ or > 0.*

If $rt - s^2 < 0$, then Δ will change with h and k and hence there is no maximum or minimum at (a, b) *i.e.*, it is a *saddle point*.

If $rt - s^2 = 0$, further investigation is required to find whether there is a maximum or minimum at (a, b) or not.

Note. Stationary value. *$f(a, b)$ is said to be a stationary value of $f(x, y)$, if $f_x(a, b) = 0$ and $f_y(a, b) = 0$ i.e.* the function is stationary at (a, b).

Thus every extreme value is a stationary value but the converse may not be true.

(3) Working rule to find the maximum and minimum values of f(x, y)

1. Find $\partial f/\partial x$ and $\partial f/\partial y$ *and equate each to zero. Solve these as simultaneous equations in x and y.* Let (a, b), (c, d), ... *be the pairs of values.*
2. *Calculate the value of $r = \partial^2 f/\partial x^2$, $s = \partial^2 f/\partial x\partial y$, $t = \partial^2 f/\partial y^2$ for each pair of values.*

3. *(i) If $rt - s^2 > 0$ and $r < 0$ at (a, b), $f(a, b)$ is a max. value.*
(ii) If $rt - s^2 > 0$ and $r > 0$ at (a, b), $f(a, b)$ is a min. value.
(iii) If $rt - s^2 < 0$ at (a, b), $f(a, b)$ is not an extreme value, i.e., (a, b) is a saddle point.
(iv) If $rt - s^2 = 0$ at (a, b), the case is doubtful and needs further investigation.

Similarly examine the other pairs of values one by one.

Example 5.42. *Examine the following function for extreme values :*

$$f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2. \qquad \text{(J.N.T.U., 2003)}$$

Solution. We have $f_x = 4x^3 - 4x + 4y$; $f_y = 4y^3 + 4x - 4y$

and $r = f_{xx} = 12x^2 - 4$, $s = f_{xy} = 4$, $t = f_{yy} = 12y^2 - 4$ ...(i)

Now $f_x = 0, f_y = 0$ give $x^3 - x + y = 0$, ...(i) $y^3 + x - y = 0$ ...(ii)

Adding these, we get $4(x^3 + y^3) = 0$ or $y = -x$.

Putting $y = -x$ in (i), we obtain $x^3 - 2x = 0$, i.e. $x = \sqrt{2}, -\sqrt{2}, 0$.

$\therefore$ Corresponding values of y are $-\sqrt{2}, \sqrt{2}, 0$.

At $(\sqrt{2}, -\sqrt{2})$, $rt - s^2 = 20 \times 20 - 4^2 = +$ ve and r is also + ve. Hence $f(\sqrt{2}, -\sqrt{2})$ is a minimum value.

At $(-\sqrt{2}, \sqrt{2})$ also both $rt - s^2$ and r are + ve.

Hence $f(-\sqrt{2}, \sqrt{2})$, is also a minimum value.

At $(0, 0)$, $rt - s^2 = 0$ and, therefore, further investigation is needed.

Now $f(0, 0) = 0$ and for points along the x-axis, where $y = 0$, $f(x, y) = x^4 - 2x^2 = x^2(x^2 - 2)$, which is negative for points in the neighbourhood of the origin.

Again for points along the line $y = x$, $f(x, y) = 2x^4$ which is positive.

Thus in the neighbourhood of $(0, 0)$ there are points where $f(x, y) < f(0, 0)$ and there are points where $f(x, y) > f(0, 0)$.

Hence $f(0, 0)$ is not an extreme value *i.e.*, it is a *saddle point*.

Example 5.43. *Discuss the maxima and minima of $f(x, y) = x^3y^2(1 - x - y)$.*

(Anna, 2009 ; J.N.T.U., 2006 ; Bhopal, 2002)

Solution. We have $f_x = 3x^2y^2 - 4x^3y^2 - 3x^2y^3$; $f_y = 2x^3y - 2x^4y - 3x^3y^2$

and $r = f_{xx} = 6xy^2 - 12x^2y^2 - 6xy^3$; $s = f_{xy} = 6x^2y - 8x^3y - 9x^2y^2$; $t = f_{yy} = 2x^3 - 2x^4 - 6x^3y$.

When $f_x = 0, f_y = 0$, we have $x^2y^2(3 - 4x - 3y) = 0$, $x^3y(2 - 2x - 3y) = 0$

Solving these, the stationary points are (1/2, 1/3), (0, 0).

Now $rt - s^2 = x^4y^2[12(1 - 2x - y)(1 - x - 3y) - (6 - 8x - 9y)^2]$

$$\text{At } (1/2, 1/3), \quad rt - s^2 = \frac{1}{16} \cdot \frac{1}{9}\left[12\left(1 - 1 - \frac{1}{3}\right)\left(1 - \frac{1}{2} - 1\right) - (6 - 4 - 3)^2\right] = \frac{1}{14} > 0$$

$$\text{Also} \qquad r = 6\left(\frac{1}{2} \cdot \frac{1}{9} - \frac{2}{4} \cdot \frac{1}{9} - \frac{1}{2} \cdot \frac{1}{27}\right) = -\frac{1}{9} < 0$$

Hence $f(x, y)$ has a maximum at (1/2, 1/3) and maximum value $= \frac{1}{8} \cdot \frac{1}{9}\left(1 - \frac{1}{2} - \frac{1}{3}\right) = \frac{1}{432}$.

At $(0, 0)$, $rt - s^2 = 0$ and therefore further investigation is needed.

For points along the line $y = x$, $f(x, y) = x^5(1 - 2x)$ which is positive for $x = 0.1$ and negative for $x = -0.1$ *i.e.*, in the neighbourhood of $(0, 0)$ there are points where $f(x, y) > f(0, 0)$ and there are points where $f(x, y) < f(0, 0)$. Hence $f(0, 0)$ is not an extreme value.

Example 5.44. *In a plane triangle, find the maximum value of $\cos A \cos B \cos C$.*

(V.T.U., 2010 ; Nagpur, 2009 ; Anna, 2005 S)

Solution. We have $A + B + C = \pi$ so that $C = \pi - (A + B)$.

$$\cos A \cos B \cos C = \cos A \cos B \cos[\pi - (A + B)]$$
$$= -\cos A \cos B \cos(A + B) = f(A, B), \text{ say.}$$

We get $\dfrac{\partial f}{\partial A} = \cos B\,[\sin A \cos (A + B) + \cos A \sin (A + B)]$

$= \cos B \sin (2A + B)$

and $\dfrac{\partial f}{\partial B} = \cos A \sin (A + 2B)$

$\dfrac{\partial f}{\partial A} = 0, \dfrac{\partial f}{\partial B} = 0$ only when $A = B = \pi/3$.

Also $r = \dfrac{\partial^2 f}{\partial A^2} = 2 \cos B \cos (2A + B), t = \dfrac{\partial^2 f}{\partial B^2} = 2 \cos A \cos (A + 2B)$

$s = \dfrac{\partial^2 f}{\partial A \partial B} = -\sin B \sin (2A + B) + \cos B \cos (2A + B) = \cos (2A + 2B)$

When $A = B = \pi/3$, $r = -1$, $s = -1/2$, $t = -1$ so that $rt - s^2 = 3/4$.

These show that $f(A, B)$ is maximum for $A = B = \pi/3$.

Then $C = \pi - (A + B) = \pi/3$.

Hence $\cos A \cos B \cos C$ is maximum when each of the angles is $\pi/3$ *i.e.*, triangle is equilateral and its maximum value = 1/8.

5.12 LAGRANGE'S METHOD OF UNDERTERMINED MULTIPLIERS

Sometimes it is required to find the stationary values of a function of several variables which are not all independent but are connected by some given relations. Ordinarily, we try to convert the given function to the one, having least number of independent variables with the help of given relations. Then solve it by the above method. When such a procedure becomes impracticable, Lagrange's method* proves very convenient. Now we explain this method.

Let $u = f(x, y, z)$ ...(1)

be a function of three variables x, y, z which are connected by the relation.

$\phi(x, y, z) = 0$ ...(2)

For u to have stationary values, it is necessary that

$\partial u/\partial x = 0, \partial u/\partial y = 0, \partial u/\partial z = 0.$

$\therefore \quad \dfrac{\partial u}{\partial x} dx + \dfrac{\partial u}{\partial y} dy + \dfrac{\partial u}{\partial z} dz = du = 0$ ...(3)

Also differentiating (2), we get $\dfrac{\partial \phi}{\partial x} dx + \dfrac{\partial \phi}{\partial y} dy + \dfrac{\partial \phi}{\partial z} dz = d\phi = 0$ ...(4)

Multiply (4) by a parameter λ and add to (3). Then

$$\left(\frac{\partial u}{\partial x} + \lambda \frac{\partial \phi}{\partial x}\right) dx + \left(\frac{\partial u}{\partial y} + \lambda \frac{\partial \phi}{\partial y}\right) dy + \left(\frac{\partial u}{\partial z} + \lambda \frac{\partial \phi}{\partial z}\right) dz = 0$$

This equation will be satisfied if $\dfrac{\partial u}{\partial x} + \lambda \dfrac{\partial \phi}{\partial x} = 0, \dfrac{\partial u}{\partial y} + \lambda \dfrac{\partial \phi}{\partial y} = 0, \dfrac{\partial u}{\partial z} + \lambda \dfrac{\partial \phi}{\partial z} = 0.$

These three equations together with (2) will determine the values of x, y, z and λ for which u is stationary.

Working rule : 1. *Write* $F = f(x, y, z) + \lambda\phi(x, y, z)$

2. *Obtain the equations* $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0, \dfrac{\partial F}{\partial z} = 0.$

3. *Solve the above equations together with* $\phi(x, y, z) = 0$.

The values of x, y, z *so obtained will give the stationary value of* $f(x, y, z)$.

Obs. Although the Lagrange's method is often very useful in application yet *the drawback is that we cannot determine the nature of the stationary point*. This can sometimes, be determined from physical considerations of the problem.

*See footnote page 142.

Example 5.45. *A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction.* (*Kurukshetra, 2006 ; P.T.U., 2006 ; U.P.T.U., 2005*)

Solution. Let x, y and z ft. be the edges of the box and S be its surface.

Then $$S = xy + 2yz + 2zx \qquad ...(i)$$

and $$xyz = 32 \qquad ...(ii)$$

Eliminating z from (*i*) with the help of (*ii*), we get $S = xy + 2(y + x)\dfrac{32}{xy} = xy + 64\left(\dfrac{1}{x} + \dfrac{1}{y}\right)$

$\therefore$ $$\partial S/\partial x = y - 64/x^2 = 0 \quad \text{and} \quad \partial S/\partial y = x - 64/y^2 = 0.$$

Solving these, we get $x = y = 4$.

Now $r = \partial^2 S/\partial x^2 = 128/x^3$, $s = \partial^2 S/\partial x \partial y = 1$, $t = \partial^2 S/\partial y^2 = 128/y^3$.

At $x = y = 4$, $rt - s^2 = 2 \times 2 - 1 = +$ ve and r is also + ve.

Hence S is minimum for $x = y = 4$. Then from (*ii*), $z = 2$.

Otherwise (*by Lagrange's method*) :

Write $$F = xy + 2yz + 2zx + \lambda(xyz - 32)$$

Then $$\frac{\partial F}{\partial x} = y + 2z + \lambda yz = 0 \qquad ...(iii)$$

$$\frac{\partial F}{\partial y} = x + 2z + \lambda zx = 0 \qquad ...(iv)$$

$$\frac{\partial F}{\partial z} = 2y + 2x + \lambda xy = 0 \qquad ...(v)$$

Multiplying (*iii*) by x and (*iv*) by y and subtracting, we get $2zx - 2zy = 0$ or $x = y$.

[The value $z = 0$ is neglected, as it will not satisfy (*ii*)]

Again multiplying (*iv*) by y and (*v*) by z and subtracting, we get $y = 2z$.

Hence the dimensions of the box are $x = y = 2z = 4$ ...(*vi*)

Now let us see what happens as z increases from a small value to a large one. When z is small, the box is flat with a large base showing that S is large. As z increases, the base of the box decreases rapidly and S also decreases. After a certain stage, S again starts increasing as z increases. Thus S must be a minimum at some intermediate stage which is given by (*vi*). Hence S is minimum when $x = y = 4$ ft and $z = 2$ ft.

Example 5.46. *Given $x + y + z = a$, find the maximum value of $x^m y^n z^p$.* (*Anna, 2009*)

Solution. Let $f(x, y, z) = x^m y^n z^p$ and $\phi(x, y, z) = x + y + z - a$.

Then $$F(x, y, z) = f(x, y, z) + \lambda\, \phi(x, y, z)$$
$$= x^m y^n z^p + \lambda(x + y + z - a).$$

For stationary values of F, $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0, \dfrac{\partial F}{\partial z} = 0$

$\therefore$ $$mx^{m-1}y^n z^p + \lambda = 0,\ nx^m y^{n-1} z^p + \lambda = 0,\ px^m y^n z^{p-1} + \lambda = 0$$

or $$-\lambda = mx^{n-1}y^n z^p = nx^m y^{n-1} z^p = px^m y^n z^{p-1}$$

i.e. $$\frac{m}{x} = \frac{n}{y} = \frac{p}{z} = \frac{m + n + p}{x + y + z} = \frac{m + n + p}{a} \qquad [\because\ x + y + z = a$$

$\therefore$ The maximum value of f occurs when

$$x = am/(m + n + p),\ y = an/(m + n + p),\ z = ap/(m + n + p)$$

Hence the maximum value of $f(x, y, z) = \dfrac{a^{m+n+p} \cdot m^m \cdot n^n \cdot p^p}{(m + n + p)^{m+n+p}}$.

Example 5.47. *Find the maximum and minimum distances of the point (3, 4, 12) from the sphere $x^2 + y^2 + z^2 = 4$.* (*U.T.U., 2010*)

Solution. Let $P(x, y, z)$ be any point on the sphere and $A(3, 4, 12)$ the given point so that

$$AP^2 = (x - 3)^2 + (y - 4)^2 + (z - 12)^2 = f(x, y, z), \text{ say} \qquad ...(i)$$

We have to find the maximum and minimum values of $f(x, y, z)$ subject to the condition

$$\phi(x, y, z) = x^2 + y^2 + z^2 - 4 = 0 \quad ...(ii)$$

Let $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$

$$= (x - 3)^2 + (y - 4)^2 + (z - 12)^2 + \lambda(x^2 + y^2 + z^2 - 4)$$

Then $$\frac{\partial F}{\partial x} = 2(x - 3) + 2\lambda x, \ \frac{\partial F}{\partial y} = 2(y - 4) + 2\lambda y, \ \frac{\partial F}{\partial z} = 2(z - 12) + 2\lambda z$$

$\therefore$ $$\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0 \text{ and } \frac{\partial F}{\partial z} = 0 \text{ give}$$

$$x - 3 + \lambda x = 0, y - 4 + \lambda y = 0, z - 12 + \lambda z = 0 \quad ...(iii)$$

which give $$\lambda = -\frac{x-3}{x} = -\frac{y-4}{y} = -\frac{z-12}{z}$$

$$= \pm\frac{\sqrt{[(x-3)^2 + (y-4)^2 + (z-12)^2]}}{\sqrt{(x^2+y^2+z^2)}} = \pm\frac{\sqrt{f}}{1}$$

Substituting for λ in (iii), we get

$$x = \frac{3}{1+\lambda} = \frac{3}{1 \pm \sqrt{f}}, y = \frac{4}{1 \pm \sqrt{f}}, z = \frac{12}{1 \pm \sqrt{f}}$$

$\therefore$ $$x^2 + y^2 + z^2 = \frac{9 + 16 + 144}{(1 \pm \sqrt{f})^2} = \frac{169}{(1 \pm \sqrt{f})^2}$$

Using (ii), $$1 = \frac{169}{(1 \pm \sqrt{f})^2} \quad \text{or} \quad 1 \pm \sqrt{f} = \pm 13, \sqrt{f} = 12, 14.$$

[We have left out the negative values of $\sqrt{f}$, because $\sqrt{f} = AP$ is + ve by (i)]

Hence maximum $AP = 14$ and minimum $AP = 12$.

Example 5.48. *Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.* *(Kurukshetra, 2006 ; U.P.T.U., 2004)*

Solution. Let $2x$, $2y$, $2z$ be the length, breadth and height of the rectangular solid so that its volume

$$V = 8xyz \quad ...(i)$$

Let R be the radius of the sphere so that $x^2 + y^2 + z^2 = R^2$ ...(ii)

Then $$F(x, y, z) = 8xyz + \lambda(x^2 + y^2 + z^2 - R^2)$$

and $$\partial F/\partial x = 0, \partial F/\partial y = 0 \text{ and } \partial F/\partial z = 0 \quad \text{give}$$

$$8yz + 2x\lambda = 0, 8zx + 2y\lambda = 0, 8xy + 2z\lambda = 0$$

or $$2x^2\lambda = -8xyz = 2y^2\lambda = 2z^2\lambda$$

Thus for a maximum volume $x = y = z$.

i.e., the rectangular solid is a cube.

Example 5.49. *A tent on a square base of side x, has its sides vertical of height y and the top is a regular pyramid of height h. Find x and y in terms of h, if the canvas required for its construction is to be minimum for the tent to have a given capacity.*

Solution. Let V be the volume enclosed by the tent and S be its surface area (Fig. 5.6).

Then V = cuboid $(ABCD, A'B'C'D')$ + pyramid $(K, A'B'C'D')$

$$= x^2y + \frac{1}{3}x^2h = x^2(y + h/3)$$

$$S = 4(ABGF) + 4\Delta KGH = 4xy + 4\frac{1}{2}(x \cdot KM)$$

$$= 4xy + x\sqrt{(x^2 + 4h^2)} \quad \left[\because KM = \sqrt{(KL^2 + LM^2)} = \sqrt{[h^2 + (x/2)^2]}\right]$$

For constant V, we have

$$\delta V = 2x(y + h/3)\,\delta x + x^2(\delta y) + \frac{x^2}{3}\,\delta h = 0$$

For minimum S, we have

$$\delta S = [4y + \sqrt{(x^2+4h^2)} + x \cdot \frac{1}{2}(x^2+4h^2)^{-1/2} \cdot 2x]\,\delta x$$
$$+ 4x\delta y + x \cdot \frac{1}{2}(x^2+4h^2)^{-1/2} \cdot 8h\delta h = 0$$

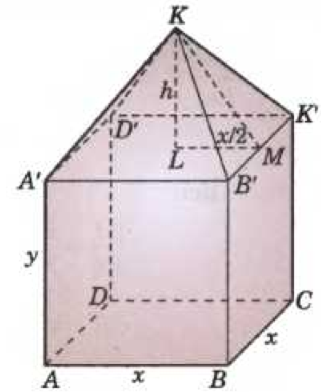

Fig. 5.6

By Lagrange's method,

$$[4y + \sqrt{(x^2+4h^2)} + x^2(x^2+4h^2)^{-1/2}] + \lambda \cdot 2x(y + h/3) = 0 \qquad ...(i)$$
$$4x + \lambda \cdot x^2 = 0 \qquad ...(ii)$$
$$4hx(x^2+4h^2)^{-1/2} + \lambda \cdot x^2/3 = 0 \qquad ...(iii)$$

(ii) gives $\lambda = -4/x$. Then (iii) becomes

$$4hx(x^2+4h^2)^{-1/2} - 4x/3 = 0 \quad \text{or} \quad x = \sqrt{5}\,h$$

Now putting $x = \sqrt{5}\,h$, $\lambda = -4/x$ in (i), we get

$$4y + 3h + \frac{5}{3}h - \frac{4}{x} \cdot 2x(y + h/3) = 0 \quad \text{or} \quad 4y + \frac{14}{3}h - 8y - \frac{8h}{3} = 0, \quad i.e., \quad y = h/2.$$

Example 5.50. *If $u = a^3x^2 + b^3y^2 + c^3z^2$ where $x^{-1} + y^{-1} + z^{-1} = 1$, show that the stationary value of u is given by $x = \Sigma a/a$, $y = \Sigma a/b$, $z = \Sigma a/c$.* *(Kerala, 2005)*

Solution. Let $u = f(x, y, z) = a^3x^2 + b^3y^2 + c^3z^2$

and $\phi(x, y, z) = x^{-1} + y^{-1} + z^{-1} - 1$

Let $F(x, y, z) = f(x, y, z) + \lambda\,\phi(x, y, z)$
$= a^3x^2 + b^3y^2 + c^3z^2 + \lambda(x^{-1} + y^{-1} + z^{-1} - 1)$

Then $\dfrac{\partial F}{\partial x} = 0, \dfrac{\partial F}{\partial y} = 0$ and $\dfrac{\partial F}{\partial z} = 0$ give

$$2a^3x^2 - \lambda/x^2 = 0, \quad 2b^3y - \lambda/y^2 = 0, \quad 2c^3z - \lambda/z^2 = 0$$

or
$$2a^3x^3 = \lambda,\ 2b^3y^3 = \lambda,\ 2c^3z^3 = \lambda$$

which give $ax = by = cz = k$ (say) *i.e.*, $x = k/a$, $y = k/b$, $z = k/c$.

Substituting these in $x^{-1} + y^{-1} + z^{-1} = 1$, we get $k = a + b + c$

Hence the stationary value of u is given by

$$x = \Sigma a/a, \quad y = \Sigma a/b \quad \text{and} \quad z = \Sigma a/c.$$

Example 5.51. *Find the volume of the greatest rectangular parallelopiped that can be inscribed in the ellipsoid*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

(U.T.U., 2010 ; Anna, 2009 ; Madras, 2006)

Solution. Let the edges of the parallelopiped be $2x$, $2y$ and $2z$ which are parallel to the axes. Then its volume $V = 8xyz$.

Now we have to find the maximum value of V subject to the condition that

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \qquad ...(i)$$

Write
$$F = 8xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$$

Then
$$\frac{\partial F}{\partial x} = 8yz + \lambda\left(\frac{2x}{a^2}\right) = 0 \qquad ...(ii)$$

$$\frac{\partial F}{\partial y} = 8zx + \lambda\left(\frac{2y}{b^2}\right) = 0 \quad ...(iii) \qquad \frac{\partial F}{\partial z} = 8xy + \lambda\left(\frac{2z}{c^2}\right) = 0 \quad ...(iv)$$

Equating the values of λ from (*ii*) and (*iii*), we get $x^2/a^2 = y^2/b^2$

Similarly from (*iii*) and (*iv*), we obtain $y^2/b^2 = z^2/c^2 \quad \therefore \quad x^2/a^2 = y^2/b^2 = z^2/c^2$

Substituting these in (*i*), we get $x^2/a^2 = \frac{1}{3}$ *i.e.* $\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} = \frac{1}{3}$

These give $x = a/\sqrt{3}$, $y = b/\sqrt{3}$, $z = c/\sqrt{3}$ ...(*v*)

When $x = 0$, the parallelopiped is just a rectangular sheet and as such its volume $V = 0$.

As x increases, V also increases continuously.

Thus V must be greatest at the stage given by (*v*).

Hence the greatest volume $= \frac{8abc}{3\sqrt{3}}$.

PROBLEMS 5.10

1. Find the maximum and minimum values of
   (*i*) $x^3 + y^3 - 3axy$ (*U.P.T.U., 2005*) (*ii*) $xy + a^3/x + a^3/y$.
   (*iii*) $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ (*Mumbai, 2007*) (*iv*) $2(x^2 - y^2) - x^4 + y^4$ (*Osmania, 2003*)
   (*v*) $\sin x \sin y \sin (x + y)$.
2. If $xyz = 8$, find the values of x, y for which $u = 5xyz/(x + 2y + 4z)$ is a maximum. (*S.V.T.U., 2007 ; Kurukshetra, 2005*)
3. Find the minimum value of $x^2 + y^2 + z^2$, given that
   (*i*) $xyz = a^3$ (*P.T.U., 2009 ; Osmania, 2003*) (*ii*) $ax + by + cz = p$. (*V.T.U., 2010 ; U.P.T.U., 2006*)
   (*iii*) $xy + yz + zx = 3a^2$ (*Anna, 2009*)
4. Find the dimensions of the rectangular box, open at the top, of maximum capacity whose surface is 432 sq. cm. (*Madras, 2000 S*)
5. The sum of three numbers is constant. Prove that their product is maximum when they are equal.
6. Find the points on the surface $z^2 = xy + 1$ nearest to the origin. (*Burdwan, 2003 ; Andhra, 2000*)
7. Show that, if the perimeter of a triangle is constant, the triangle has maximum area when it is equilateral.
8. Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.
9. The temperature T at any point (x, y, z) in space is $T = 400\,xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (*V.T.U., 2009 ; Hissar, 2005 S*)
10. Divide 24 into three parts such that the continued product of the first, square of the second and the cube of the third may be maximum. (*Bhillai, 2005*)
11. Find the stationary values of $u = x^2 + y^2 + z^2$ subject to $ax^2 + by^2 + cz^2 = 1$ and $lx + my + nz = 0$. (*S.V.T.U., 2008*)

5.13 DIFFERENTIATION UNDER THE INTEGRAL SIGN

If a function $f(x, \alpha)$ of two variables x and α (called a parameter), be integrated with respect to x between the limits a and b, then $\int_a^b f(x, \alpha)\,dx$ is a function of α : $F(\alpha)$, say. To find the derivative of $F(\alpha)$, when it exists, it is not always possible to first evaluate this integral and then to find the derivative. Such problems are solved by the following rules :

(1) Leibnitz's rule*

If $f(x, \alpha)$ and $\frac{\partial f(x, \alpha)}{\partial \alpha}$ *be continuous functions of x and* α, *then*

$$\frac{d}{d\alpha}\left[\int_a^b f(x, \alpha)\,dx\right] = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha}\,dx \quad \textit{where, } a, b \textit{ are constants independent of } \alpha.$$

*See foot note on p. 139.

CHAPTER 17

Partial Differential Equations

17.1 INTRODUCTION

The reader has, already been introduced to the notion of partial differential equations. Here, we shall begin by studying the ways in which partial differential equations are formed. Then we shall investigate the solutions of special types of partial differential equations of the first and higher orders.

In what follows x and y will, usually be taken as the independent variables and z, the dependent variable so that $z = f(x, y)$ and we shall employ the following notation :

$$\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial^2 z}{\partial x^2} = r, \frac{\partial^2 z}{\partial x \partial y} = s, \frac{\partial^2 z}{\partial y^2} = t.$$

17.2 FORMATION OF PARTIAL DIFFERENTIAL EQUATIONS

Unlike the case of ordinary differential equations which arise from the elimination of arbitrary constants; *the partial differential equations can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions from a relation involving three or more variables.* The method is best illustrated by the following examples :

Example 17.1. *Derive a partial differential equation (by eliminating the constants) from the equation*

$$2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}. \qquad ...(i)$$

Solution. Differentiating (*i*) partially with respect to x and y, we get

$$2\frac{\partial z}{\partial x} = \frac{2x}{a^2} \quad \text{or} \quad \frac{1}{a^2} = \frac{1}{x}\frac{\partial z}{\partial x} = \frac{p}{x}$$

and

$$\frac{2\partial z}{\partial y} = \frac{2y}{b^2} \quad \text{or} \quad \frac{1}{b^2} = \frac{1}{y}\frac{\partial z}{\partial y} = \frac{q}{y}$$

Substituting these values of $1/a^2$ and $1/b^2$ in (*i*), we get

$$2z = xp + yq$$

as the desired partial differential equation of the first order.

Example 17.2. *Form the partial differential equations (by eliminating the arbitrary functions) from*
(*a*) $z = (x + y)\,\phi\,(x^2 - y^2)$ (*P.T.U., 2009*)
(*b*) $z = f(x + at) + g(x - at)$ (*V.T.U., 2009*) (*c*) $f(x^2 + y^2, z - xy) = 0$ (*S.V.T.U., 2007*)

Solution. (*a*) We have $z = (x + y)\,\phi\,(x^2 - y^2)$

Differentiating z partially with respect to x and y,

$$p = \frac{\partial z}{\partial x} = (x + y)\,\phi'\,(x^2 - y^2)\,.\,2x + \phi\,(x^2 - y^2), \quad ...(i)$$

$$q = \frac{\partial z}{\partial y} = (x + y)\,\phi'\,(x^2 - y^2)\,.\,(-2y) + \phi\,(x^2 - y^2) \quad ...(ii)$$

From (*i*), $$p - \frac{z}{x + y} = 2x\,(x + y)\,\phi'\,(x^2 - y^2)$$

From (*ii*), $$q - \frac{z}{x + y} = -2y\,(x + y)\,\phi'\,(x^2 - y^2)$$

Division gives $\dfrac{p - z/(x + y)}{q - z/(x + y)} = -\dfrac{x}{y}$

i.e., $[p(x + y) - z]y + [q\,(x + y) - z]x$

i.e., $(x + y)\,(py + qx) - z(x + y) = 0$

Hence $py + qz = z$ is required equation.

(*b*) We have $z = f\,(x + at) + g(x - at)$ ...(*i*)

Differentiating z partially with respect to x and t,

$$\frac{\partial z}{\partial x} = f'\,(x + at) + g'(x - at),\ \frac{\partial^2 z}{\partial x^2} = f''\,(x + at) + g''(x - at) \quad ...(ii)$$

$$\frac{\partial z}{\partial t} = af'\,(x + at) - ag'(x - at),\ \frac{\partial^2 z}{\partial t^2} = a^2 f''\,(x + at) + a^2 g''(x - at) = a^2\,\frac{\partial^2 z}{\partial x^2} \quad [\text{By } (ii)]$$

Thus the desired partial differential equation is $\dfrac{\partial^2 z}{\partial t^2} = a^2\,\dfrac{\partial^2 z}{\partial x^2}$

which is an equation of the second order and (*i*) is its solution.

(*c*) Let $x^2 + y^2 = u$ and $z - xy = v$ so that $f\,(u, v) = 0$.

Differentiating partially w.r.t. x and y, we have

$$\frac{\partial f}{\partial u}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z}p\right) + \frac{\partial f}{\partial v}\left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z}p\right) = 0$$

or $$\frac{\partial f}{\partial u}(2x) + \frac{\partial f}{\partial v}(-y + p) = 0 \quad ...(i)$$

and $$\frac{\partial f}{\partial u}\left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}q\right) + \frac{\partial f}{\partial v}\left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z}q\right) = 0 \quad \text{or} \quad \frac{\partial f}{\partial u}(2y) + \frac{\partial f}{\partial v}(-x + q) = 0 \quad ...(ii)$$

Eliminating $\dfrac{\partial f}{\partial u}$ and $\dfrac{\partial f}{\partial v}$ from (*i*) and (*ii*), we get

$$\begin{vmatrix} 2x & -y + p \\ 2y & -x + q \end{vmatrix} = 0 \quad \text{or} \quad xq - yp = x^2 - y^2.$$

Example 17.3. *Find the differential equation of all planes which are at a constant distance a from the origin.* (*V.T.U., 2009 S ; Kurukshetra, 2006*)

Solution. The equation of the plane in 'normal form' is

$$lx + my + nz = a \quad ...(i)$$

where l, m, n are the d.c.s of the normal from the origin to the plane.

Then $\quad l^2 + m^2 + n^2 = 1$ or $n = \sqrt{(1 - l^2 - m^2)}$

$\therefore$ (*i*) becomes $\quad lx + my + \sqrt{(1 - l^2 - m^2)}\ z = a \qquad$ ...(*ii*)

Differentiating partially w.r.t. x, we get

$$l + \sqrt{(1 - l^2 - m^2)}\ .\ p = 0 \qquad ...(iii)$$

Differentiating partially w.r.t. y, we get

$$m + \sqrt{(1 - l^2 - m^2)}\ .\ q = 0 \qquad ...(iv)$$

Now we have to eliminate l, m from (*ii*), (*iii*) and (*iv*).

From (*iii*), $l = -\sqrt{(1 - l^2 - m^2)}\ .\ p$ and $m = -\sqrt{(1 - l^2 - m^2)}\ .\ q$

Squaring and adding, $l^2 + m^2 = (1 - l^2 - m^2)(p^2 + q^2)$

or $\quad (l^2 + m^2)(1 + p^2 + q^2) = p^2 + q^2$ or $1 - l^2 - m^2 = 1 - \dfrac{p^2 + q^2}{1 + p^2 + q^2} = \dfrac{1}{1 + p^2 + q^2}$

Also $\quad l = -\dfrac{p}{\sqrt{(1 + p^2 + q^2)}}$ and $m = -\dfrac{q}{\sqrt{(1 + p^2 + q^2)}}$

Substituting the values of l, m and $1 - l^2 - m^2$ in (*ii*), we obtain

$$\frac{-px}{\sqrt{(1 + p^2 + q^2)}} - \frac{qy}{\sqrt{(1 + p^2 + q^2)}} + \frac{1}{\sqrt{(1 + p^2 + q^2)}}\ z = a$$

or $\quad z = px + qy + a\sqrt{(1 + p^2 + q^2)}$ which is the required partial differential equation.

PROBLEMS 17.1

From the partial differential equation (by eliminating the arbitrary constants from :

1. $z = ax + by + a^2 + b^2$.
2. $(x - a)^2 + (y - b)^2 + z^2 = c^2$. (*Kottayam, 2005*)
3. $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (*Anna, 2009*)
4. $z = a \log \left\{\dfrac{b(y-1)}{1-x}\right\}$ (*J.N.T.U., 2002 S*)
5. Find the differential equation of all spheres of fixed radius having their centres in the xy-plane. (*Madras 2000 S*)
6. Find the differential equation of all spheres whose centres lie on the z-axis. (*Kerala, 2005*)

Form the partial differential equations (by eliminating the arbitrary functions) from :

7. $z = f(x^2 - y^2)$ (*S.V.T.U., 2008*)
8. $z = f(x^2 + y^2) + x + y$ (*Anna, 2009*)
9. $z = yf(x) + xg(y)$. (*V.T.U., 2004*)
10. $z = x^2 f(y) + y^2 g(x)$. (*Anna, 2003*)
11. $z = f(x) + e^y g(x)$.
12. $xyz = \phi(x + y + z)$.
13. $z = f_1(x) f_2(y)$.
14. $z = e^{my} \phi(x - y)$. (*P.T.U., 2002*)
15. $z = y^2 + 2f\left(\dfrac{1}{x} + \log y\right)$. (*V.T.U., 2010 ; J.N.T.U., 2010 ; Madras, 2000*)
16. $z = f_1(y + 2x) + f_2(y - 3x)$. (*Kurukshetra, 2005*)
17. $v = \dfrac{1}{r}[f(r - at) + F(r + at)]$.
18. $z = xf_1(x + t) + f_2(x + t)$.
19. $F(xy + z^2, x + y + z) = 0$. (*V.T.U., 2006*)
20. $F(x + y + z, x^2 + y^2 + z^2) = 0$. (*S.V.T.U., 2007*)
21. If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that $(y^2 - zx)\dfrac{\partial u}{\partial x} + (x^2 - yz)\dfrac{\partial u}{\partial y} + (z^2 - xy)\dfrac{\partial u}{\partial z} = 0$.

17.3 SOLUTIONS OF A PARTIAL DIFFERENTIAL EQUATION

It is clear from the above examples that a partial differential equation can result both from elimination of arbitrary constants and from the elimination of arbitrary functions.

The solution $\quad f(x, y, z, a, b) = 0 \qquad$ ...(1)

of a first order partial differential equation which contains two arbitrary constants is called a *complete integral*.

A solution obtained from the complete integral by assigning particular values to the arbitrary constants is called a particular integral.

If we put $b = \phi(a)$ in (1) and find the envelope of the family of surfaces $f[x, y, z, \phi(a)] = 0$, then we get a solution containing an arbitrary function ϕ, which is called the *general integral*.

The envelope of the family of surfaces (1), with parameters a and b, if it exists, is called a *singular integral*. The singular integral differs from the particular integral in that it is not obtained from the complete integral by giving particular values to the constants.

17.4 EQUATIONS SOLVABLE BY DIRECT INTEGRATION

We now consider such partial differential equations which can be solved by direct integration. In place of the usual constants of integration, we must, however, use arbitrary functions of the variable held fixed.

Example 17.4. *Solve* $\dfrac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x - y) = 0.$ (V.T.U., 2010)

Solution. Integrating twice with respect to x (keeping y fixed),

$$\frac{\partial^2 z}{\partial x \partial y} + 9x^2 y^2 - \frac{1}{2}\cos(2x - y) = f(y)$$

$$\frac{\partial z}{\partial y} + 3x^3 y^2 - \frac{1}{4}\sin(2x - y) = xf(y) + g(y).$$

Now integrating with respect to y (keeping x fixed)

$$z + x^3y^3 - \frac{1}{4}\cos(2x - y) = x\int f(y)dy + \int g(y)dy + w(x)$$

The result may be simplified by writing

$$\int f(y)dy = u(y) \text{ and } \int g(y)dy = v(y).$$

Thus $z = \dfrac{1}{4}\cos(2x - y) - x^3y^3 + xu(y) + v(y) + w(x)$ where u, v, w are arbitrary functions.

Example 17.5. *Solve* $\dfrac{\partial^2 z}{\partial x^2} + z = 0$, *given that when* $x = 0$, $z = e^y$ *and* $\dfrac{\partial z}{\partial x} = 1$.

Solution. If z were function of x alone, the solution would have been $z = A \sin x + B \cos x$, where A and B are constants. Since z is a function of x and y, A and B can be arbitrary functions of y. Hence the solution of the given equation is $z = f(y) \sin x + \phi(y) \cos x$

$$\therefore \quad \frac{\partial z}{\partial x} = f(y)\cos x - \phi(y)\sin x$$

When $x = 0$; $z = e^y$, $\quad \therefore e^y = \phi(y)$. $\quad$ When $x = 0$, $\dfrac{\partial z}{\partial x} = 1$, $\quad \therefore \; 1 = f(y)$.

Hence the desired solution is $z = \sin x + e^y \cos x$.

Example 17.6. *Solve* $\dfrac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$, *for which* $\dfrac{\partial z}{\partial y} = -2\sin y$ *when* $x = 0$ *and* $z = 0$ *when* y *is an odd multiple of* $\pi/2$. (V.T.U., 2010 S)

Solution. Given equation is $\dfrac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$

Integrating w.r.t. x, keeping y constant, we get

$$\frac{\partial z}{\partial y} = -\cos x \sin y + f(y) \qquad ...(i)$$

When $x = 0$, $\frac{\partial z}{\partial y} = -2 \sin y$, $\quad \therefore \quad -2 \sin y = -\sin y + f(y)$ or $f(y) = -\sin y$

$\therefore$ (*i*) becomes $\frac{\partial z}{\partial y} = -\cos x \sin y - \sin y$

Now integrating w.r.t. y, keeping x constant, we get

$$z = \cos x \cos y + \cos y + g(x) \qquad \text{...(ii)}$$

When y is an odd multiple of $\pi/2$, $z = 0$.

$\therefore \qquad 0 = 0 + 0 + g(x)$ or $g(x) = 0 \qquad [\because \cos(2n+1)\pi/2 = 0]$

Hence from (*ii*), the complete solution is $z = (1 + \cos x) \cos y$.

PROBLEMS 17.2

Solve the following equations :

1. $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a.$
2. $\frac{\partial^2 z}{\partial x^2} = xy.$
3. $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x.$
4. $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y).$
5. $\frac{\partial^2 z}{\partial y^2} = z$, gives that when $y = 0$, $z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$
6. $\frac{\partial^2 z}{\partial x^2} = a^2 z$ given that when $x = 0$, $\frac{\partial z}{\partial x} = a \sin y$ and $\frac{\partial z}{\partial y} = 0.$

17.5 LINEAR EQUATIONS OF THE FIRST ORDER

A linear partial differential equation of the first order, commonly known as *Lagrange's Linear equation**, is of the form

$$Pp + Qq = R \qquad \text{...(1)}$$

where P, Q and R are functions of x, y, z. This equation is called a *quasi-linear equation*. When P, Q and R are independent of z it is known as *linear equation*.

Such an equation is obtained by eliminating an arbitrary function ϕ from $\phi(u, v) = 0 \qquad$ ...(2)

where u, v are some functions of x, y, z.

Differentiating (2) partially with respect to x and y.

$$\frac{\partial \phi}{\partial u}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p\right) + \frac{\partial \phi}{\partial v}\left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} p\right) = 0 \text{ and } \frac{\partial \phi}{\partial u}\left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q\right) + \frac{\partial \phi}{\partial v}\left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} q\right) = 0.$$

Eliminating $\frac{\partial \phi}{\partial u}$ and $\frac{\partial \phi}{\partial v}$, we get $\begin{vmatrix} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p & \frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} p \\ \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q & \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} q \end{vmatrix} = 0$

which simplifies to $\left(\frac{\partial u}{\partial y}\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\frac{\partial v}{\partial y}\right) p + \left(\frac{\partial u}{\partial z}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial v}{\partial z}\right) q = \left(\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}\right) \qquad$ ...(3)

This is of the same form as (1).

Now suppose $u = a$ and $v = b$, where a, b are constants, so that

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = du = 0$$

$$\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz = dv = 0.$$

*See footnote p. 142.

By cross-multiplication, we have

$$\frac{dx}{\frac{\partial u}{\partial y}\frac{\partial v}{\partial z}-\frac{\partial u}{\partial z}\frac{\partial v}{\partial y}}=\frac{dy}{\frac{\partial u}{\partial z}\frac{\partial v}{\partial x}-\frac{\partial u}{\partial x}\frac{\partial v}{\partial z}}=\frac{dz}{\frac{\partial u}{\partial x}\frac{\partial v}{\partial y}-\frac{\partial u}{\partial y}\frac{\partial v}{\partial x}}.$$

or $$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R} \qquad \text{...(4) [By virtue of (1) and (3)]}$$

The solutions of these equations are $u = a$ and $v = b$.

$\therefore$ $\phi(u, v) = 0$ is the required solution of (1).

Thus to solve the equation $Pp + Qq = R$.

(*i*) *form the subsidiary equations* $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$.

(*ii*) *solve these simultaneous equations by the method of § 16.10 giving $u = a$ and $v = b$ as its solutions.*

(*iii*) *write the complete solution as $\phi(u, v) = 0$* or $u = f(v)$.

Example 17.7. *Solve* $\frac{y^2z}{x}\,p + xzq = y^2$. *(Kottayam, 2005)*

Solution. Rewriting the given equation as

$$y^2zp + x^2zq = y^2x,$$

The subsidiary equations are $\frac{dx}{y^2z}=\frac{dy}{x^2z}=\frac{dz}{y^2x}$

The first two fractions give $x^2dx = y^2dy$.

Integrating, we get $x^3 - y^3 = a$ ...(*i*)

Again the first and third fractions give $xdx = zdz$

Integrating, we get $x^2 - z^2 = b$ ...(*ii*)

Hence from (*i*) and (*ii*), the complete solution is

$$x^3 - y^3 = f(x^2 - z^2).$$

Example 17.8. *Solve* $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx.$ *(V.T.U., 2010 ; S.V.T.U., 2009)*

Solution. Here the subsidiary equations are $\frac{dx}{mz-ny}=\frac{dy}{mx-lz}=\frac{dz}{ly-mx}$

Using multipliers x, y, and z, we get each fraction $= \frac{xdx + ydy + zdz}{0}$

$\therefore$ $xdx + ydy + zdz = 0$ which on integration gives $x^2 + y^2 + z^2 = a$ ...(*i*)

Again using multipliers l, m and n, we get each fraction $= \frac{ldx + mdy + ndz}{0}$

$\therefore$ $ldx + mdy + ndx = 0$ which on integration gives $lx + my + nz = b$ ...(*ii*)

Hence from (*i*) and (*ii*), the required solution is $x^2 + y^2 + z^2 = f(lx + my + nz)$.

Example 17.9. *Solve* $(x^2 - y^2 - z^2)\,p + 2xyq = 2xz.$ *(V.T.U., 2010 ; Anna, 2009 ; S.V.T.U., 2008)*

Solution. Here the subsidiary equations are $\frac{dx}{x^2-y^2-z^2}=\frac{dy}{2xy}=\frac{dz}{2xz}$

From the last two fractions, we have $\frac{dy}{y}=\frac{dz}{z}$

which on integration gives $\log y = \log z + \log a$ or $y/z = a$ ...(*i*)

Using multipliers x, y and z, we have

$$\text{each fraction} = \frac{xdx + ydy + zdz}{x(x^2+y^2+z^2)} \quad \therefore \quad \frac{2xdx + 2ydy + 2zdz}{x^2+y^2+z^2}=\frac{dz}{z}$$

which on integration gives $\log(x^2+y^2+z^2) = \log z + \log b$

or $$\frac{x^2+y^2+z^2}{z} = b \qquad ...(ii)$$

Hence from (*i*) and (*ii*), the required solution is $x^2+y^2+z^2 = zf(y/z)$.

Example 17.10. *Solve* $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$. *(P.T.U., 2009 ; Bhopal, 2008 ; S.V.T.U. 2007)*

Solution. Here the subsidiary equations are

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

Using the multipliers $1/x$, $1/y$ and $1/z$, we have

$$\text{each fraction} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\therefore \quad \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0 \text{ which on integration gives}$$

$$\log x + \log y + \log z = \log a \quad \text{or} \quad xyz = a \qquad ...(i)$$

Using the multipliers $\frac{1}{x^2}, \frac{1}{y^2}$ and $\frac{1}{z^2}$, we get

$$\text{each fraction} = \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{0}$$

$$\therefore \quad \frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2} = 0, \text{ which on integrating gives}$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0 \qquad ...(ii)$$

Hence from (*i*) and (*ii*), the complete solution is

$$xyz = f\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right).$$

Example 17.11. *Solve* $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. *(Bhopal, 2008 ; V.T.U., 2006 ; Madras, 2000)*

Solution. Here the subsidiary equations are

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} \qquad ...(i)$$

$$\text{Each of these equations} = \frac{dx - dy}{x^2 - y^2 - (y - x)z} = \frac{dy - dz}{y^2 - z^2 - x(z - y)}$$

i.e., $$\frac{d(x-y)}{(x-y)(x+y+z)} = \frac{d(y-z)}{(y-z)(x+y+z)} \quad \text{or} \quad \frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$$

Integrating, $$\log(x-y) = \log(y-z) + \log c \quad \text{or} \quad \frac{x-y}{y-z} = c \qquad ...(ii)$$

$$\text{Each of the subsidiary equations } (i) = \frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz}$$

$$= \frac{xdx + ydy + zdz}{(x+y+z)(x^2+y^2+z^2-yz-zx-xy)} \qquad ...(iii)$$

$$\text{Also each of the subsidiary equations} = \frac{dx + dy + dz}{x^2+y^2+z^2-yz-zx-xy} \qquad ...(iv)$$

Equating (*iii*) and (*iv*) and cancelling the common factor, we get

$$\frac{xdx + ydy + zdz}{x + y + z} = dx + dy + dz$$

or $$\int(xdx + ydy + zdz) = \int(x + y + z)d\,(x + y + z) + c'$$

or $$x^2 + y^2 + z^2 = (x + y + z)^2 + 2c' \quad \text{or} \quad xy + yz + zx + c' = 0 \qquad ...(v)$$

Combining (*ii*) and (*v*), the general solution is

$$\frac{x - y}{y - z} = f\,(xy + yz + zx).$$

PROBLEMS 17.3

Solve the following equations :

1. $xp + yq = 3z$.
2. $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$.
3. $(z - y)p + (x - z)q = y - x$.
4. $p \cos(x + y) + q \sin(x + y) = z$.
5. $pyz + qzx = xy$.
6. $p \tan x + q \tan y = \tan z$.
7. $p - q = \log(x + y)$.
8. $xp - yq = y^2 - x^2$ (*J.N.T.U., 2002 S*)
9. $(y + z)p - (z + x)q = x - y$.
10. $x(y - z)p + y(z - x)q = z(x - y)$. (*Bhopal, 2007*)
11. $x(y^2 - z^2)p + y(z^2 - x^2)q - z(x^2 - y^2) = 0$. (*V.T.U., 2010 ; Anna, 2008*)
12. $y^2p - xyq = x(z - 2y)$. (*S.V.T.U., 2008*)
13. $(y^2 + z^2)p - xyq + zx = 0$. (*P.T.U., 2009 ; V.T.U., 2009*)
14. $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$. (*Kerala, 2005*)
15. $px(z - 2y^2) = (z - qy)(z - y^2 - 2x^3)$.

17.6 NON-LINEAR EQUATIONS OF THE FIRST ORDER

Those equations in which p and q occur other than in the first degree are called *non-linear partial differential equations of the first order*. The *complete solution* of such an equation contains only two arbitrary constants (*i.e.*, equal to the number of independent variables involved) and the particular integral is obtained by giving particular values to the constants.]

Here we shall discuss four standard forms of these equations.

Form I. f(p, q) = 0, *i.e., equations containing p and q only.*

Its complete solution is $z = ax + by + c$ ...(1)

where a and b are connected by the relation $f(a, b) = 0$ ...(2)

[Since from (1), $p = \dfrac{\partial z}{\partial x} = a$ and $q = \dfrac{\partial z}{\partial y} = b$, which when substituted in (2) give $f(p, q) = 0$].

Expressing (2) as $b = \phi(a)$ and substituting this value of b in (1), we get the required solution as $z = ax + \phi(a)y + c$ in which a and c are arbitrary constants.

Example 17.12. *Solve* $p - q = 1$. (*Anna, 2009*)

Solution. The complete solution is $z = ax + by + c$ where $a - b = 1$

Hence $z = ax + a - 1y + c$ is the desired solution.

Example 17.13. *Solve* $x^2p^2 + y^2q^2 = z^2$. (*Anna, 2008 ; Bhopal, 2008 ; Kerala, 2005 ; Kurukshetra, 2005*)

Solution. Given equation can be reduced to the above form by writing it as

$$\left(\frac{x}{z}\cdot\frac{\partial z}{\partial x}\right)^2 + \left(\frac{y}{z}\cdot\frac{\partial z}{\partial y}\right)^2 = 1 \qquad ...(i)$$

and setting $\dfrac{dx}{x} = du, \dfrac{dy}{y} = dv, \dfrac{dz}{z} = dw$ so that $u = \log x, v = \log y, w = \log z$.

Then (*i*) becomes $$\left(\frac{\partial w}{\partial u}\right)^2 + \left(\frac{\partial w}{\partial v}\right)^2 = 1$$

i.e., $P^2 + Q^2 = 1$ where $P = \dfrac{\partial w}{\partial u}$ and $Q = \dfrac{\partial w}{\partial v}$.

Its complete solution is $w = au + bv + c$ ...(*ii*)

where $a^2 + b^2 = 1$ or $b = \sqrt{(1 - a^2)}$.

$\therefore$ (*ii*) becomes $w = au + \sqrt{(1-a^2)}\, v + c$

or $\log z = a \log x + \sqrt{(1-a^2)} \log y + c$ which is the required solution.

Form II. f(z, p, q) = 0, *i.e., equations not containing x and y.*

As a trial solution, assume that z is a function of $u = x + ay$, where a is an arbitrary constant.

$$\therefore \quad p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du} \qquad q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = a\frac{dz}{du}$$

Substituting the values of p and q in $f(z, p, q) = 0$, we get

$$f\left(z, \frac{\partial z}{\partial u}, a\frac{dz}{du}\right) = 0$$ which is an ordinary differential equation of the first order.

Rewiting it as $\dfrac{dz}{du} = \phi(z, a)$ it can be easily integrated giving

$F(z, a) = u + b$, or $x + ay + b = F(z, a)$ which is the desired complete solution.

Thus to solve $f(z, p, q) = 0$,

(*i*) *assume* $u = x + ay$ *and substitute* $p = dz/du$, $q = a\, dz/du$ *in the given equation;*

(*ii*) *solve the resulting ordinary differential equation in z and u;*

(*iii*) *replace u by* $x + ay$.

Example 17.14. *Solve* $p(1 + q) = qz$. (*Madras, 2000 S*)

Solution. Let $u = x + ay$, so that $p = dz/du$ and $q = a\, dz/du$.

Substituting these values of p and q in the given equation, we have

$$\frac{dz}{du}\left(1 + a\frac{dz}{du}\right) = az\frac{dz}{du} \text{ or } a\frac{dz}{du} = az - 1 \quad \text{or} \quad \int \frac{a\, dz}{az - 1} = \int du + b$$

or $\log(az - 1) = u + b$ or $\log(az - 1) = x + ay + b$

which is the required complete solution.

Example 17.15. *Solve* $q^2 = z^2p^2(1 - p^2)$. (*J.N.T.U., 2005 ; Kerala, 2005*)

Solution. Setting $u = y + ax$ and $z = f(u)$, we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = a\frac{dz}{du} \text{ and } q = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \frac{dz}{du}$$

$\therefore$ The given equation becomes $\left(\dfrac{dz}{du}\right)^2 = a^2z^2\left(\dfrac{dz}{du}\right)^2\left\{1 - a^2\left(\dfrac{dz}{du}\right)^2\right\}$ ...(*i*)

or $$a^4z^2\left(\frac{dz}{du}\right)^2 = a^2z^2 - 1 \quad \text{or} \quad \frac{dz}{du} = \frac{\sqrt{(a^2z^2 - 1)}}{a^2z}$$

Integrating, $$\int \frac{a^2z}{\sqrt{(a^2z^2 - 1)}}\, dz = \int du + c \quad \text{or} \quad (a^2z^2 - 1)^{1/2} = u + c$$

i.e., $a^2z^2 = (y + ax + c)^2 + 1$ $[\because u = y + ax]$

The second factor in (*i*) is $dz/du = 0$. Its solution is $z = c'$.

Example 17.16. *Solve* $z^2(p^2x^2 + q^2) = 1$. (*Bhopal, 2008 S*)

Solution. Given equation can be reduced to the above form by writing it as

$$z^2\left[\left(x\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\right]=1 \qquad ...(i)$$

Putting $X = \log x$, so that $x\dfrac{\partial z}{\partial x}=\dfrac{\partial z}{\partial X}$, (i) takes the standard form

$$z^2\left[\left(\frac{\partial z}{\partial X}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\right]=1 \qquad ...(ii)$$

Let $u = X + ay$ and put $\dfrac{\partial z}{\partial X}=\dfrac{dz}{du}$ and $\dfrac{\partial z}{\partial y}=a\dfrac{dz}{du}$ in (ii), so that

$$z^2\left[\left(\frac{dz}{du}\right)^2+a^2\left(\frac{dz}{du}\right)^2\right]=1 \quad \text{or} \quad \sqrt{(1+a^2)}\,zdz=\pm du$$

Integrating, $\sqrt{(1+a^2)}\,z^2 = \pm 2u + b = \pm 2(X + ay) + b$

or $z^2\sqrt{(1+a^2)} = \pm 2(\log x + ay) + b$

which is the complete solution required.

Form III. $f(x, p) = F(y, q)$, *i.e., equations in which z is absent and the terms containing x and p can be separated from those containing y and q.*

As a trial solution assume that $f(x, p) = F(y, q) = a$, say

Then solving for p, we get $p = \phi(x)$

and solving for q, we get $q = \psi(y)$

Since $dz = \dfrac{\partial z}{\partial x}dx + \dfrac{\partial z}{\partial y}dy = pdx + qdy$

$\therefore$ $dz = \phi(x)dx + \psi(y)dy$

Integrating, $z = \int \phi(x)dx + \int \psi(y)dy + b$

which is the desired complete solution containing two constants a and b.

Example 17.17. *Solve $p^2 + q^2 = x + y$.* *(Bhopal, 2006 ; Madras, 2003)*

Solution. Given equation is $p^2 - x = y - q^2 = a$, say

$\therefore$ $p^2 - x = a$ gives $p = \sqrt{(a + x)}$

and $y - q^2 = a$ gives $q = \sqrt{(y - a)}$

Substituting these values of p and q in $dz = pdx + qdy$, we get

$$dz = \sqrt{(a+x)}\,dx + \sqrt{(y-a)}\,dy$$

$\therefore$ integrating gives, $z = \dfrac{2}{3}(a+x)^{3/2} + \dfrac{2}{3}(y-a)^{3/2} + b$

which is the required complete solution.

Example 17.18. *Solve $z^2(p^2 + q^2) = x^2 + y^2$.* *(Bhopal, 2008)*

Solution. The equation can be reduced to the above form by writing it as

$$\left(z\frac{\partial z}{\partial x}\right)^2+\left(z\frac{\partial z}{\partial y}\right)^2=x^2+y^2 \qquad ...(i)$$

and putting $zdz = dZ$, *i.e.*, $Z = \dfrac{1}{2}z^2$

$$\therefore \quad \frac{\partial Z}{\partial x}=\frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial x}=z\frac{\partial z}{\partial x}=P$$

and
$$\frac{\partial Z}{\partial y}=\frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial y}=z\frac{\partial z}{\partial y}=Q$$

$\therefore$ (i) becomes $$P^2 + Q^2 = x^2 + y^2$$

or $$P^2 - x^2 = y^2 - Q^2 = a, \text{ say.}$$

$\therefore$ $$P = \sqrt{(x^2 + a)} \text{ and } Q = \sqrt{(y^2 - a)}.$$

$\therefore$ $$dZ = Pdx + Qdy \text{ gives}$$

$$dZ = \sqrt{(x^2 + a)}\, dx + \sqrt{(y^2 - a)}\, dy$$

Integrating, we have $$Z = \frac{1}{2} x \sqrt{(x^2 + a)} + \frac{1}{2} a \log [x + \sqrt{(x^2 + a)}]$$

$$+ \frac{1}{2} y \sqrt{(y^2 - a)} - \frac{1}{2} a \log [y + \sqrt{(y^2 - a)}] + b$$

or $$z^2 = x \sqrt{(x^2 + a)} + y \sqrt{(y^2 - a} + a \log \frac{x + \sqrt{(x^2 + a)}}{y + \sqrt{(y^2 - a)}} + 2b$$

which is the required complete solution.

Example 17.19. *Solve* $(x + y)(p + q)^2 + (x - y)(p - q)^2 = 1.$

(Bhopal, 2006 ; Rajasthan, 2006 ; V.T.U., 2003)

Solution. This equation can be reduced to the form $f(x, q) = F(y, q)$ by putting $u = x + y, v = x - y$ and taking $z = z(u, v)$.

Then $$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = P + Q$$

and $$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = P - Q, \text{ where } P = \frac{\partial z}{\partial u}, Q = \frac{\partial z}{\partial v}$$

Substituting these, the given equation reduces to

$$u(2P)^2 + v(2Q)^2 = 1 \quad \text{or} \quad 4P^2u = 1 - 4Q^2v = a \text{ (say)}$$

$$P = \pm \frac{1}{2}\sqrt{\frac{a}{u}}, Q = \pm \frac{1}{2}\sqrt{\frac{1-a}{v}}$$

$\therefore$ $$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv = Pdu + Qdv$$

$$= \pm \frac{\sqrt{a}}{2} \frac{du}{\sqrt{u}} \pm \frac{\sqrt{1-a}}{2} \frac{dv}{\sqrt{v}}$$

Integrating, we have $$z = \pm \sqrt{a}\sqrt{u} \pm \sqrt{1-a}\sqrt{v} + b$$

or $$z = \pm \sqrt{a(x+y)} \pm \sqrt{(1-a)(x-y)} + b$$

which is the required complete solution.

Form IV. z = px + qy + f(p, q): *an equation analogous to the Clairaut's equation (§ 11.14).*

Its complete solution is $z = ax + by + f(a, b)$ which is obtained by writing a for p and b for q in the given equation.

Example 17.20. *Solve* $z = px + qy + \sqrt{(1 + p^2 + q^2)}$. *(Anna, 2009)*

Solution. Given equation is of the form $z = px + qy + f(p, q)$ where $f(p, q) = \sqrt{(1 + p^2 + q^2)}$

$\therefore$ Its complete solution is $z = ax + by + \sqrt{(1 + a^2 + b^2)}$.

PROBLEMS 17.4

Obtain the complete solution of the following equations :

1. $pq + p + q = 0$.
2. $p^2 + q^2 = 1$. *(Osmania, 2000)*
3. $z = p^2 + q^2$. *(Anna, 2005 S ; J.N.T.U., 2002 S)*
4. $p(1 - q^2) = q(1 - z)$ *(Anna, 2006)*
5. $yp + xq + pq = 0$.
6. $p + q = \sin x + \sin y$.

7. $p^2 - q^2 = x - y$.
8. $\sqrt{p} + \sqrt{q} = x + y$.
9. $p^2 + q^2 = x^2 + y^2$. (*Osmania, 2003*)
10. $z = px + qy + \sin(x + y)$.
11. $\sqrt{p} + \sqrt{q} = 2x$. (*J.N.T.U., 2006*)
12. $z = px + qy - 2\sqrt{(pq)}$.
13. $(x - y)(px - qy) = (p - q)^2$. [**Hint.** Use $x + y = u, xy = v$]

17.7 CHARPIT'S METHOD*

We now explain a general method for finding the complete integral of a non-linear partial differential equation which is due to Charpit.

Consider the equation

$$f(x, y\ z, p, q) = 0 \qquad ...(1)$$

Since z depends on x and y, we have

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = pdx + qdy \qquad ...(2)$$

Now if we can find another relation involving x, y, z, p, q such as $\phi(x, y, z, p, q) = 0$ ...(3)
then we can solve (1) and (3) for p and q and substitute in (2). This will give the solution provided (2) is integrable.

To determine ϕ, we differentiate (1) and (3) with respect to x and y giving

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = 0 \qquad ...(4)$$

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z} p + \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial \phi}{\partial q}\frac{\partial q}{\partial x} = 0 \qquad ...(5)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q + \frac{\partial f}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial f}{\partial q}\frac{\partial q}{\partial y} = 0 \qquad ...(6)$$

$$\frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z} q + \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial \phi}{\partial q}\frac{\partial q}{\partial y} = 0 \qquad ...(7)$$

Eliminating $\frac{\partial p}{\partial x}$ between the equations (4) and (5), we get

$$\left(\frac{\partial f}{\partial x}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial x}\frac{\partial f}{\partial p}\right) + \left(\frac{\partial f}{\partial z}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial z}\frac{\partial f}{\partial p}\right) p + \left(\frac{\partial f}{\partial q}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial q}\frac{\partial f}{\partial p}\right)\frac{\partial q}{\partial x} = 0 \qquad ...(8)$$

Also eliminating $\frac{\partial q}{\partial y}$ between the equations (6) and (7), we obtain

$$\left(\frac{\partial f}{\partial y}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial y}\frac{\partial f}{\partial q}\right) + \left(\frac{\partial f}{\partial z}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial z}\frac{\partial f}{\partial q}\right) q + \left(\frac{\partial f}{\partial p}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial p}\frac{\partial f}{\partial q}\right)\frac{\partial p}{\partial y} = 0 \qquad ...(9)$$

Adding (8) and (9) and using $\frac{\partial q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y}$,

we find that the last terms in both cancel and the other terms, on rearrangement, give

$$\left(\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial p} + \left(\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial q} + \left(-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial z} + \left(-\frac{\partial f}{\partial p}\right)\frac{\partial \phi}{\partial x} + \left(-\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial y} = 0 \qquad ...(10)$$

i.e.,

$$\left(-\frac{\partial f}{\partial p}\right)\frac{\partial \phi}{\partial x} + \left(-\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial y} + \left(-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial z} + \left(\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial p} + \left(\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial q} = 0 \qquad ...(11)$$

This is Lagrange's linear equation (§ 17.5) with x, y, z, p, q as independent variables and ϕ as the dependent variable. Its solution will depend on the solution of the subsidiary equations

*Charpit's memoir containing this method was presented to the Paris Academy of Sciences in 1784.

$$\frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}} = \frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}} = \frac{\partial \phi}{0}$$

An integral of these equations involving p or q or both, can be taken as the required relation (3), which alongwith (1) will give the values of p and q to make (2) integrable. Of course, we should take the simplest of the integrals so that it may be easier to solve for p and q.

Example 17.21. *Solve* $(p^2 + q^2)y = qz$. *(V.T.U., 2007 ; Hissar, 2005)*

Solution. Let $f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0$ ...(*i*)

Charpit's subsidiary equations are

$$\frac{dx}{-2py} = \frac{dy}{z-2qy} = \frac{dz}{-qz} = \frac{dp}{-pq} = \frac{dq}{p^2}$$

The last two of these give $pdp + qdq = 0$

Integrating, $p^2 + q^2 = c^2$ ...(*ii*)

Now to solve (*i*) and (*ii*), put $p^2 + q^2 = c^2$ in (*i*), so that $q = c^2y/z$

Substituting this value of q in (*ii*), we get $p = c\sqrt{(z^2 - c^2y^2)}/z$

Hence $$dz = pdx + qdy = \frac{c}{z}\sqrt{(z^2 - c^2y^2)}\,dx + \frac{c^2y}{z}dy$$

or $$zdz - c^2y\,dy = c\sqrt{(z^2 - c^2y^2)}\,dx \quad \text{or} \quad \frac{\frac{1}{2}d(z^2 - c^2y^2)}{\sqrt{(z^2 - c^2y^2)}} = c\,dx$$

Integrating, we get $\sqrt{(z^2 - c^2y^2)} = cx + a$ or $z^2 = (a + cx)^2 + c^2y^2$ which is the required complete integral.

Example 17.22. *Solve* $2xz - px^2 - 2qxy + pq = 0$. *(Rajasthan, 2006)*

Solution. Let $f(x, y, z, p, q) = 2xz - px^2 - 2qxy + pq = 0$ ...(*i*)

Charpit's subsidiary equations are

$$\frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 - 2pq + 2qxy} = \frac{dp}{2z - 2qy} = \frac{dq}{0}$$

$\therefore$ $dq = 0$ or $q = a$.

Putting $q = a$ in (*i*), we get $p = \dfrac{2x(z - ay)}{x^2 - a}$

$$\therefore \quad dz = pdx + qdy = \frac{2x(z - ay)}{x^2 - a}dx + ady \quad \text{or} \quad \frac{dz - ady}{z - ay} = \frac{2x}{x^2 - a}dx$$

Integrating, $\log(z - ay) = \log(x^2 - a) + \log b$

or $z - ay = b(x^2 - a)$ or $z = ay + b(x^2 - a)$

which is the required complete solution.

Example 17.23. *Solve* $2z + p^2 + qy + 2y^2 = 0$. *(J.N.T.U., 2005 ; Kurukshetra, 2005)*

Solution. Let $f(x, y, z, p, q) = 2z + p^2 + qy + 2y^2$

Charpit's subsidiary equations are

$$\frac{dx}{-2p} = \frac{dy}{-y} = \frac{dz}{-(2p^2 + qy)} = \frac{dp}{2p} = \frac{dq}{4y + 3q}$$

From first and fourth ratios,

$$dp = -dx \quad \text{or} \quad p = -x + a$$

Substituting $p = a - x$ in the given equation, we get

$$q = \frac{1}{y}[-2z - 2y^2 - (a - x)^2]$$

$\therefore \qquad dz = pdx + qdy = (a - x)dx - \dfrac{1}{y}[2z + 2y^2 + (a - x)^2]dy$

Multiplying both sides by $2y^2$,

$$2y^2dz + 4yz\,dy = 2y^2(a - x)dx - 4y^3dy - 2y(a - x)^2dy$$

Integrating $\qquad 2zy^2 = -[y^2(a - x)^2 + y^4] + b$

or $\qquad y^2[(x - a)^2 + 2z + y^2] = b$, which is the desired solution.

PROBLEMS 17.5

Solve the following equations :

1. $z = p^2x + q^2x.$
2. $z^2 = pq\,xy.$ (*Anna, 2009 ; V.T.U., 2004*)
3. $1 + p^2 = qz.$
4. $pxy + pq + qy = yz.$ (*J.N.T.U., 2006 ; Kurukshetra, 2006*)
5. $p(p^2 + 1) + (b - z)q = 0.$
6. $q + xp = p^2.$ (*Osmania, 2003*)

17.8 HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

An equation of the form

$$\frac{\partial^n z}{\partial x^n} + k_1\frac{\partial^n z}{\partial x^{n-1}\partial y} + \dots + k_n\frac{\partial^n z}{\partial y^n} = F(x, y) \qquad ...(1)$$

in which k's are constants, is called a *homogeneous linear partial differential equation of the nth order with constant coefficients.* It is called homogeneous because all terms contain derivatives of the same order.

On writing, $\dfrac{\partial^r}{\partial x^r} = D^r$ and $\dfrac{\partial^r}{\partial y^r} = D'^r$. (1) becomes $(D^n + k_1D^{n-1}D'^r + D' + \dots + k_nD'^n)\,z = F(x, y)$

or briefly $\qquad f(D, D')z = F(x, y) \qquad ...(2)$

As in the case of ordinary linear equations with constant coefficients the complete solution of (1) consists of two parts, namely : the *complementary function* and *the particular integral.*

The complementary function is the complete solution of the equation $f(D, D')z = 0$, which must contain n arbitrary functions. The particular integral is the particular solution of equation (2).

17.9 RULES FOR FINDING THE COMPLEMENTARY FUNCTION

Consider the equation $\dfrac{\partial^2 z}{\partial x^2} + k_1\dfrac{\partial^2 z}{\partial x\partial y} + k_2\dfrac{\partial^2 z}{\partial y^2} = 0 \qquad ...(1)$

which in *symbolic form is* $(D^2 + k_1DD' + k_2D'^2)z = 0 \qquad ...(2)$

Its symbolic operator equated to zero, *i.e.*, $D^2 + k_1DD' + k_2D'^2 = 0$ is called the *auxiliary equation* (A.E.)

Let its root be $\qquad D/D' = m_1, m_2.$

Case I. *If the roots be real and distinct* then (2) is equivalent to

$$(D - m_1D')(D - m_2D')z = 0 \qquad ...(3)$$

It will be satisfied by the solution of

$$(D - m_2D')z = 0,\ i.e.,\ p - m_2q = 0.$$

This is a Lagrange's linear and the subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{-m_2} = \frac{dz}{0}, \text{ whence } y + m_2x = a \text{ and } z = b.$$

$\therefore$ its solution is $z = \phi(y + m_2x)$.

Similarly (3) will also be satisfied by the solution of

$$(D - m_1D')z = 0, \quad i.e., \quad \text{by } z = f(y + m_1x)$$

Hence the complete solution of (1) is $z = f(y + m_1x) + \phi(y + m_2x)$.